

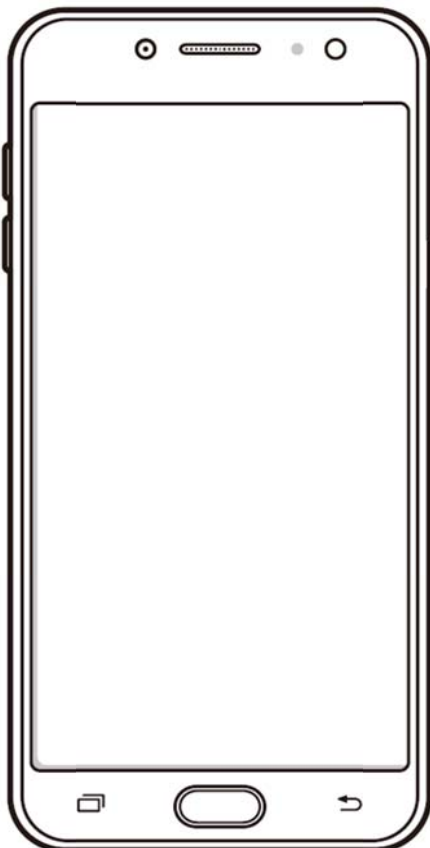
# SAMSUNG

## Mobile Device SM-C7108

# *SERVICE* **Manual**

Mobile Device

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# 1. Safety Precautions

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## 1-1. Repair Precaution

Before attempting any repair or detailed tuning, shield the device from RF noise or static electricity discharges.

Use only demagnetized tools that are specifically designed for small electronic repairs, as most electronic parts are sensitive to electromagnetic forces.

Use only high quality screwdrivers when servicing products. Low quality screwdrivers can easily damage the heads of screws.

Use only conductor wire of the properly gauge and insulation for low resistance, because of the low margin of error of most testing equipment.

We recommend 22-gauge twisted copper wire.

Hand-soldering is not recommended, because printed circuit boards (PCBs) can be easily damaged, even with relatively low heat. Never use a soldering iron with a power rating of more than 100 watts and use only lead-free solder with a melting point below 250°C (482°F).

Prior to disassembling the battery charger for repair, ensure that the AC power is disconnected. Always use the replacement parts that are registered in the SEC system. Third-party replacement parts may not function properly.

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# 1. Safety Precautions

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## 1-2. ESD(Electrostatically Sensitive Devices) Precaution

Many semiconductors and ESDs in electronic devices are particularly sensitive to static discharge and can be easily damaged by it. We recommend protecting these components with conductive anti-static bags when you store or transport them.

Always use an anti-static strap or wristband and remove electrostatic buildup or dissipate static electricity from your body before repairing ESDs.

Ensure that soldering irons have AC adapter with ground wires and that the ground wires are properly connected.

Use only desoldering tools with plastic tips to prevent static discharge.

Properly shield the work environment from accidental electrostatic discharge before opening packages containing ESDs.

The potential for static electricity discharge may be increased in low humidity environments, such as air-conditioned rooms. Increase the airflow to the working area to decrease the chance of accidental static electricity discharges.

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## 2. Specification

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### 2-1. CDMA & GPS General Specification

| ITEM                  | CDMA               | GPS          |
|-----------------------|--------------------|--------------|
| Tx Freq. range        | 824.025~848.985MHz | -            |
| Rx Freq. range        | 869.025~893.985MHz | 1575.42MHz   |
| Channel Bandwidth     | 1.23MHz            | 2MHz         |
| Channel Spacing       | 30KHz              | Not Used     |
| Number of Channel     | 832                | 1            |
| Duplex Separation     | 45MHz              | -            |
| In/Output Impedance   | 50Ohm              | 50Ohm        |
| Tx Local Frequency    | $F_{Tx} * 1.6666$  | -            |
| Rx Local Frequency    | $F_{Rx} * 2$       | $F_{Rx} * 2$ |
| TCXO Frequency        | 26MHz              | 26MHz        |
| Freq. Stability       | (FRX-45MHz)±300Hz  | -            |
| Operating Temperature | -30℃ ~ +60℃        | -30℃ ~ +60℃  |



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## 2. Specification

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| Item                               |              | GSM 850                | EGSM 900               | DCS1800                | PCS1900                |
|------------------------------------|--------------|------------------------|------------------------|------------------------|------------------------|
| Freq. Band[MHz]<br>Uplink/Downlink |              | 824~849<br>869~894     | 880~915<br>925~960     | 1710~1785<br>1805~1880 | 1850~1910<br>1930~1990 |
| ARFCN range                        |              | 128~251                | 0~124 &<br>975~1023    | 512~885                | 512~810                |
| Tx/Rx spacing                      |              | 45MHz                  | 45MHz                  | 95MHz                  | 80MHz                  |
| Mod. Bit rate/<br>Bit Period       |              | 270.833kbps<br>3.692us | 270.833kbps<br>3.692us | 270.833kbps<br>3.692us | 270.833kbps<br>3.692us |
| Time Slot Period/<br>Frame Period  |              | 576.9us<br>4.615ms     | 576.9us<br>4.615ms     | 576.9us<br>4.615ms     | 576.9us<br>4.615ms     |
| Modulation                         | GSM/<br>GPRS | 0.3GMSK                | 0.3GMSK                | 0.3GMSK                | 0.3GMSK                |
| MS Power                           |              | 33dBm<br>~5dBm         | 33dBm<br>~5dBm         | 30dBm<br>~0dBm         | 30dBm<br>~0dBm         |
| Power Class                        |              | 5pcl ~ 19pcl           | 5pcl ~ 19pcl           | 0pcl ~ 15pcl           | 0pcl ~ 15pcl           |
| Sensitivity                        |              | -102dBm                | -102dBm                | -100dBm                | -100dBm                |
| TDMA Mux                           |              | 8                      | 8                      | 8                      | 8                      |
| Cell Radius                        |              | 35Km                   | 35Km                   | 2Km                    | 2Km                    |

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## 2. Specification

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### 2-2. GSM Tx Power Class

| <b>TX Power<br/>control level</b> | <b>GSM850</b> | <b>TX Power<br/>control level</b> | <b>EGSM900</b> | <b>TX Power<br/>control level</b> | <b>DCS1800</b> | <b>TX Power<br/>control level</b> | <b>PCS1900</b> |
|-----------------------------------|---------------|-----------------------------------|----------------|-----------------------------------|----------------|-----------------------------------|----------------|
| 5                                 | 33±2 dBm      | 5                                 | 33±2 dBm       | 0                                 | 30±3 dBm       | 0                                 | 30±3 dBm       |
| 6                                 | 31±2 dBm      | 6                                 | 31±2 dBm       | 1                                 | 28±3 dBm       | 1                                 | 28±3 dBm       |
| 7                                 | 29±2 dBm      | 7                                 | 29±2 dBm       | 2                                 | 26±3 dBm       | 2                                 | 26±3 dBm       |
| 8                                 | 27±2 dBm      | 8                                 | 27±2 dBm       | 3                                 | 24±3 dBm       | 3                                 | 24±3 dBm       |
| 9                                 | 25±2 dBm      | 9                                 | 25±2 dBm       | 4                                 | 22±3 dBm       | 4                                 | 22±3 dBm       |
| 10                                | 23±2 dBm      | 10                                | 23±2 dBm       | 5                                 | 20±3 dBm       | 5                                 | 20±3 dBm       |
| 11                                | 21±2 dBm      | 11                                | 21±2 dBm       | 6                                 | 18±3 dBm       | 6                                 | 18±3 dBm       |
| 12                                | 19±2 dBm      | 12                                | 19±2 dBm       | 7                                 | 16±3 dBm       | 7                                 | 16±3 dBm       |
| 13                                | 17±2 dBm      | 13                                | 17±2 dBm       | 8                                 | 14±3 dBm       | 8                                 | 14±3 dBm       |
| 14                                | 15±2 dBm      | 14                                | 15±2 dBm       | 9                                 | 12±4 dBm       | 9                                 | 12±4 dBm       |
| 15                                | 13±2 dBm      | 15                                | 13±2 dBm       | 10                                | 10±4 dBm       | 10                                | 10±4 dBm       |
| 16                                | 11±3 dBm      | 16                                | 11±3 dBm       | 11                                | 8±4 dBm        | 11                                | 8±4 dBm        |
| 17                                | 9±3 dBm       | 17                                | 9±3 dBm        | 12                                | 6±4 dBm        | 12                                | 6±4 dBm        |
| 18                                | 7±3 dBm       | 18                                | 7±3 dBm        | 13                                | 4±4 dBm        | 13                                | 4±4 dBm        |
| 19                                | 5±3 dBm       | 19                                | 5±3 dBm        | 14                                | 2±5 dBm        | 14                                | 2±5 dBm        |
| -                                 | -             | -                                 | -              | 15                                | 0±5 dBm        | 15                                | 0±5 dBm        |

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## 2. Specification

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### 2-3. WCDMA General Specification

|                                    | WCDMA2100                              | WCDMA1900                              | WCDMA850                               | WCDMA900                               |
|------------------------------------|----------------------------------------|----------------------------------------|----------------------------------------|----------------------------------------|
| Freq. Band[MHz]<br>Uplink/Downlink | 1922~1977<br>2112~2167                 | 1852~1907<br>1932~1987                 | 824~849<br>869~894                     | 880~915<br>925~960                     |
| ARFCN range                        | UL: 9612~9888<br>DL: 10562~10838       | UL: 9262~9538<br>DL: 9662~9938         | UL: 4132~4233<br>DL: 4357~4458         | UL: 2712~2863<br>DL: 2937~3088         |
| Tx/Rx spacing                      | 190MHz                                 | 80MHz                                  | 45MHz                                  | 45MHz                                  |
| Mod. Bit rate/<br>Bit Period       | 3.84 Mcps                              | 3.84 Mcps                              | 3.84 Mcps                              | 3.84 Mcps                              |
| Time Slot Period/Frame<br>Period   | FrameLength:10ms<br>Slotlength:0.667ms | FrameLength:10ms<br>Slotlength:0.667ms | FrameLength:10ms<br>Slotlength:0.667ms | FrameLength:10ms<br>Slotlength:0.667ms |
| Modulation                         | QPSK/HQPSK                             | QPSK/HQPSK                             | QPSK/HQPSK                             | QPSK/HQPSK                             |
| MS Power                           | 24dBm~-50dBm                           | 21dBm~-50dBm                           | 24dBm~-50dBm                           | 24dBm~-50dBm                           |
| Power Class                        | 3(max+24dBm)                           | 4(max+21dBm)                           | 3(max+24dBm)                           | 3(max+24dBm)                           |
| Sensitivity                        | -106.7dBm                              | -106.7dBm                              | -106.7dBm                              | -106.7dBm                              |
| TDMA Mux                           | 8                                      | 8                                      | 8                                      | 8                                      |
| Cell Radius                        | 2Km                                    | 2Km                                    | 2Km                                    | 2Km                                    |

## 2. Specification

### 2-4. TD-SCDMA General Specification

|                                    | Band34               | Band39               |
|------------------------------------|----------------------|----------------------|
| Freq. Band[MHz]<br>Uplink/Downlink | 2010~2025            | 1880~1920            |
| ARFCN range                        | 10054~10121          | 9404~9596            |
| Tx/Rx spacing                      | /                    | /                    |
| TCXO Frequency                     | 26MHz                | 26MHz                |
| Modulation                         | QPSK/16-QAM          | QPSK/16-QAM          |
| MS Power                           | -49dBm~ 25dBm        | -49dBm~ 25dBm        |
| Power Class                        | 2 (max: 24 +1/-3dBm) | 2 (max: 24 +1/-3dBm) |
| Sensitivity                        | -108dBm/1.28MHz      | -108dBm/1.28MHz      |
| Cell Radius                        | >10Km                | >10Km                |
| In/Output Impedance                | 50Ω                  | 50Ω                  |
| Operating Temperature              | -30°C ~ +60°C        | -30°C ~ +60°C        |

## 2. Specification

### 2-5. LTE General Specification (FDD)

|                          | Band1 (FDD)                                                                       | Band3 (FDD)                                                                                                              | Band5 (FDD)                                                                        | Band7 (FDD)                                                                       | Band8 (FDD)                                                                             |
|--------------------------|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Freq. Band[MHz]<br>UL/DL | 1920~1980<br>2110~2170                                                            | 1710~1785<br>1805~1880                                                                                                   | 824~849<br>869~894                                                                 | 2500~2570<br>2620~2690                                                            | 880~915<br>925~960                                                                      |
| ARFCN range<br>UL/DL     | 18000~18599<br>0~599                                                              | 19200~19949<br>1200~1949                                                                                                 | 20400~20649<br>2400~2649                                                           | 20750~20449<br>2750~3449                                                          | 21450~21799<br>3450~3799                                                                |
| Tx/Rx spacing            | 190MHz                                                                            | 95MHz                                                                                                                    | 45MHz                                                                              | 120MHz                                                                            | 45MHz                                                                                   |
| TCXO Frequency           | 26MHz                                                                             | 26MHz                                                                                                                    | 26MHz                                                                              | 26MHz                                                                             | 26MHz                                                                                   |
| Modulation               | QPSK/16-QAM/64-QAM                                                                | QPSK/16-QAM/64-QAM                                                                                                       | QPSK/16-QAM/64-QAM                                                                 | QPSK/16-QAM/64-QAM                                                                | QPSK/16-QAM/64-QAM                                                                      |
| MS Power                 | -40dBm~ 25dBm                                                                     | -40dBm~ 25dBm                                                                                                            | -40dBm~ 25dBm                                                                      | -40dBm~ 25dBm                                                                     | -40dBm~ 25dBm                                                                           |
| Power Class              | 3 (max: 23 ±2dBm)                                                                 | 3 (max: 23 ±2dBm)                                                                                                        | 3 (max: 23 ±2dBm)                                                                  | 3 (max: 23 ±2dBm)                                                                 | 3 (max: 23 ±2dBm)                                                                       |
| Sensitivity              | -99.3(BW : 5 MHz)<br>-96.3(BW : 10 MHz)<br>-94.5(BW : 15MHz)<br>-93.3(BW : 20MHz) | -101(BW : 1.4MHz)<br>-98(BW : 3MHz)<br>-96.3(BW : 5 MHz)<br>-93.3(BW : 10 MHz)<br>-91.5(BW : 15MHz)<br>-90.3(BW : 20MHz) | -102.5(BW : 1.4MHz)<br>-99.5(BW : 3MHz)<br>-97.3(BW : 5 MHz)<br>-94.3(BW : 10 MHz) | -97.3(BW : 5 MHz)<br>-94.3(BW : 10 MHz)<br>-92.5(BW : 15MHz)<br>-91.3(BW : 20MHz) | -<br>101.5(BW : 1.4MHz)<br>-98.5(BW : 3MHz)<br>-96.3 (BW : 5 MHz)<br>-93.3(BW : 10 MHz) |
| Cell Radius              | >5Km                                                                              | >5Km                                                                                                                     | >5Km                                                                               | >5Km                                                                              | >5Km                                                                                    |
| In/Output Impedance      | 50Ω                                                                               | 50Ω                                                                                                                      | 50Ω                                                                                | 50Ω                                                                               | 50Ω                                                                                     |
| Operating Temperature    | -30℃ ~ +60℃                                                                       | -30℃ ~ +60℃                                                                                                              | -30℃ ~ +60℃                                                                        | -30℃ ~ +60℃                                                                       | -30℃ ~ +60℃                                                                             |

## 2. Specification

|                                    | Band38 (TDD)                                                                      | Band39 (TDD)                                                                      | Band40 (TDD)                                                                      | Band41 (TDD)                                                                      |
|------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Freq. Band[MHz]<br>Uplink/Downlink | 2570~2620                                                                         | 1880~1920                                                                         | 2300~2400                                                                         | 2555~2655                                                                         |
| ARFCN range                        | 37750~38249                                                                       | 38250~38649                                                                       | 38650~39649                                                                       | 40240~41240                                                                       |
| Tx/Rx spacing                      | /                                                                                 | /                                                                                 | /                                                                                 | /                                                                                 |
| TCXO Frequency                     | 26MHz                                                                             | 26MHz                                                                             | 26MHz                                                                             | 26MHz                                                                             |
| Modulation                         | QPSK/16-QAM<br>/64-QAM                                                            | QPSK/16-QAM<br>/64-QAM                                                            | QPSK/16-QAM<br>/64-QAM                                                            | QPSK/16-QAM<br>/64-QAM                                                            |
| MS Power                           | -40dBm~ 25dBm                                                                     | -40dBm~ 25dBm                                                                     | -40dBm~ 25dBm                                                                     | -40dBm~ 25dBm                                                                     |
| Power Class                        | 3 (max: 23 ±2dBm)                                                                 | 3 (max: 23 ±2dBm)                                                                 | 3 (max: 23 ±2dBm)                                                                 | 3 (max: 23 ±2dBm)                                                                 |
| Sensitivity                        | -99.3(BW : 5 MHz)<br>-96.3(BW : 10 MHz)<br>-94.5(BW : 15MHz)<br>-93.3(BW : 20MHz) | -99.3(BW : 5 MHz)<br>-96.3(BW : 10 MHz)<br>-94.5(BW : 15MHz)<br>-93.3(BW : 20MHz) | -99.3(BW : 5 MHz)<br>-96.3(BW : 10 MHz)<br>-94.5(BW : 15MHz)<br>-93.3(BW : 20MHz) | -97.3(BW : 5 MHz)<br>-94.3(BW : 10 MHz)<br>-92.5(BW : 15MHz)<br>-91.3(BW : 20MHz) |
| Cell Radius                        | >5Km                                                                              | >5Km                                                                              | >5Km                                                                              | >5Km                                                                              |
| In/Output Impedance                | 50Ω                                                                               | 50Ω                                                                               | 50Ω                                                                               | 50Ω                                                                               |
| Operating Temperature              | -30℃ ~ +60℃                                                                       | -30℃ ~ +60℃                                                                       | -30℃ ~ +60℃                                                                       | -30℃ ~ +60℃                                                                       |

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## 3. Operation Instruction and Installation

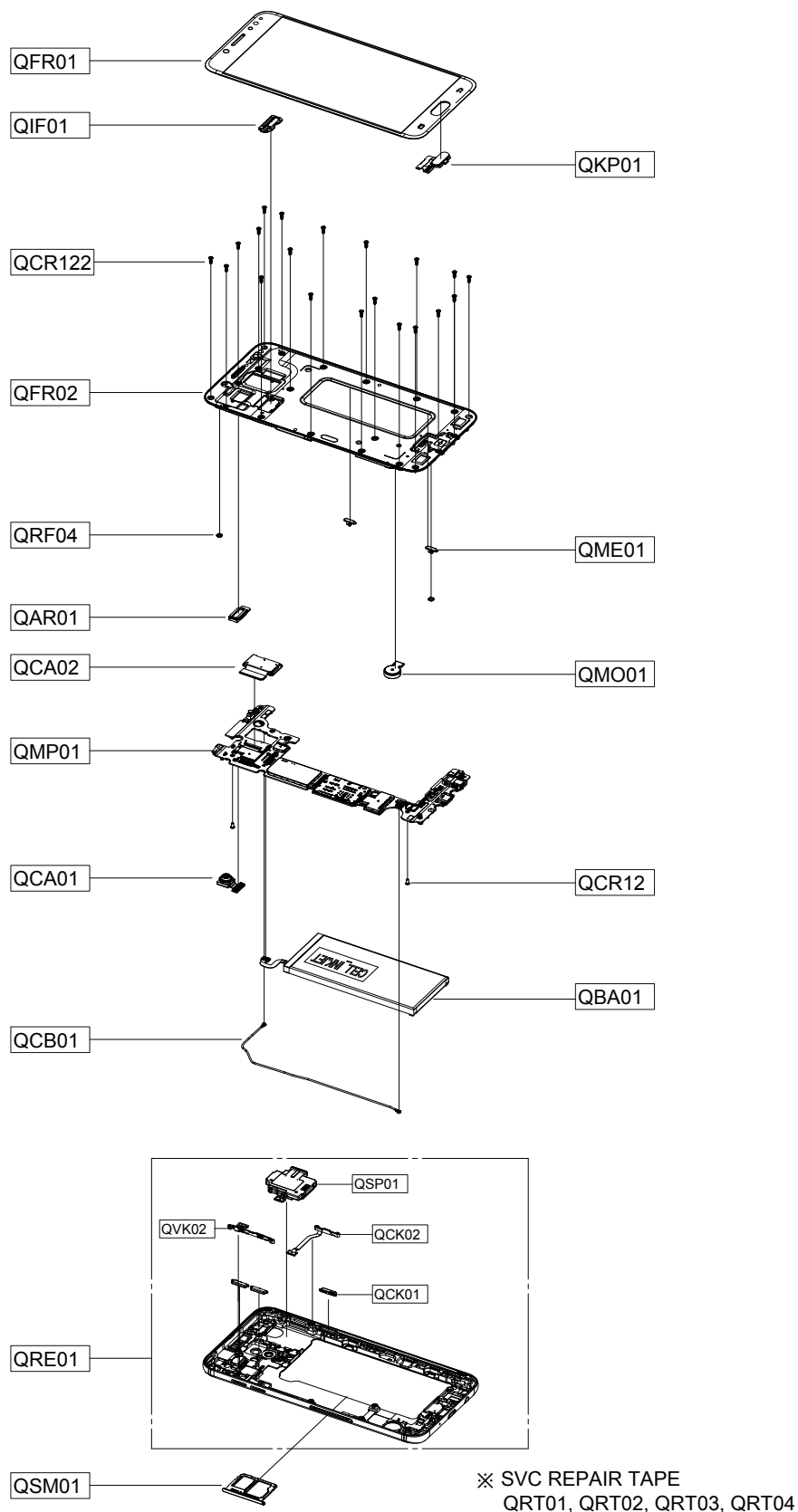
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### Main Function

| Item      | Description                                                                                                                             |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------|
| OS        | Android V7.1                                                                                                                            |
| RF        | GSM850 /GSM900/ DCS1800/ PCS1900<br>CDMA:BC0<br>WCDMA: B1/ B2/ B5/ B8<br>TDSCDMA: B34/B39<br>LTE: B1/ B3/ B5/B7/ B8/ B38/ B39/ B40/ B41 |
| Battery   | 3000mAh                                                                                                                                 |
| Base Band | Octa-Core 2.39Ghz+1.69GHz                                                                                                               |
| Other RF  | GPS, Glonass, BT4.2, USB2.0, WIFI 802.11 b/g/n, FM                                                                                      |
| Camera    | 13MP+5MP Main CAM; 16MP Sub CAM                                                                                                         |
| LCD       | 5.5." AMOLED, 1920*1080(FHD)                                                                                                            |
| RAM       | 3GB                                                                                                                                     |
| Sensor    | Accelerometer, Proximity, RGB light sensor, Fingerprint sensor, Geomagnetic sensor, Gyro Sensor                                         |
| Accessory | Charger: 5V/2A                                                                                                                          |

## 4. Exploded View and Parts List

### 4-1. Cellular phone Exploded View





## 5. MAIN Electrical Parts List

| Parts Code  | Design Loc           | Description        |
|-------------|----------------------|--------------------|
| 0401-001110 | D5002                | SDS511Q            |
| 0404-001250 | D5000,D5001          | RB520G-30          |
| 0406-001459 | ZD6000,ZD6001,ZD6006 | RD1301B            |
| 0406-001459 | ZD6007               | RD1301B            |
| 0406-001506 | ZD5004,ZD5005,ZD5006 | UCLAMP0541Z        |
| 0406-001506 | ZD5007,ZD7000,ZD7001 | UCLAMP0541Z        |
| 0406-001506 | ZD7002,ZD7003,ZD7004 | UCLAMP0541Z        |
| 0406-001506 | ZD7005,ZD7006,ZD7008 | UCLAMP0541Z        |
| 0406-001506 | ZD7009,ZD7010,ZD7011 | UCLAMP0541Z        |
| 0406-001592 | ZD6009,ZD6010,ZD6011 | PESD5V0V1BDSF      |
| 0406-001592 | ZD6012,ZD6013,ZD6014 | PESD5V0V1BDSF      |
| 0406-001592 | ZD6016,ZD7007        | PESD5V0V1BDSF      |
| 0406-001633 | D5004                | RDL3081            |
| 0406-001640 | ZD5002,ZD5003        | RCLAMP3521P.TCT    |
| 0406-001682 | ZD6020               | PS05TBUL2          |
| 0406-001709 | ZD5010,ZD6017,ZD6021 | RDL1151B           |
| 0406-001709 | ZD6022               | RDL1151B           |
| 0406-001728 | ZD5000,ZD5001,ZD5009 | ES45D3HP           |
| 0406-001740 | ZD6002,ZD6005        | UCLAMP1211P        |
| 0406-001743 | D6000,D6001,D6006    | RCLAMP3331ZA       |
| 0406-001743 | D6007,ZD6003,ZD6004  | RCLAMP3331ZA       |
| 0406-001763 | D6008,D6009,D6010    | VS3V3BC1HS         |
| 0406-001763 | D6011,D6012,D6013    | VS3V3BC1HS         |
| 0406-001763 | D6014                | VS3V3BC1HS         |
| 0406-001781 | ZD6018,ZD6019,ZD6023 | D3V3H1B2LP         |
| 0406-001781 | ZD6024               | D3V3H1B2LP         |
| 0505-002088 | Q5000,Q5002          | 2SK3541            |
| 0505-002341 | Q5001                | SI1013X-T1-GE3     |
| 0505-002388 | Q6001                | RUE003N02          |
| 0601-003338 | LED5002              | SCF0165G           |
| 0601-003569 | LED5001              | SPFCW02201AR       |
| 0601-003611 | LED5000              | SPFCW13331CRS1D4T0 |
| 1001-001650 | U5001,U6004          | DG2750DN-T1-E4     |
| 1001-001844 | U1000                | BGS12PL6           |
| 1001-001911 | U1004,U1007          | BGSA14GN10-1       |
| 1001-001956 | U2010,U2017          | BGS15MA12          |
| 1001-001969 | U2006,U2008,U2009    | MXD8631            |
| 1001-001970 | U2014,U2015          | MXD8621            |

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## 5. MAIN Electrical Parts List

|             |                   |                    |
|-------------|-------------------|--------------------|
| 1001-001992 | U1002,U1003,U1006 | RF1646             |
| 1001-001999 | U2002             | MXD8643            |
| 1001-002030 | U1001             | MXD8631SF          |
| 1108-000624 | UME5000           | KMRC10014M-B809    |
| 1201-003786 | U2003             | BGU6009            |
| 1201-003915 | PAM1000           | SKY77621-31        |
| 1201-003975 | U1011             | SFMLB7RB002        |
| 1201-003985 | U2011             | BGS8L2             |
| 1201-004100 | PAM1001           | WIPS33232-01       |
| 1201-004122 | U1008,U1009,U1012 | MXD8011HU          |
| 1201-004122 | U1013,U2000,U2001 | MXD8011HU          |
| 1201-004122 | U2012,U2013       | MXD8011HU          |
| 1201-004148 | U6000             | RT5509WSC          |
| 1202-001120 | U6006             | NCX2200GM          |
| 1203-008249 | U7009             | AP7341D-33FS4-7    |
| 1203-008250 | U6002,U7012       | AP7340D-18FS4-7    |
| 1203-008326 | U7001,U7003       | AP7340D-30FS4-7-71 |
| 1203-008603 | U6001             | AP7346             |
| 1203-008711 | U5009             | ET9530L            |
| 1203-008751 | U7002             | SM3004B            |
| 1203-008859 | U7007             | AP7341D-12FS4-7R   |
| 1203-008865 | U5005,U7004,U7005 | ET51528YB          |
| 1203-008867 | U7006             | ET51518YB          |
| 1203-008985 | U5003             | MT6355W            |
| 1203-008995 | U5008             | S2MU005X03-6030    |
| 1205-005217 | U7008             | FSA644BUCX         |
| 1205-005711 | U2004             | MT6625L            |
| 1205-005712 | U5004             | A96T316AU          |
| 1205-005813 | U1010             | MT6176V            |
| 1205-005820 | UCP3000           | MT6757V/CD         |
| 1209-002413 | U7010             | YAS539             |
| 1209-002450 | U7013             | LSM6DSL            |
| 1404-001664 | TH3000,TH3001     | NCP15WF104F03RC    |
| 1404-001724 | TH1000,TH5000     | NCP03WF104F05RL    |
| 1405-001395 | U7016,V1000,V1001 | ESPC02039ES1       |
| 1405-001395 | V1003,V1008,V1009 | ESPC02039ES1       |
| 1405-001395 | V2001,V2005,V2006 | ESPC02039ES1       |
| 1405-001395 | V2007,V2008,V2009 | ESPC02039ES1       |
| 1405-001395 | V2010,V2011,V2012 | ESPC02039ES1       |

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## 5. MAIN Electrical Parts List

|             |                      |                 |
|-------------|----------------------|-----------------|
| 1405-001395 | V6000,V6001,V6002    | ESPC02039ES1    |
| 1405-001395 | VR2000,VR2004,VR2005 | ESPC02039ES1    |
| 1405-001415 | C2110,V1005,V2003    | IMG0330C101JFR  |
| 1405-001415 | V2004                | IMG0330C101JFR  |
| 1405-001423 | V6003                | ESPC014R7JS1    |
| 2007-003015 | R5035,R5036          | MCR01MZP5J2R2   |
| 2007-007142 | R6032,R7013          | RC1005F103CS    |
| 2007-007199 | R6033,R6046          | TSR16GJ2R7V     |
| 2007-007528 | R3020,R3021          | RK73H1ETP1501F  |
| 2007-007698 | R5012                | RK73H1ETP5101F  |
| 2007-007741 | R1009,R3042,R3043    | M5534204B10E0R  |
| 2007-007741 | R5007,R5013,R5029    | M5534204B10E0R  |
| 2007-007741 | R5042,R5044,R5049    | M5534204B10E0R  |
| 2007-007741 | R5054,R6013,R7015    | M5534204B10E0R  |
| 2007-007766 | R1011                | RK73H1ETP2001F  |
| 2007-007946 | R6096                | 470K,1%         |
| 2007-007981 | R6094                | ERJ2RKF1803X    |
| 2007-008045 | R3051,R5075,R5076    | ERJ1GEJ101C     |
| 2007-008137 | R1010                | RC1005F243CS    |
| 2007-008167 | R3023                | RC0402FR-07120K |
| 2007-008210 | R1000,R6004          | RMC1/20-180JTP  |
| 2007-008294 | R3028                | RC1005F330CS    |
| 2007-008304 | R5027                | RC1005F304CS    |
| 2007-008312 | R1008,R3030,R3031    | RC1005F394CS    |
| 2007-008403 | R3026                | RC1005F364CS    |
| 2007-008531 | R5037,R7001,R7008    | MCR006MZPJ0100  |
| 2007-008531 | R7022                | MCR006MZPJ0100  |
| 2007-008582 | R1002,R1003          | RC0603J301CS    |
| 2007-008587 | R5045,R5047          | ERJ1GEJ561C     |
| 2007-008588 | R7000                | MCR006YZPJ222   |
| 2007-008788 | R3006                | RC0603J270CS    |
| 2007-008816 | R6038                | MCR006MZPJ822   |
| 2007-009111 | R7014                | RC0603J220CS    |
| 2007-009157 | R5003,R5017,R5019    | MCR006YZPF1003  |
| 2007-009157 | R5020,R5024,R5080    | MCR006YZPF1003  |
| 2007-009157 | R6034,R6048          | MCR006YZPF1003  |
| 2007-009158 | R5004                | MCR006MZPF1503  |
| 2007-009168 | R5022,R6036          | MCR006MZPJ563   |
| 2007-009171 | R5023                | MCR006YZPJ513   |

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## 5. MAIN Electrical Parts List

|             |                   |                    |
|-------------|-------------------|--------------------|
| 2007-009212 | R5009,R5071,R6017 | MCR006YZPF1001     |
| 2007-009315 | R5000,R6001,R6002 | RC0603J4R7CS       |
| 2007-009315 | R6003,R6005,R6006 | RC0603J4R7CS       |
| 2007-009315 | R6007,R6008,R6009 | RC0603J4R7CS       |
| 2007-009315 | R6010,R6011,R6012 | RC0603J4R7CS       |
| 2007-009352 | R3022,R3027       | MCR006YZPF4701     |
| 2007-009354 | R6039             | ERJ1GEF3301C       |
| 2007-009408 | R3000,R3001,R3007 | ERJ1GEF222C        |
| 2007-009408 | R3008,R3009,R3010 | ERJ1GEF222C        |
| 2007-009408 | R3011,R3012,R3013 | ERJ1GEF222C        |
| 2007-009408 | R3014,R3015,R3016 | ERJ1GEF222C        |
| 2007-009408 | R5034             | ERJ1GEF222C        |
| 2007-009801 | R5016,R5018,R5056 | MCR006YZPF1000     |
| 2007-009801 | R6031             | MCR006YZPF1000     |
| 2007-009805 | R1001             | RC0603F510CS       |
| 2007-009866 | R5021             | MCR006YZPF2003     |
| 2007-009879 | R3047             | ERJ1GEF1101C       |
| 2007-009920 | R5058,R5060       | RC0603F241CS       |
| 2007-009954 | R5028             | MCR006YZPF6802     |
| 2007-009969 | R6024             | MCR006YZPF1201     |
| 2007-010029 | R5032             | MCR006YZP5F5104    |
| 2007-010202 | R5063,R5064,R6028 | RC0603F473CS       |
| 2007-010685 | R6000             | RC0201FR-071ML     |
| 2007-011377 | R6037             | RC0603F433CS       |
| 2007-011532 | R6091             | RC0603F121CS       |
| 2007-011648 | R5025,R5026,R5031 | RC0603F1R0CS       |
| 2007-012038 | R5005             | ESR03EZPF10R0      |
| 2007-012068 | R5033             | RES                |
| 2203-000995 | C5104             | GRP1555C1H470J     |
| 2203-001153 | C5110             | GRP1555C1H680J     |
| 2203-005682 | C2001,C2046,C2067 | GRP0335C1E330JD01E |
| 2203-005682 | C2081,C5074,C5111 | GRP0335C1E330JD01E |
| 2203-005682 | C5146,C5921,C6000 | GRP0335C1E330JD01E |
| 2203-005682 | C6001,C6006,C6013 | GRP0335C1E330JD01E |
| 2203-005682 | C6017,C7012,C7020 | GRP0335C1E330JD01E |
| 2203-005682 | C7053,C7054       | GRP0335C1E330JD01E |
| 2203-005717 | C1027,C1038,C1045 | GRP0335C1E270JD01E |
| 2203-005719 | C1039,C1055,C6034 | GRM0335C1E390JD01E |
| 2203-005725 | C5075             | GRP0335C1E220JD01E |

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## 5. MAIN Electrical Parts List

|             |                   |                    |
|-------------|-------------------|--------------------|
| 2203-005726 | C1060,C5151       | GRM0335C1E560J     |
| 2203-005727 | C1135,C5136       | CL03C120JA3GNNC    |
| 2203-005729 | C5044,C6023,C6024 | CL03C150JAGC       |
| 2203-005729 | C6030,C7014       | CL03C150JAGC       |
| 2203-005731 | C1044,C6041,C6046 | GRM0335C1E180J     |
| 2203-005732 | C5105,C5900,C6018 | GRP0335C1E680JD01E |
| 2203-005732 | C6020,C6029       | GRP0335C1E680JD01E |
| 2203-005734 | C2047,C2095,C2107 | GRM0335C1E820JD01D |
| 2203-005736 | C1002,C1003,C1008 | GRP0335C1E101JD01E |
| 2203-005736 | C1009,C1011,C1013 | GRP0335C1E101JD01E |
| 2203-005736 | C1015,C1021,C1022 | GRP0335C1E101JD01E |
| 2203-005736 | C1023,C1026,C1029 | GRP0335C1E101JD01E |
| 2203-005736 | C1033,C1040,C1053 | GRP0335C1E101JD01E |
| 2203-005736 | C1057,C1061,C1064 | GRP0335C1E101JD01E |
| 2203-005736 | C1065,C1066,C1070 | GRP0335C1E101JD01E |
| 2203-005736 | C1072,C1076,C1092 | GRP0335C1E101JD01E |
| 2203-005736 | C1110,C1113,C1125 | GRP0335C1E101JD01E |
| 2203-005736 | C1126,C1127,C1128 | GRP0335C1E101JD01E |
| 2203-005736 | C1129,C1130,C1131 | GRP0335C1E101JD01E |
| 2203-005736 | C1134,C1136,C1137 | GRP0335C1E101JD01E |
| 2203-005736 | C1139,C1141,C1142 | GRP0335C1E101JD01E |
| 2203-005736 | C1149,C1151,C1153 | GRP0335C1E101JD01E |
| 2203-005736 | C1154,C1155,C1157 | GRP0335C1E101JD01E |
| 2203-005736 | C1158,C1159,C1163 | GRP0335C1E101JD01E |
| 2203-005736 | C1165,C1166,C1167 | GRP0335C1E101JD01E |
| 2203-005736 | C1168,C1169,C1170 | GRP0335C1E101JD01E |
| 2203-005736 | C1186,C1190,C1198 | GRP0335C1E101JD01E |
| 2203-005736 | C1203,C1207,C1210 | GRP0335C1E101JD01E |
| 2203-005736 | C1213,C1214,C1216 | GRP0335C1E101JD01E |
| 2203-005736 | C1218,C1219,C1220 | GRP0335C1E101JD01E |
| 2203-005736 | C1229,C2002,C2003 | GRP0335C1E101JD01E |
| 2203-005736 | C2004,C2006,C2007 | GRP0335C1E101JD01E |
| 2203-005736 | C2010,C2013,C2014 | GRP0335C1E101JD01E |
| 2203-005736 | C2015,C2018,C2020 | GRP0335C1E101JD01E |
| 2203-005736 | C2021,C2023,C2030 | GRP0335C1E101JD01E |
| 2203-005736 | C2039,C2050,C2052 | GRP0335C1E101JD01E |
| 2203-005736 | C2069,C2080,C2091 | GRP0335C1E101JD01E |
| 2203-005736 | C2094,C2098,C2099 | GRP0335C1E101JD01E |
| 2203-005736 | C2103,C2104,C6033 | GRP0335C1E101JD01E |

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## 5. MAIN Electrical Parts List

|             |                   |                    |
|-------------|-------------------|--------------------|
| 2203-005736 | C6039,C7026,C7031 | GRP0335C1E101JD01E |
| 2203-005736 | C7034             | GRP0335C1E101JD01E |
| 2203-005739 | C1095             | GRM0335C1E4ROCD01D |
| 2203-005777 | C1006,C1083,C1114 | CL03C010CA3GNNC    |
| 2203-005777 | C2008,L1050       | CL03C010CA3GNNC    |
| 2203-005789 | C1012,C1042,C1046 | GRM0335C1ER50B     |
| 2203-005789 | C1090,C1176,C1212 | GRM0335C1ER50B     |
| 2203-005789 | C2025             | GRM0335C1ER50B     |
| 2203-005792 | C1050,C1161,C1208 | CL03C1R5CA3GNNC    |
| 2203-005806 | C1140,C1147,C1164 | GRP033R71C102KD01E |
| 2203-005806 | C2019,C2071       | GRP033R71C102KD01E |
| 2203-006120 | C5039,C5041       | GRP033R71C221KD01E |
| 2203-006123 | C1025,C1051       | GRM0335C1E5R6CD01D |
| 2203-006194 | C1175,C2017,C6027 | GRP033R70J103KA01E |
| 2203-006305 | C1047,R6052       | CM03CG100J25AH     |
| 2203-006318 | C1007             | GRP0335C1E1R2CD01  |
| 2203-006379 | C2024             | GRP033R70J472KA01  |
| 2203-006399 | C5071             | GRM155R60J105KE19D |
| 2203-006400 | C1063,C1152,C1173 | GRM033R71E102K     |
| 2203-006400 | C2033,C2049,C2059 | GRM033R71E102K     |
| 2203-006400 | C5000,C5030,C5149 | GRM033R71E102K     |
| 2203-006400 | C6037             | GRM033R71E102K     |
| 2203-006410 | C2012,C6016       | CL03C1R8CA3GNNC    |
| 2203-006556 | C1014,C1049,C1056 | GRM0335C1E470JD01D |
| 2203-006556 | C1084,C1086,C1132 | GRM0335C1E470JD01D |
| 2203-006556 | C1150,C5139,C5904 | GRM0335C1E470JD01D |
| 2203-006556 | C5906             | GRM0335C1E470JD01D |
| 2203-006562 | C1143,C2117,C5027 | CV05X5R105K10AH    |
| 2203-006562 | C5052,C5073,C5080 | CV05X5R105K10AH    |
| 2203-006562 | C5081,C5098,C5108 | CV05X5R105K10AH    |
| 2203-006562 | C5142,C5144,C5911 | CV05X5R105K10AH    |
| 2203-006562 | C5912,C5914,C6051 | CV05X5R105K10AH    |
| 2203-006562 | C7001,C7008,C7023 | CV05X5R105K10AH    |
| 2203-006562 | C7035,C7037,C7039 | CV05X5R105K10AH    |
| 2203-006562 | C7041,C7043,C7046 | CV05X5R105K10AH    |
| 2203-006562 | C7050             | CV05X5R105K10AH    |
| 2203-006604 | C6015             | GRM0335C1E2R4BD01D |
| 2203-006611 | C1124,C2115       | GRM0335C1E2R7CD01D |
| 2203-006647 | C5132             | GRM033R60J473KE19D |

## 5. MAIN Electrical Parts List

|             |                   |                    |
|-------------|-------------------|--------------------|
| 2203-006665 | C1019,C1034       | GRM0335C1E2R2CD01D |
| 2203-006707 | C1099,C1100,C5006 | GRM0335C1E4R7C     |
| 2203-006707 | C5008,C5011       | GRM0335C1E4R7C     |
| 2203-006841 | C1079,C5002,C5072 | CV05X5R105K16AH    |
| 2203-006841 | C5097,C5913       | CV05X5R105K16AH    |
| 2203-006846 | C1111             | ECJZEC1E2R0B       |
| 2203-006872 | C4066,C5042,C5048 | GRM155R60J225ME15D |
| 2203-006872 | C5920,C7045       | GRM155R60J225ME15D |
| 2203-006978 | C5109             | GMK107BJ105KA-T    |
| 2203-007165 | C1101,C5009,C5019 | CL05A475KQ5NRNC    |
| 2203-007165 | C5083,C5117,C5129 | CL05A475KQ5NRNC    |
| 2203-007165 | C6010,C6045,C7047 | CL05A475KQ5NRNC    |
| 2203-007194 | C1191,C7061       | CL03C6R8BA3GNNC    |
| 2203-007210 | C5116,C5128       | CV03X5R224K06AH    |
| 2203-007240 | C5026,C5915       | CL10A226MQ8NRNE    |
| 2203-007269 | C7025,C7028,C7029 | CL21A226MPCLRNC    |
| 2203-007269 | C7032,C7033       | CL21A226MPCLRNC    |
| 2203-007270 | C5102,C5103       | CL10A106KP8NNNC    |
| 2203-007271 | C1080,C1091,C1146 | CL05A225KP5NSNC    |
| 2203-007271 | C5022,C5051,C5077 | CL05A225KP5NSNC    |
| 2203-007271 | C5084,C5085,C5086 | CL05A225KP5NSNC    |
| 2203-007271 | C5087,C5091,C5092 | CL05A225KP5NSNC    |
| 2203-007271 | C5101,C5106,C5107 | CL05A225KP5NSNC    |
| 2203-007271 | C5134,C5138,C5908 | CL05A225KP5NSNC    |
| 2203-007317 | C1093,C4006,C4007 | CV05X5R475M06AH    |
| 2203-007317 | C4028,C4033,C4041 | CV05X5R475M06AH    |
| 2203-007317 | C4055,C4092,C5018 | CV05X5R475M06AH    |
| 2203-007317 | C5053,C5070,C7015 | CV05X5R475M06AH    |
| 2203-007391 | C2048,C2056,C2064 | CL03A474KQ3NNNH    |
| 2203-007391 | C4001,C4002,C4003 | CL03A474KQ3NNNH    |
| 2203-007391 | C4004,C4030,C4031 | CL03A474KQ3NNNH    |
| 2203-007391 | C4034,C4035,C4036 | CL03A474KQ3NNNH    |
| 2203-007391 | C4067,C4068,C4069 | CL03A474KQ3NNNH    |
| 2203-007391 | C4070,C5121       | CL03A474KQ3NNNH    |
| 2203-007393 | C2016,C5082,C5088 | CL05A475KP5NRNC    |
| 2203-007393 | C5089,C5090,C5093 | CL05A475KP5NRNC    |
| 2203-007393 | C5901,C5916       | CL05A475KP5NRNC    |
| 2203-007393 | C7000,C7022       | CL05A475KP5NRNC    |
| 2203-007474 | C7059             | CL05A106MQ5NUNC    |

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|             |                   |                 |
|-------------|-------------------|-----------------|
| 2203-007775 | C5094             | CV05X5R475M10AH |
| 2203-007781 | C5023,C5024,C5079 | CL10A226MP8NUNE |
| 2203-007796 | C1088,C2031,C2032 | CL03A105MP3NSNC |
| 2203-007796 | C2034,C3006,C4005 | CL03A105MP3NSNC |
| 2203-007796 | C4009,C4010,C4011 | CL03A105MP3NSNC |
| 2203-007796 | C4012,C4013,C4014 | CL03A105MP3NSNC |
| 2203-007796 | C4015,C4016,C4017 | CL03A105MP3NSNC |
| 2203-007796 | C4018,C4019,C4020 | CL03A105MP3NSNC |
| 2203-007796 | C4022,C4023,C4024 | CL03A105MP3NSNC |
| 2203-007796 | C4025,C4026,C4027 | CL03A105MP3NSNC |
| 2203-007796 | C4038,C4042,C4043 | CL03A105MP3NSNC |
| 2203-007796 | C4044,C4045,C4046 | CL03A105MP3NSNC |
| 2203-007796 | C4047,C4052,C4057 | CL03A105MP3NSNC |
| 2203-007796 | C4059,C4060,C4061 | CL03A105MP3NSNC |
| 2203-007796 | C4065,C4086,C4097 | CL03A105MP3NSNC |
| 2203-007796 | C5004,C5007,C5010 | CL03A105MP3NSNC |
| 2203-007796 | C5012,C5013,C5014 | CL03A105MP3NSNC |
| 2203-007796 | C5015,C5016,C5017 | CL03A105MP3NSNC |
| 2203-007796 | C5025,C5034,C5035 | CL03A105MP3NSNC |
| 2203-007796 | C5036,C5038,C5045 | CL03A105MP3NSNC |
| 2203-007796 | C5046,C5047,C5054 | CL03A105MP3NSNC |
| 2203-007796 | C5055,C5056,C5057 | CL03A105MP3NSNC |
| 2203-007796 | C5058,C5059,C5060 | CL03A105MP3NSNC |
| 2203-007796 | C5061,C5062,C5063 | CL03A105MP3NSNC |
| 2203-007796 | C5065,C5066,C5067 | CL03A105MP3NSNC |
| 2203-007796 | C5078,C5112,C5113 | CL03A105MP3NSNC |
| 2203-007796 | C5114,C5118,C5119 | CL03A105MP3NSNC |
| 2203-007796 | C5123,C5145,C5148 | CL03A105MP3NSNC |
| 2203-007796 | C5155,C5156,C6012 | CL03A105MP3NSNC |
| 2203-007796 | C6014,C6021,C6022 | CL03A105MP3NSNC |
| 2203-007796 | C6032,C6035,C6038 | CL03A105MP3NSNC |
| 2203-007796 | C6050,C6052,C7004 | CL03A105MP3NSNC |
| 2203-007796 | C7006,C7007,C7009 | CL03A105MP3NSNC |
| 2203-007796 | C7013,C7016,C7017 | CL03A105MP3NSNC |
| 2203-007796 | C7024,C7036,C7038 | CL03A105MP3NSNC |
| 2203-007796 | C7040,C7042,C7044 | CL03A105MP3NSNC |
| 2203-007796 | C7049,C7051,C7058 | CL03A105MP3NSNC |
| 2203-008097 | C1078,C1081,C1085 | CL03A104KA3NNNC |
| 2203-008097 | C1096,C1098,C1102 | CL03A104KA3NNNC |

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|-------------|-------------------|--------------------|
| 2203-008097 | C1106,C1109,C1112 | CL03A104KA3NNNC    |
| 2203-008097 | C1119,C1120,C1156 | CL03A104KA3NNNC    |
| 2203-008097 | C2005,C3000,C3002 | CL03A104KA3NNNC    |
| 2203-008097 | C3003,C3004,C3005 | CL03A104KA3NNNC    |
| 2203-008097 | C4049,C4050,C4051 | CL03A104KA3NNNC    |
| 2203-008097 | C4053,C4054,C4056 | CL03A104KA3NNNC    |
| 2203-008097 | C4058,C4062,C4063 | CL03A104KA3NNNC    |
| 2203-008097 | C4064,C4071,C4073 | CL03A104KA3NNNC    |
| 2203-008097 | C4074,C4075,C4076 | CL03A104KA3NNNC    |
| 2203-008097 | C4077,C4078,C4082 | CL03A104KA3NNNC    |
| 2203-008097 | C4083,C4084,C4085 | CL03A104KA3NNNC    |
| 2203-008097 | C5005,C5020,C5122 | CL03A104KA3NNNC    |
| 2203-008097 | C5127,C5130,C5131 | CL03A104KA3NNNC    |
| 2203-008097 | C5133,C5157,C5158 | CL03A104KA3NNNC    |
| 2203-008097 | C5903,C5926,C5927 | CL03A104KA3NNNC    |
| 2203-008097 | C5928,C6002,C6003 | CL03A104KA3NNNC    |
| 2203-008097 | C6011,C6028,C6049 | CL03A104KA3NNNC    |
| 2203-008097 | C7018,C7019,C7052 | CL03A104KA3NNNC    |
| 2203-008097 | C7055             | CL03A104KA3NNNC    |
| 2203-008242 | C4029,C4048       | CL03A225MQ3CRNC    |
| 2203-008860 | C5068,C5069,C5096 | CL03A105MO3NRNC    |
| 2203-008860 | C6019             | CL03A105MO3NRNC    |
| 2203-009223 | C2043,C2119,C2120 | CAP                |
| 2203-009328 | C5115             | CL05A226MQ5N6J8    |
| 2203-009537 | C4000,C4008,C4021 | GRM155R60J156ME47D |
| 2203-009537 | C4032,C4037,C4039 | GRM155R60J156ME47D |
| 2203-009537 | C4040,C4079,C4080 | GRM155R60J156ME47D |
| 2203-009537 | C4081,C4087,C4088 | GRM155R60J156ME47D |
| 2203-009537 | C4089,C4090,C4091 | GRM155R60J156ME47D |
| 2203-009537 | C4093,C4094,C4095 | GRM155R60J156ME47D |
| 2203-009537 | C4096             | GRM155R60J156ME47D |
| 2203-009617 | C2121             | CAP                |
| 2203-009618 | C5001,C5003,C6004 | CAP                |
| 2203-009618 | C6005             | CAP                |
| 2203-009665 | C6008,C6009,C6042 | CAP                |
| 2203-009733 | C5037,C5099,C5100 | CL05A106MP6NUN8    |
| 2203-009733 | C5902,C5905,C6007 | CL05A106MP6NUN8    |
| 2203-009733 | C7010,C7011,C7030 | CL05A106MP6NUN8    |
| 2203-009733 | C7056,C7057       | CL05A106MP6NUN8    |

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## 5. MAIN Electrical Parts List

|             |                   |                 |
|-------------|-------------------|-----------------|
| 2203-009858 | C1087,C7027       | CL05A106MP9NUB8 |
| 2703-002313 | L6000,L6001       | CIH05TR10JNC    |
| 2703-002649 | C1178,L1046,L1106 | LLV0603-F6N8J   |
| 2703-002858 | L1062,L1112       | HK0603-4N7S-T   |
| 2703-002901 | L1019,L1029,L2037 | LLV0603-F15NJ   |
| 2703-002907 | L1057,L1095,L2021 | LLV0603-F8N2J   |
| 2703-002951 | L1013,L1048,L1111 | CIH03T12NJNC    |
| 2703-002951 | L2002,L2045       | CIH03T12NJNC    |
| 2703-002953 | C1115             | CIH03T18NJNC    |
| 2703-002958 | L1011,L1012,L6037 | CIH03T56NJNC    |
| 2703-002958 | L6038             | CIH03T56NJNC    |
| 2703-002959 | L6015             | MLG0603S68NJT   |
| 2703-002960 | L1023,L1051,L2003 | MLG0603S82NJT   |
| 2703-002960 | L2008,L2060,L6026 | MLG0603S82NJT   |
| 2703-002960 | L6029,L6031       | MLG0603S82NJT   |
| 2703-002961 | L2056             | MLG0603SR10JT   |
| 2703-002999 | C1031             | CIH03T22NJNC    |
| 2703-003004 | L1037             | LLV0603-F47NJ   |
| 2703-003878 | L6022             | CIG10WR27MNC    |
| 2703-003904 | L1002,L1004,L1027 | CML03061N0BNH   |
| 2703-003904 | L1038,L2018,L2022 | CML03061N0BNH   |
| 2703-003921 | L1052,L1107,L2031 | CML0306-1N2-BNH |
| 2703-004001 | L1021,L1054,L6006 | LQP03TN39NJ02D  |
| 2703-004012 | L1001,L1033,L1047 | LQP03TN2N2B02D  |
| 2703-004012 | L1053,L1071,L1074 | LQP03TN2N2B02D  |
| 2703-004012 | L1089,L1094,L1096 | LQP03TN2N2B02D  |
| 2703-004012 | L1108,L2020,L2047 | LQP03TN2N2B02D  |
| 2703-004013 | L1070,L1080,L1092 | LQP03TN1N8B02D  |
| 2703-004013 | L1093,L2029,L2046 | LQP03TN1N8B02D  |
| 2703-004013 | L2052             | LQP03TN1N8B02D  |
| 2703-004014 | L1040             | LQP03TN1N2B02D  |
| 2703-004018 | C1018,L1028,L1066 | LQP03TN5N6J02D  |
| 2703-004018 | L1084,L2011,L2034 | LQP03TN5N6J02D  |
| 2703-004018 | L2035,L2036,L2054 | LQP03TN5N6J02D  |
| 2703-004018 | L2057             | LQP03TN5N6J02D  |
| 2703-004033 | C1068,L1006,L1091 | LQP03TN2N4B02D  |
| 2703-004033 | L2053             | LQP03TN2N4B02D  |
| 2703-004034 | L1035,L2000,L2004 | LQP03TN1N0B02D  |
| 2703-004034 | L2016,L2017,L2023 | LQP03TN1N0B02D  |

## 5. MAIN Electrical Parts List

|             |                   |                    |
|-------------|-------------------|--------------------|
| 2703-004034 | L2041,L2049,L2050 | LQP03TN1N0B02D     |
| 2703-004035 | L1015,L1017,L1065 | LQP03TN3N3B02D     |
| 2703-004036 | L1032,L1056,L1099 | LQP03TN12NJ02D     |
| 2703-004036 | L1100             | LQP03TN12NJ02D     |
| 2703-004037 | L1034,L1079,L1087 | LQP03TN2N7B02D     |
| 2703-004037 | L1090,L2007,L2012 | LQP03TN2N7B02D     |
| 2703-004037 | L2028,L2030       | LQP03TN2N7B02D     |
| 2703-004038 | L1007,L1044       | LQP03TN1N5B02D     |
| 2703-004230 | L6018             | LQG15HS68NJ02D     |
| 2703-004259 | L1043             | AML0603E9N1-T      |
| 2703-004286 | L1058,L1064,L1102 | CIH03S3N3BNC       |
| 2703-004328 | L1042,L2040       | LQP03TN4N7H02D     |
| 2703-004363 | L1024             | LQP03TG18NJ02D     |
| 2703-004366 | L1030,L1055,L1101 | LQP03TG6N8J02D     |
| 2703-004408 | L1061             | LQP03TN15NJ02      |
| 2703-004763 | L1049,L1076,L2001 | LQP03TN4N3J02D     |
| 2703-004763 | L2032             | LQP03TN4N3J02D     |
| 2703-004764 | L1010,L1082       | LQP03TN10NH02D     |
| 2703-004853 | L1031,L1036,L2058 | LQP03TNR12J02D     |
| 2703-004853 | L2059             | LQP03TNR12J02D     |
| 2703-004862 | L1072,L1083,L2025 | LQP03TN3N6B02D     |
| 2703-004911 | L1113             | LQP03TN5N1J02D     |
| 2703-005058 | L1022             | LQP03TN5N6J02D     |
| 2703-005085 | L1003,L1069,L1075 | LQP03HQ2N7B02D     |
| 2703-005085 | L1081,L1103,L1109 | LQP03HQ2N7B02D     |
| 2703-005085 | L1110,L2026,L2038 | LQP03HQ2N7B02D     |
| 2703-005086 | L2015,L2051       | LQP03HQ3N3B02D     |
| 2703-005087 | L1063,L2044       | LQP03HQ3N9B02D     |
| 2703-005089 | L1000,L1039       | LQP03HQ15NJ02D     |
| 2703-005109 | L5025             | CIGT252010LM1R0SNE |
| 2703-005117 | L7004,L7006       | SDEM25201T-4R7MD   |
| 2703-005118 | L7003             | SDEM25201T-100MD   |
| 2703-005136 | L5012,L5013,L5016 | CIGT252010LMR47SNE |
| 2703-005136 | L5017             | CIGT252010LMR47SNE |
| 2703-005296 | L6005             | IND                |
| 2703-005438 | L5018,L5019,L5029 | CIH03T68NJNC       |
| 2703-005505 | L5003,L5004,L5011 | CIGT201610EH1R0SNE |
| 2703-005505 | L5014,L5015,L6002 | CIGT201610EH1R0SNE |
| 2801-005393 | OSC5000           | TFX-04             |

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## 5. MAIN Electrical Parts List

|             |                   |                      |
|-------------|-------------------|----------------------|
| 2809-001434 | OSC1000           | KT2016K26000ZAW23ZMS |
| 2901-001673 | F7006             | ACFE1A2G900E         |
| 2901-001674 | F7002,F7004       | ACFE2B4G101E         |
| 2904-002134 | F1004             | SAFFB2G14AA0F0A      |
| 2904-002143 | F2007             | SAFFB2G65AA0F0A      |
| 2904-002253 | F1012,F2005       | SFWG00DB002          |
| 2904-002257 | F2010             | SFHG52AA002          |
| 2904-002262 | F2014             | SFH881AA002          |
| 2904-002263 | F2006             | SFH942AA002          |
| 2904-002344 | F2011             | SFWG42CBB02          |
| 2904-002349 | F2013             | SFHG93AQ102          |
| 2904-002356 | F2001             | SFHG89XA402          |
| 2904-002365 | F1007             | SAFRH2G59MA0F3K      |
| 2904-002367 | F1006             | SFDG35AQ102          |
| 2910-000390 | F1005             | SAYEY2G53BD0F3K      |
| 2911-000448 | F1002             | LMSWKQGR-J80         |
| 3003-001236 | MIC6000,MIC6001   | SM0103B-DA381-S01    |
| 3301-001812 | L6033,L6036       | BLM15HD102SN1        |
| 3301-001885 | L6007,L6008,L6010 | BLM15HD182SN1D       |
| 3301-001885 | L6012,L6014,L6016 | BLM15HD182SN1D       |
| 3301-001885 | L6019,L6020,L6025 | BLM15HD182SN1D       |
| 3301-002223 | L7000             | BLM15PX601SN1D       |
| 3301-002228 | L1041             | BLM15PX330SN1        |
| 3301-002236 | L5020,L5022       | BLM03AX102SN1D       |
| 3301-002237 | C6043,L1020,L1025 | BLM03AX601SN1D       |
| 3301-002237 | L2005,L2006,L2013 | BLM03AX601SN1D       |
| 3301-002240 | L5000,L5005,L5006 | BLM03AX100SN1D       |
| 3301-002240 | L5007,L5008,L5009 | BLM03AX100SN1D       |
| 3301-002240 | L5010,L6013       | BLM03AX100SN1D       |
| 3301-002244 | L6003,L6004,L6034 | BLM15PX330SN1D       |
| 3301-002244 | L6035             | BLM15PX330SN1D       |
| 3301-002254 | L5024             | BLM18KG260TN1D09     |
| 3301-002279 | L7001,L7008       | BLM15PX331SN1        |
| 3301-002286 | L6021             | BKH0603LM102-T       |
| 3301-002305 | L6017,L6023       | NFZ15SG462SN10D      |
| 3301-002312 | L5023,L7002,L7005 | BLM15EX121SN1D       |
| 3301-002312 | L7007             | BLM15EX121SN1D       |
| 3301-002331 | L5001             | BLM18FD220SN1        |
| 3705-001708 | RFS1002,RFS1004   | MS-156C              |

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## 5. MAIN Electrical Parts List

|             |                   |                      |
|-------------|-------------------|----------------------|
| 3705-001937 | RFS1000,RFS1001   | 20449-001E           |
| 3705-001937 | RFS2000           | 20449-001E           |
| 3709-001891 | SIM6000           | DS254-C22B-C14       |
| 3711-007107 | HDC7001           | AXE630124AW1         |
| 3711-008182 | HDC7000           | 503776-4051          |
| 3711-008847 | HDC5000           | 505006-0811          |
| 3711-008969 | HDC7002           | BM30B-54DP-0.35V(46) |
| 3711-009058 | HDC6000           | GB355-16P-H06-E15000 |
| 3712-001634 | ANT2006,ANT2007   | CS-074A              |
| 3712-001634 | ANT2023,ANT5004   | CS-074A              |
| 3712-001634 | ANT5006           | CS-074A              |
| 3712-001666 | ANT6000,ANT6001   | CS-090A              |
| 3712-001694 | ANT2003,ANT2004   | CS-070B              |
| 3712-001694 | ANT2005,ANT2016   | CS-070B              |
| 3712-001694 | ANT5001,ANT5002   | CS-070B              |
| 3712-001694 | ANT5005,ANT5007   | CS-070B              |
| 3712-001694 | ANT5008,ANT6002   | CS-070B              |
| 3712-001694 | ANT6003           | CS-070B              |
| 3712-001749 | ANT1000,ANT1001   | CS-111A              |
| 3712-001749 | ANT1002,ANT1003   | CS-111A              |
| 3712-001749 | ANT1005,ANT2000   | CS-111A              |
| 3712-001749 | ANT2001,ANT2008   | CS-111A              |
| 3712-001749 | ANT2010,ANT2011   | CS-111A              |
| 3712-001749 | ANT2012,ANT2018   | CS-111A              |
| 3712-001749 | ANT2019,ANT2020   | CS-111A              |
| 3712-001749 | ANT2024,ANT2025   | CS-111A              |
| 3712-001749 | ANT2026,ANT2027   | CS-111A              |
| 3712-001750 | ANT5000,ANT5003   | CS-112A              |
| 3722-003954 | IFC5000           | HY07-AB1790          |
| 3722-004081 | EAR6000           | 7234-383000          |
| 4709-002196 | F1008             | DPX202700DT-4162A1   |
| 4709-002226 | F1001             | DPX202700DT-4062A1   |
| 4709-002412 | F1000,F1010,F1011 | HHM22137A3K          |
| 4709-002464 | F1009             | DPX252690DT-5225A1K  |
| 4709-002566 | F2002             | SFTG45AU002          |
| GH60-00009A | FPC2001,FPC2002   | PCC1011              |
| GH60-00009A | FPC2003           | PCC1011              |
| GH60-00010A | FPC2004           | PCC1007AW            |
| GH61-07836A | FPC2005,FPC2006   | PCC1006A             |

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## 5. MAIN Electrical Parts List

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|             |        |                    |
|-------------|--------|--------------------|
| GH63-11154A | SC6000 | SC-SMG928V-SUBPBA  |
| GH63-14647A | SC6005 | SC-SMC7100-BTWIFI  |
| GH63-14648A | SC6001 | SC-SMC7100-PMIC    |
| GH63-14649A | SC6002 | SC-SMC7100-TRV     |
| GH63-14650A | SC6003 | SC-SMC7100-AP      |
| GH63-14659A | SC6006 | SC-SMC7100-SPK     |
| GH63-14674A | SC6004 | SC-SMC7100-RF      |
| GH63-14675A | SC6008 | SC-SMC7100-FILTER  |
| GH63-14676A | SC6007 | SC-SMC7100-CHARGER |
| 1209-002510 | U7011  | CM36657VM3OH-A3    |

Please consult the GSPN website (Samsung Portal) for the most recent version of the product's part list.

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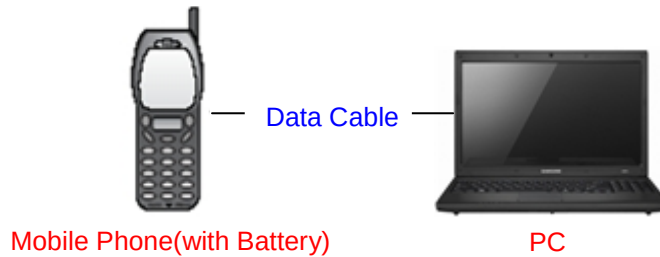
## 6. Level 1 Repair

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### 6-1. S/W Download

#### 6-1-1. Prepare for S/W Downloading

- Diagram of connection



#### 6-1-2. How to download S/W

##### 1) Downloading Binary Files

- Binary file for downloading SM-C7100
  - [xxx.pit](#)
  - [BL\\_XXXX.tar.md5](#)
  - [AP\\_XXXX.tar.md5](#)
  - [CP\\_XXXX.tar.md5](#)
  - [CSC\\_xxxx.tar.md5](#)

(file size is about 3.8GB )

##### 2) Prepare for Downloading

- Downloader Program ([Odin3 v3.12.3.exe](#))
- SM-C7100 Mobile Phone
- Data Cable
- Binary file

3) Boot the mobile phone by pressing 'Home + Vol Down + Power key at the same time, If you do properly, you can see the following message on the main LCD as the following.

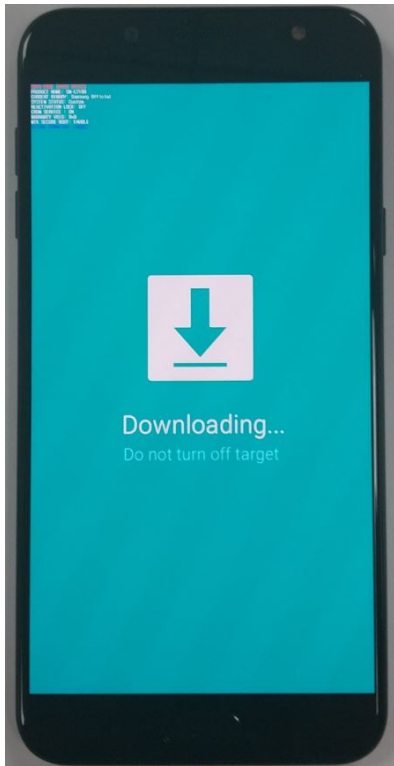


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## 6. Level 1 Repair

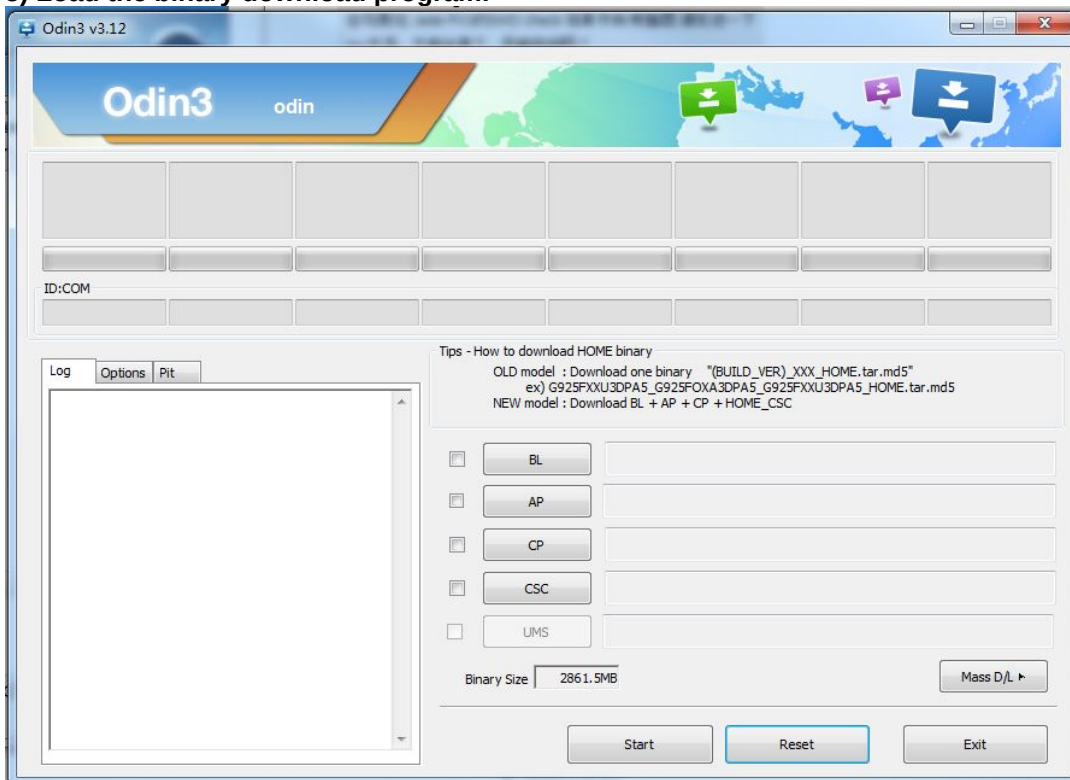
---

4) Press the Vol Up Key again, and you will see below message on Main LCD.



“ Downloading...”

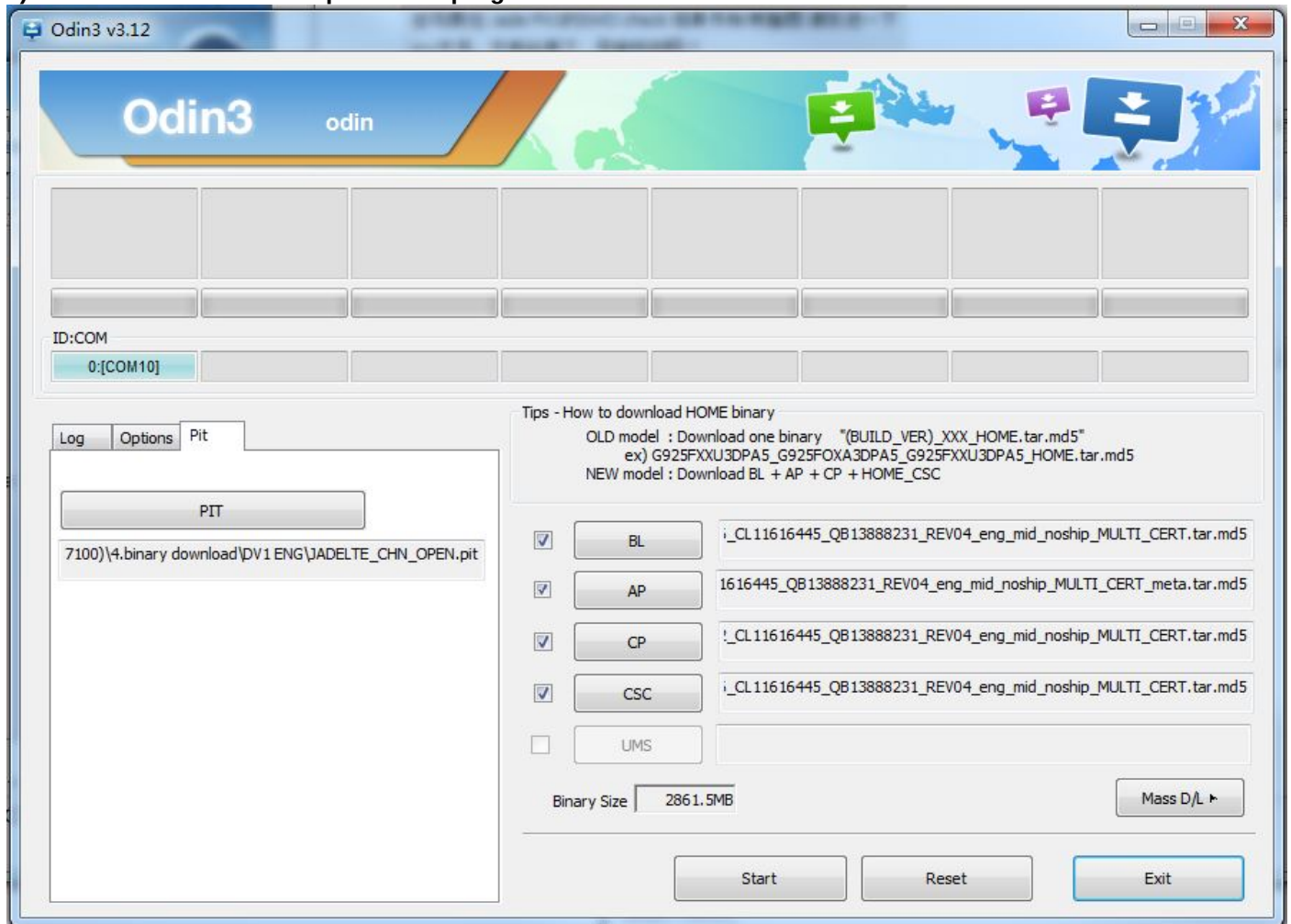
5) Load the binary download program.





## 6. Level 1 Repair

6) Connect mobile and computer. The program show as follow.



Select the file as above:

- BL\_XXXX.tar.md5
- AP\_XXXX.tar.md5
- CP\_XXXX.tar.md5
- CSC\_xxxx.tar.md5

7) Now press the button "Start".

8) Now it's time to take a rest and finish the downloading.

9) After finished downloading of phone binary, the mobile phone will restart automatically.

※ **Caution. Never disconnect during the S/W downloading.**

---

## 6. Level 1 Repair

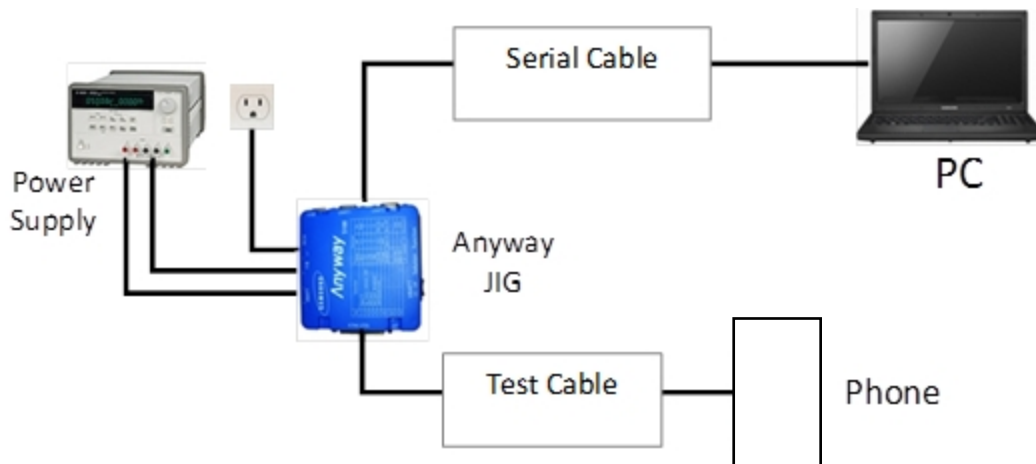
---

### 6-2. IMEI writing

#### 6-2-1. Preparation

- New IMEI writing Program has been released.
- Supported Model: Models which CAB files are uploaded on HHP svc INI File category, instead of ini file.
- Refer to below IMEI writing procedure.

#### - H/W



#### - S/W

|                   |                                                                                                                                                              |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ① Library Install | To use Daseul, library files should be installed.<br>Refer to SVC Bulletin<br>“(11-82) Daseul (New IMEI writing Program) Library Install guide_rev1.0”       |
| ② Launcher        | DASEUL_SVC_Launcher_v3_0_25 or higher<br>-Uploaded on HHPsvc Notice                                                                                          |
| ③ Runtime File    | 1. DASEUL_Runtime_Ver_3.1.139.0.CAB or higher<br>-Uploaded on HHPsvc Notice<br>2. Make 'ModelName' folder at the same position with launcher & Runtime file. |
| ④ Model File      | Copy Model File under the 'Model Name' folder                                                                                                                |

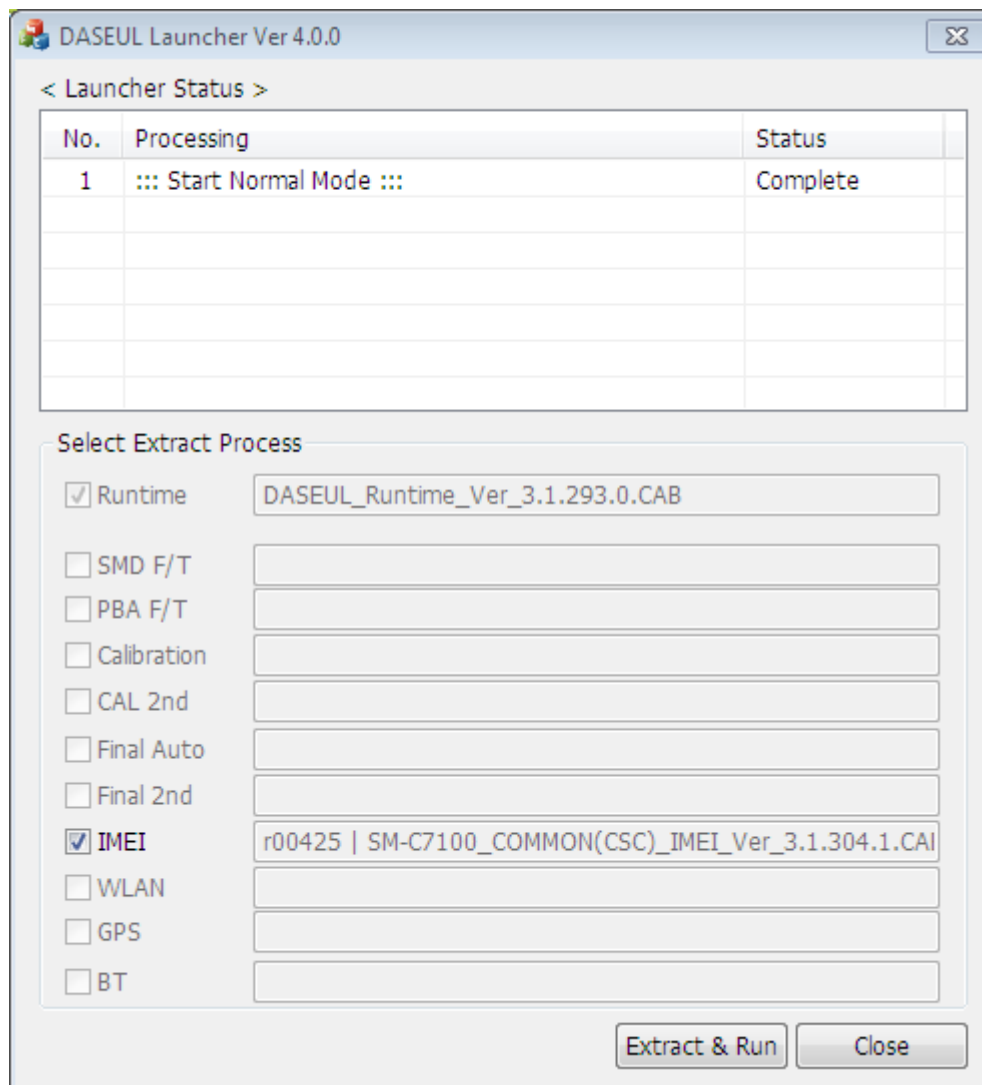
## 6. Level 1 Repair

### 6-2-2. IMEI writingProcess

#### 1. 1. Run DASEUL\_Launcher\_v4.0.0

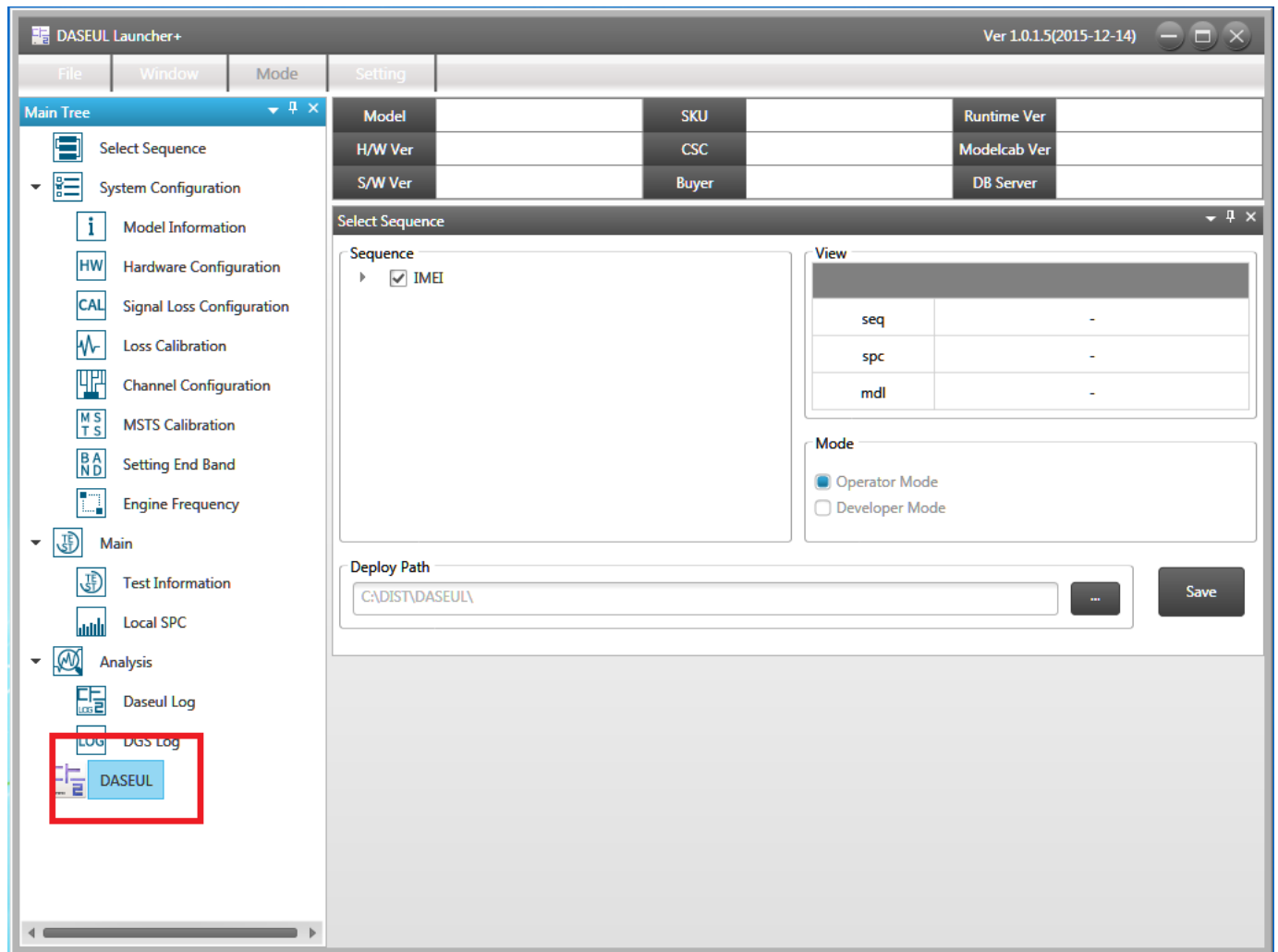
|                                                                                                                                |                  |                |            |
|--------------------------------------------------------------------------------------------------------------------------------|------------------|----------------|------------|
|  DASEUL_IMEI_ALL_Component_r00425.CAB         | 2017/4/10 10:17  | ALZip cab File | 195,594 KB |
|  DASEUL_IMEI_ALL_Runtime_3.1.293.0_r00425.CAB | 2017/4/11 9:16   | ALZip cab File | 231,261 KB |
|  DASEUL_Launcher_v4.0.0.exe                   | 2016/10/21 15:11 | Application    | 5,896 KB   |
|  DASEUL_Runtime_Ver_3.1.293.0.CAB             | 2017/4/8 7:14    | ALZip cab File | 37,075 KB  |
|  SM-C7100_COMMON(CSC)_IMEI_Ver_3.1.304.1.CAB  | 2017/6/8 10:25   | ALZip cab File | 42 KB      |

#### 2. Select Service Mode



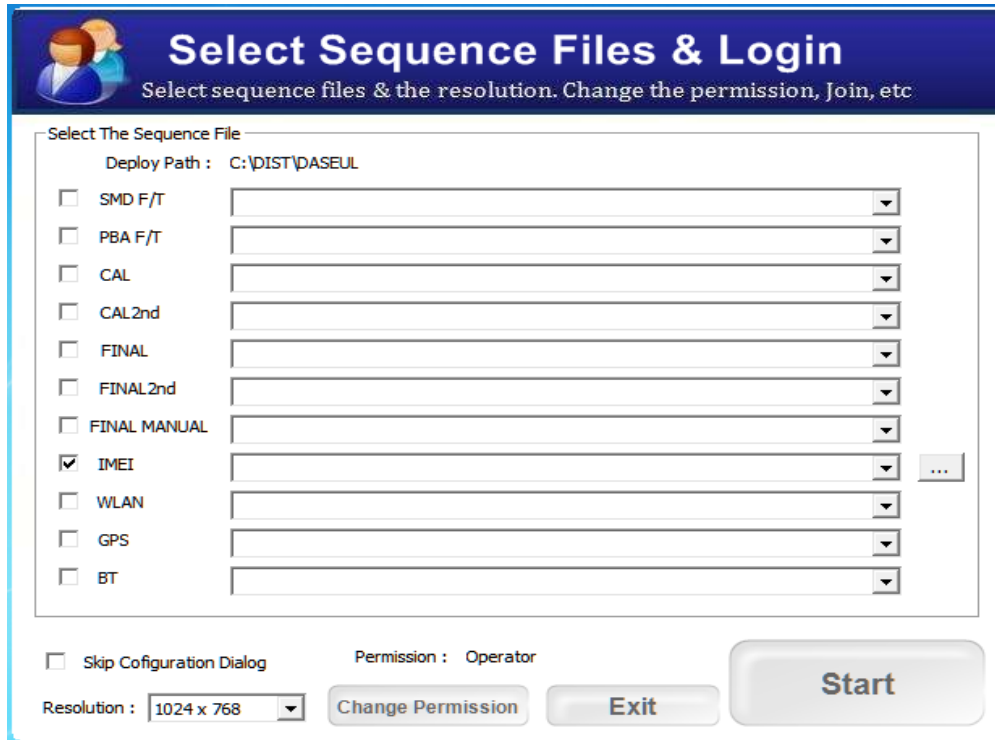
## 6. Level 1 Repair

3. Click DASEUL Select folder where the Launcher exists



## 6. Level 1 Repair

4.  Select folder where the Launcher exists



**Select Sequence Files & Login**  
Select sequence files and the resolution. Change the permission, Join, etc

Select The Sequence File

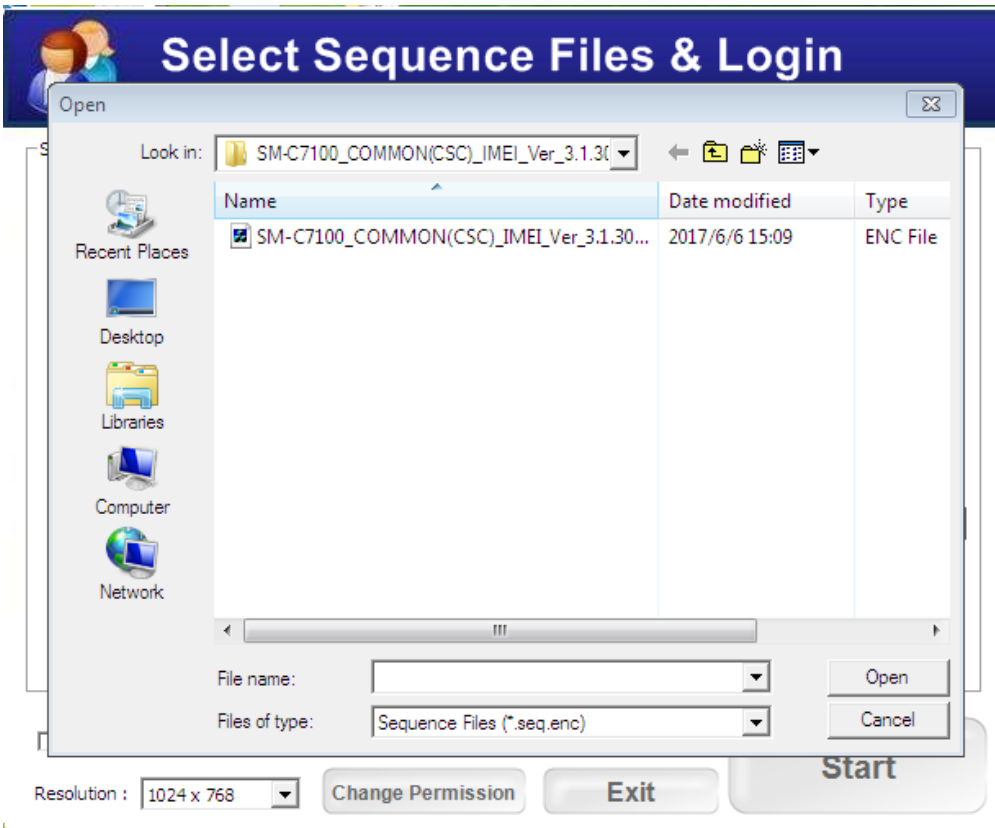
Deploy Path : C:\DIST\DASEUL

|                                     |              |     |
|-------------------------------------|--------------|-----|
| <input type="checkbox"/>            | SMD F/T      |     |
| <input type="checkbox"/>            | PBA F/T      |     |
| <input type="checkbox"/>            | CAL          |     |
| <input type="checkbox"/>            | CAL2nd       |     |
| <input type="checkbox"/>            | FINAL        |     |
| <input type="checkbox"/>            | FINAL2nd     |     |
| <input type="checkbox"/>            | FINAL MANUAL |     |
| <input checked="" type="checkbox"/> | IMEI         | ... |
| <input type="checkbox"/>            | WLAN         |     |
| <input type="checkbox"/>            | GPS          |     |
| <input type="checkbox"/>            | BT           |     |

☐ Skip Configuration Dialog      Permission : Operator

Resolution : 1024 x 768                 

5. Select Mode



**Select Sequence Files & Login**

Open

Look in: SM-C7100\_COMMON(CSC)\_IMEI\_Ver\_3.1.30...

| Name                                    | Date modified  | Type     |
|-----------------------------------------|----------------|----------|
| SM-C7100_COMMON(CSC)_IMEI_Ver_3.1.30... | 2017/6/6 15:09 | ENC File |

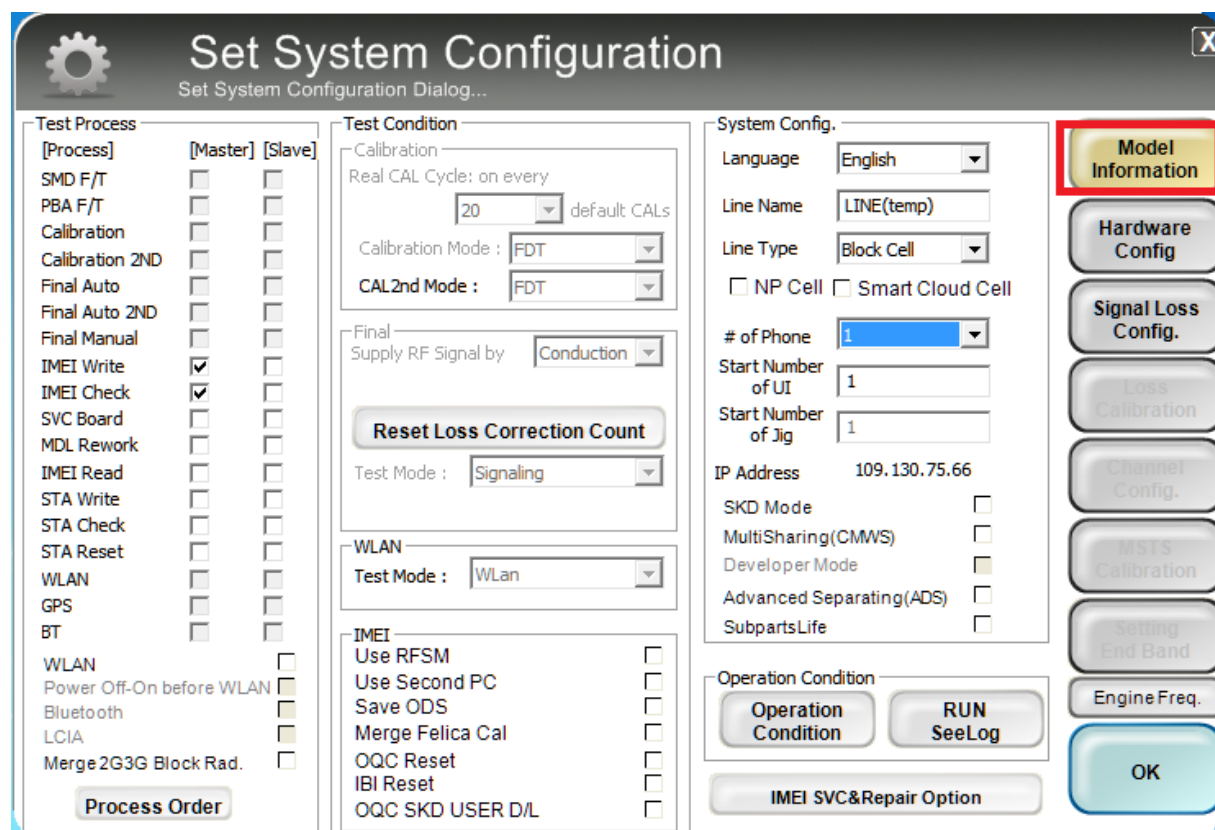
File name:           

Files of type: Sequence Files (\*.seq.enc)

Resolution : 1024 x 768

## 6. Level 1 Repair

### 6. Check 'Model information'



**Set System Configuration**  
Set System Configuration Dialog...

**Test Process**

| [Process]                | [Master]                            | [Slave]                  |
|--------------------------|-------------------------------------|--------------------------|
| SMD F/T                  | <input type="checkbox"/>            | <input type="checkbox"/> |
| PBA F/T                  | <input type="checkbox"/>            | <input type="checkbox"/> |
| Calibration              | <input type="checkbox"/>            | <input type="checkbox"/> |
| Calibration 2ND          | <input type="checkbox"/>            | <input type="checkbox"/> |
| Final Auto               | <input type="checkbox"/>            | <input type="checkbox"/> |
| Final Auto 2ND           | <input type="checkbox"/>            | <input type="checkbox"/> |
| Final Manual             | <input type="checkbox"/>            | <input type="checkbox"/> |
| IMEI Write               | <input checked="" type="checkbox"/> | <input type="checkbox"/> |
| IMEI Check               | <input checked="" type="checkbox"/> | <input type="checkbox"/> |
| SVC Board                | <input type="checkbox"/>            | <input type="checkbox"/> |
| MDL Rework               | <input type="checkbox"/>            | <input type="checkbox"/> |
| IMEI Read                | <input type="checkbox"/>            | <input type="checkbox"/> |
| STA Write                | <input type="checkbox"/>            | <input type="checkbox"/> |
| STA Check                | <input type="checkbox"/>            | <input type="checkbox"/> |
| STA Reset                | <input type="checkbox"/>            | <input type="checkbox"/> |
| WLAN                     | <input type="checkbox"/>            | <input type="checkbox"/> |
| GPS                      | <input type="checkbox"/>            | <input type="checkbox"/> |
| BT                       | <input type="checkbox"/>            | <input type="checkbox"/> |
| WLAN                     | <input type="checkbox"/>            | <input type="checkbox"/> |
| Power Off-On before WLAN | <input type="checkbox"/>            | <input type="checkbox"/> |
| Bluetooth                | <input type="checkbox"/>            | <input type="checkbox"/> |
| LCIA                     | <input type="checkbox"/>            | <input type="checkbox"/> |
| Merge 2G3G Block Rad.    | <input type="checkbox"/>            | <input type="checkbox"/> |

**Test Condition**

Calibration  
Real CAL Cycle: on every  default CALs

Calibration Mode :

CAL2nd Mode :

Final  
Supply RF Signal by

**Reset Loss Correction Count**

Test Mode :

WLAN  
Test Mode :

IMEI  
Use RFSM ☐  
Use Second PC ☐  
Save ODS ☐  
Merge Felica Cal ☐  
OQC Reset ☐  
IBI Reset ☐  
OQC SKD USER D/L ☐

**System Config.**

Language

Line Name

Line Type

☐ NP Cell ☐ Smart Cloud Cell

# of Phone

Start Number of UI

Start Number of Jig

IP Address

SKD Mode ☐

MultiSharing(CMWS) ☐

Developer Mode ☐

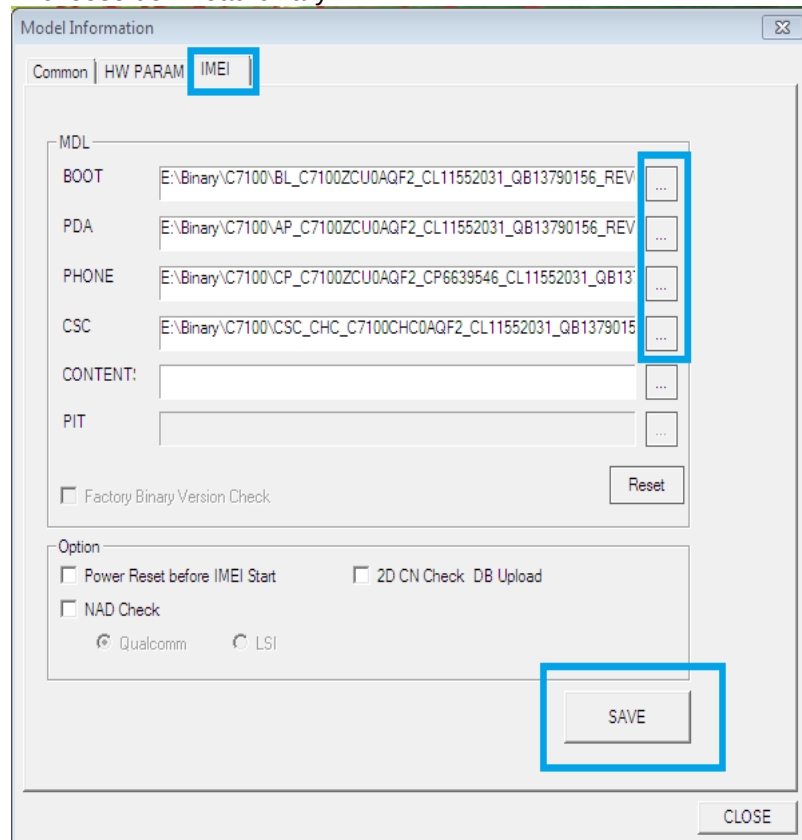
Advanced Separating(ADS) ☐

SubpartsLife ☐

**Operation Condition**

**Model Information**

### 7. choose download binary



**Model Information**

Common | HW PARAM | **IMEI**

MDL

BOOT

PDA

PHONE

CSC

CONTENT:

PIT

☐ Factory Binary Version Check

**Option**

☐ Power Reset before IMEI Start ☐ 2D CN Check DB Upload

☐ NAD Check

☒ Qualcomm ☐ LSI

## 6. Level 1 Repair

8. Click IMEI, Then choose binary, click SAVE.

Model Information

Common | HW PARAM | **IMEI**

MDL

BOOT: E:\Binary\C7100\BL\_C7100ZCU0AQF2\_CL11552031\_QB13790156\_REV ...

PDA: E:\Binary\C7100\AP\_C7100ZCU0AQF2\_CL11552031\_QB13790156\_REV ...

PHONE: E:\Binary\C7100\CP\_C7100ZCU0AQF2\_CP6639546\_CL11552031\_QB13 ...

CSC: E:\Binary\C7100\CSC\_CHC\_C7100CHC0AQF2\_CL11552031\_QB1379015 ...

CONTENT: ...

PIT: ...

☐ Factory Binary Version Check

Reset

Option

☐ Power Reset before IMEI Start ☐ 2D CN Check DB Upload

☐ NAD Check

☒ Qualcomm ☐ LSI

SAVE

CLOSE

9. Click 'Hardware Config'

Set System Configuration

Set System Configuration Dialog...

Test Process

| [Process]                | [Master]                            | [Slave]                  |
|--------------------------|-------------------------------------|--------------------------|
| SMD F/T                  | <input type="checkbox"/>            | <input type="checkbox"/> |
| PBA F/T                  | <input type="checkbox"/>            | <input type="checkbox"/> |
| Calibration              | <input type="checkbox"/>            | <input type="checkbox"/> |
| Calibration 2ND          | <input type="checkbox"/>            | <input type="checkbox"/> |
| Final Auto               | <input type="checkbox"/>            | <input type="checkbox"/> |
| Final Auto 2ND           | <input type="checkbox"/>            | <input type="checkbox"/> |
| Final Manual             | <input type="checkbox"/>            | <input type="checkbox"/> |
| IMEI Write               | <input checked="" type="checkbox"/> | <input type="checkbox"/> |
| IMEI Check               | <input checked="" type="checkbox"/> | <input type="checkbox"/> |
| SVC Board                | <input type="checkbox"/>            | <input type="checkbox"/> |
| MDL Rework               | <input type="checkbox"/>            | <input type="checkbox"/> |
| IMEI Read                | <input type="checkbox"/>            | <input type="checkbox"/> |
| STA Write                | <input type="checkbox"/>            | <input type="checkbox"/> |
| STA Check                | <input type="checkbox"/>            | <input type="checkbox"/> |
| STA Reset                | <input type="checkbox"/>            | <input type="checkbox"/> |
| WLAN                     | <input type="checkbox"/>            | <input type="checkbox"/> |
| GPS                      | <input type="checkbox"/>            | <input type="checkbox"/> |
| BT                       | <input type="checkbox"/>            | <input type="checkbox"/> |
| WLAN                     | <input type="checkbox"/>            | <input type="checkbox"/> |
| Power Off-On before WLAN | <input type="checkbox"/>            | <input type="checkbox"/> |
| Bluetooth                | <input type="checkbox"/>            | <input type="checkbox"/> |
| LCIA                     | <input type="checkbox"/>            | <input type="checkbox"/> |
| Merge 2G3G Block Rad.    | <input type="checkbox"/>            | <input type="checkbox"/> |

Process Order

Test Condition

Calibration

Real CAL Cycle: on every 20 default: CALs

Calibration Mode: FDT

CAL2nd Mode: FDT

Final

Supply RF Signal by: Conduction

Reset Loss Correction Count

Test Mode: Signaling

WLAN

Test Mode: WLAN

IMEI

Use RFSM ☐

Use Second PC ☐

Save ODS ☐

Merge Felica Cal ☐

OQC Reset ☐

IBI Reset ☐

OQC SKD USER D/L ☐

System Config.

Language: English

Line Name: LINE(temp)

Line Type: Block Cell

☐ Smart Cloud Cell

# of Phone: 1

Start Number of UI: 1

Start Number of Jig: 1

IP Address: 109.131.10.178

SKD Mode ☐

MultiSharing(CMWS) ☐

Developer Mode ☐

Advanced Separating(ADS) ☐

Operation Condition

Operation Condition RUN SeeLog

IMEI SVC&Repair Option

Model Information

Hardware Config

Signal Loss Config.

Loss Calibration

Channel Config.

WSS Calibration

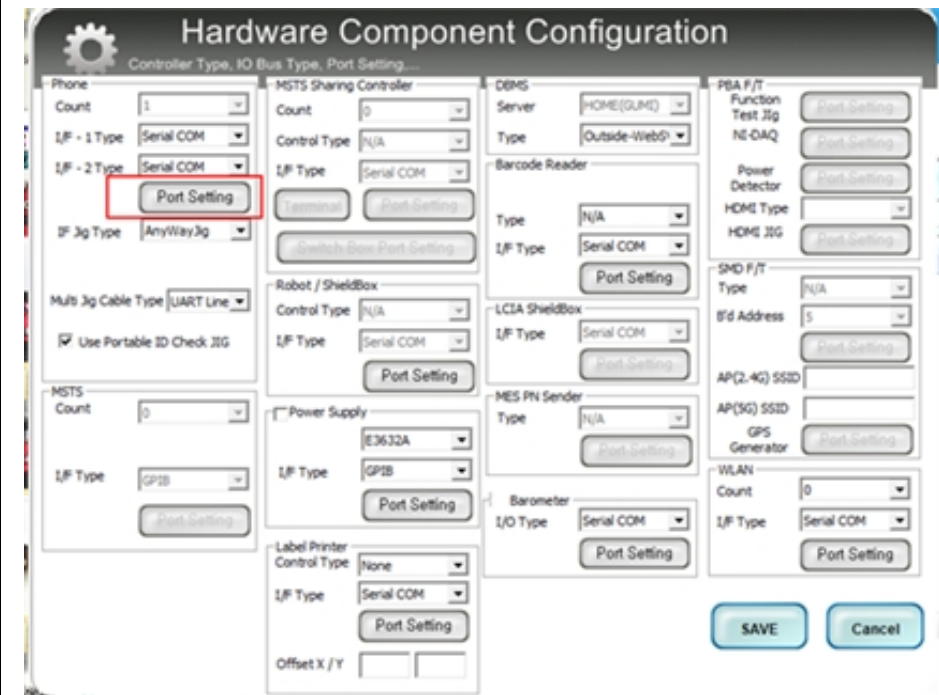
Setting End Band

Engine Freq.

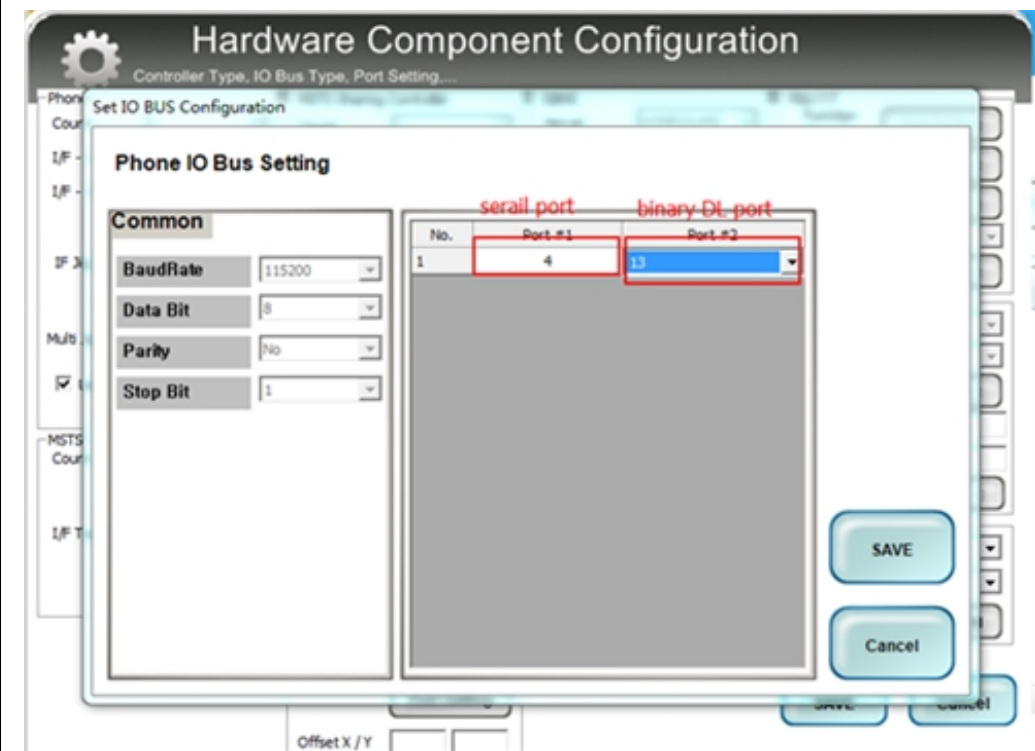
OK

## 6. Level 1 Repair

10. Click 'Port Setting'



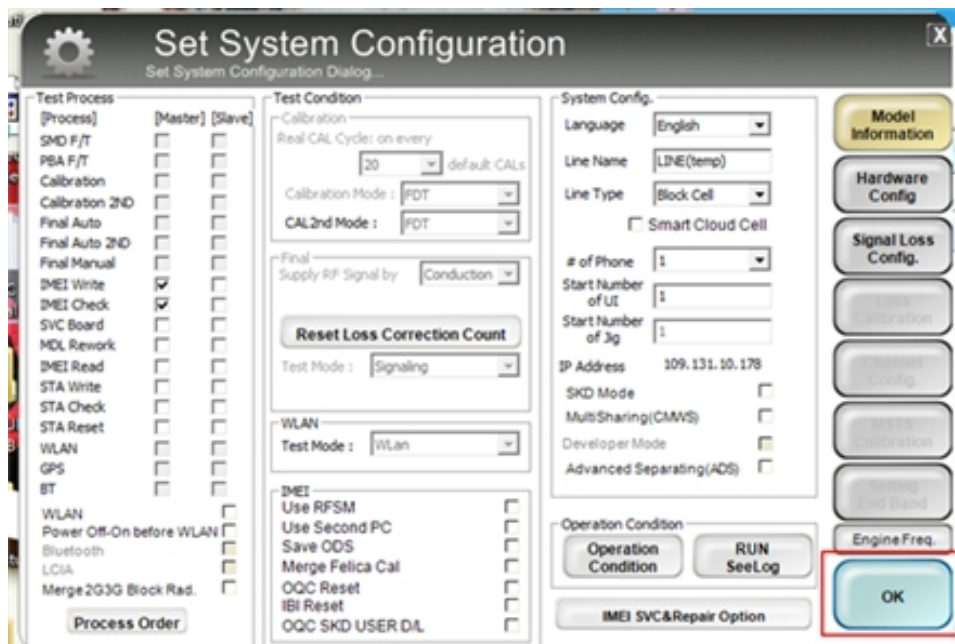
11. Select Port Number and SAVE



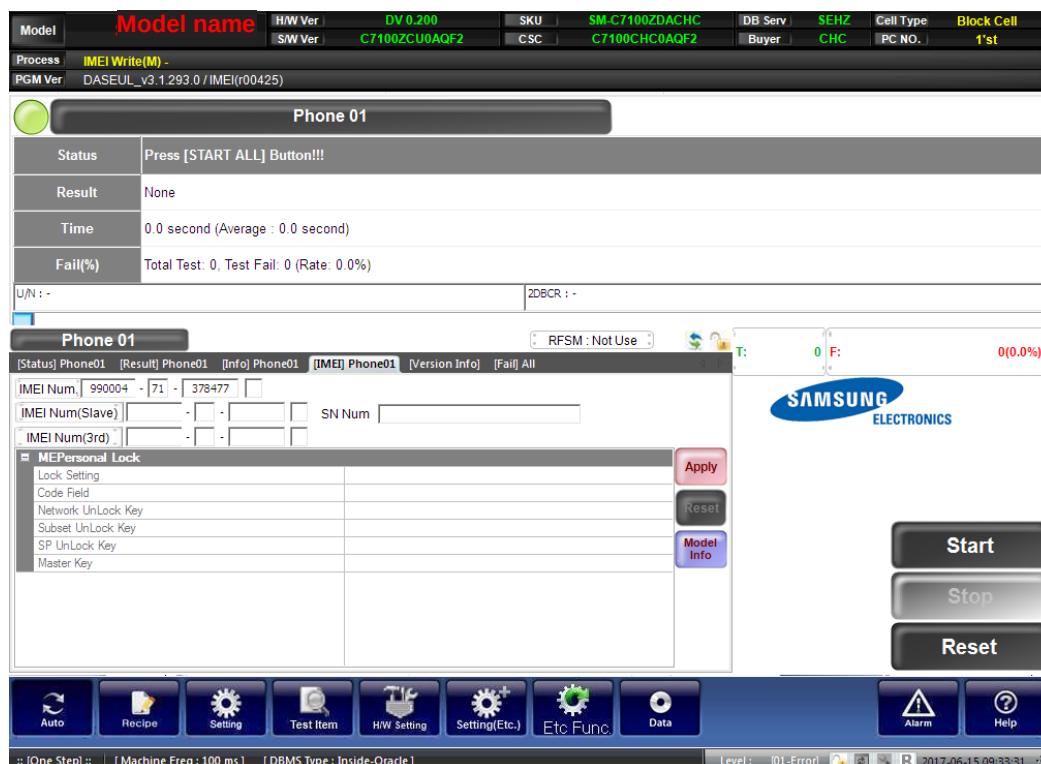


## 6. Level 1 Repair

12. Click OK to proceed



13. Click Model Info and OK when pop-up shows

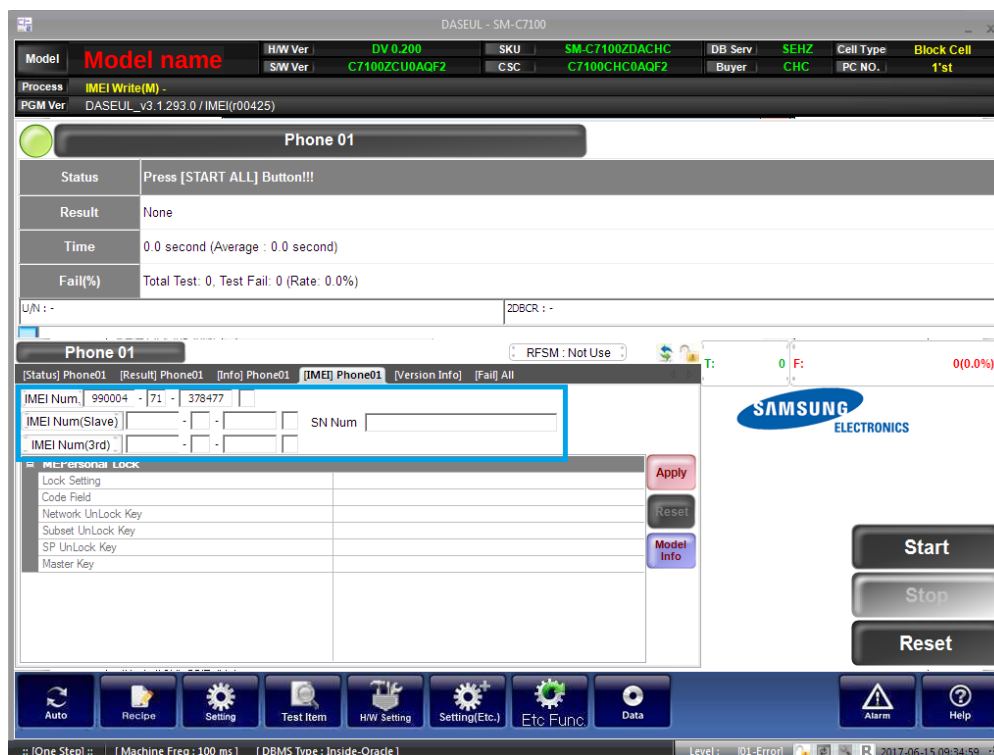


## 6. Level 1 Repair

14. Click OK

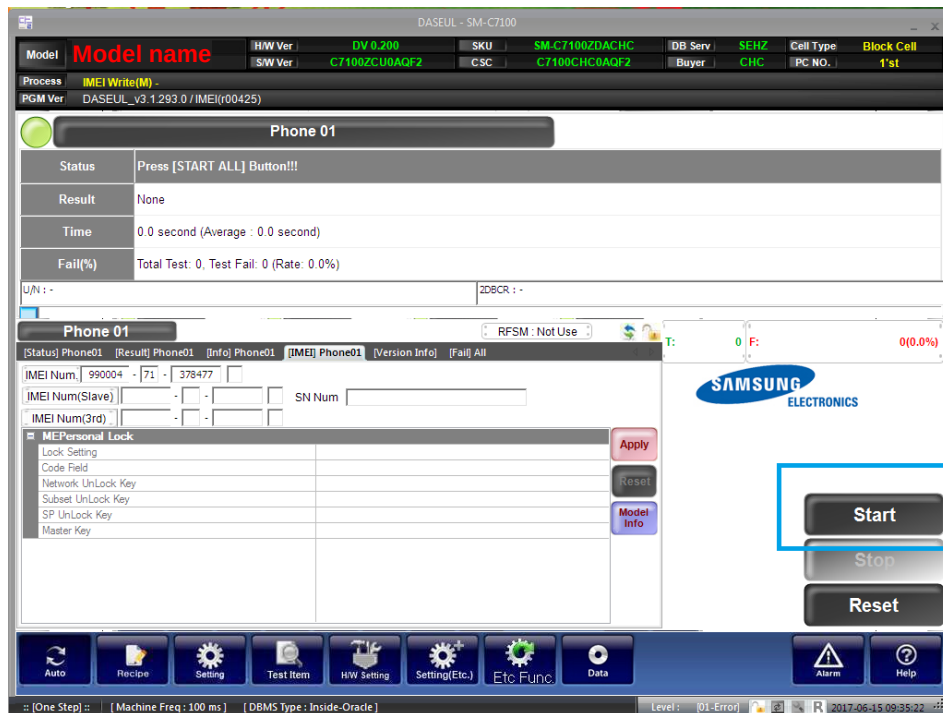


15. Input IMEI Number and click apply.



## 6. Level 1 Repair

## 16 Click Start

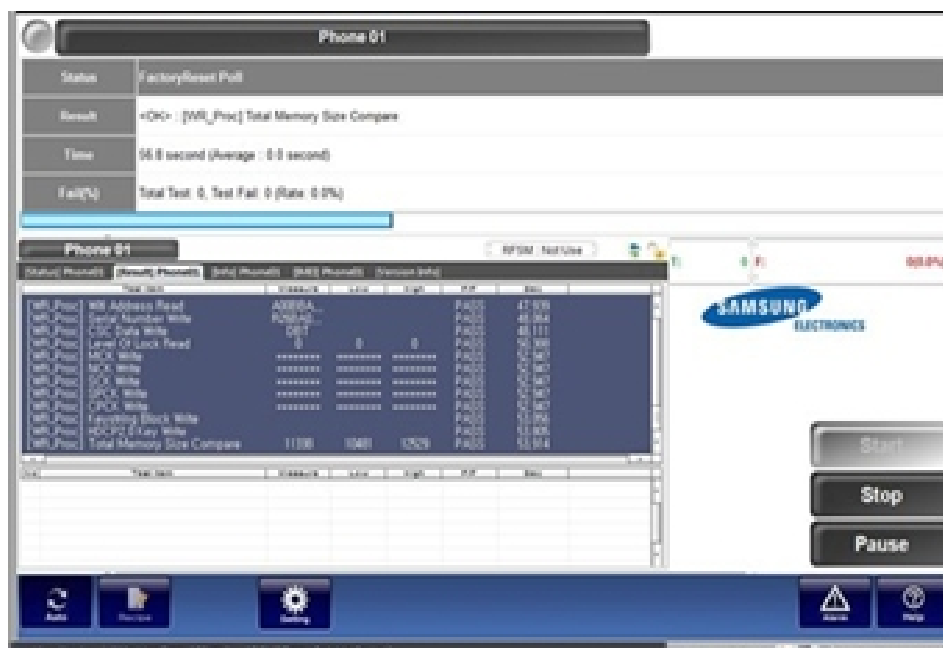


## 17. Connect the phone to Anyway JIG

※ When you connect the phone, the phone should be turned off.

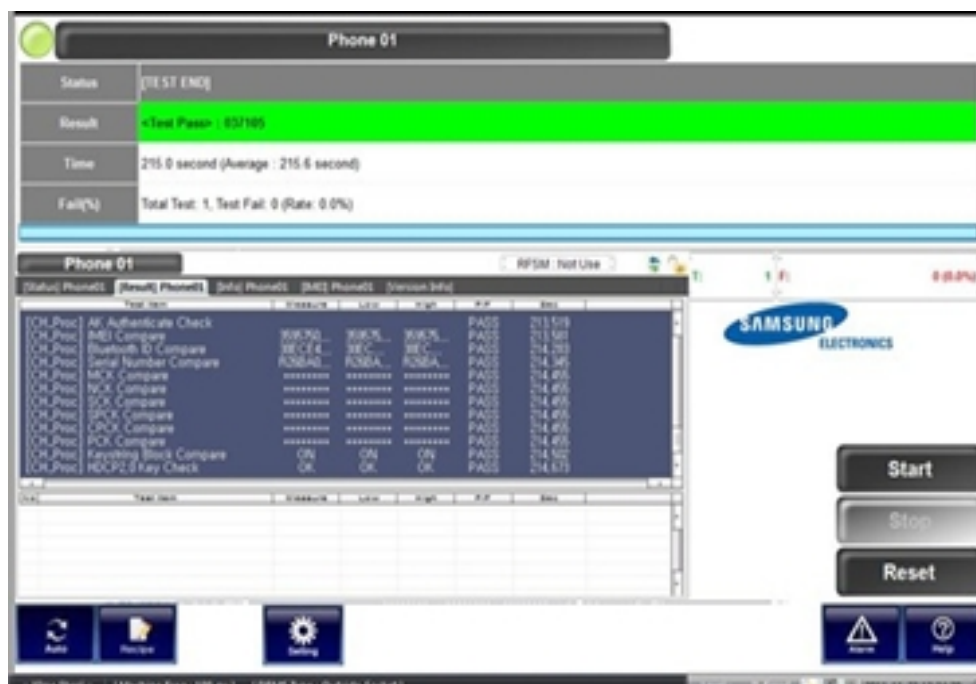
After connecting the phone, the phone will be booted automatically.

## 18. IMEI Writing Proceeding



## 6. Level 1 Repair

## 19. IMEI Writing Success



## 6. Level 1 Repair

### 6-3. RF Calibration

#### 6-3-1. Required items in order to calibrate RF

- Installation program :RF Calibration Program
  - Daseul\_Launcher\_vx.x.xx.exe
  - Daseul\_CAL\_ALL\_Runtime\_x.x.xxx.x.CAB
  - Model File ([SM-C7100\\_OPEN\\_CALIBRATION\\_Ver\\_3.1.300.1.CAB](#))

※ It is required to use the latest program.

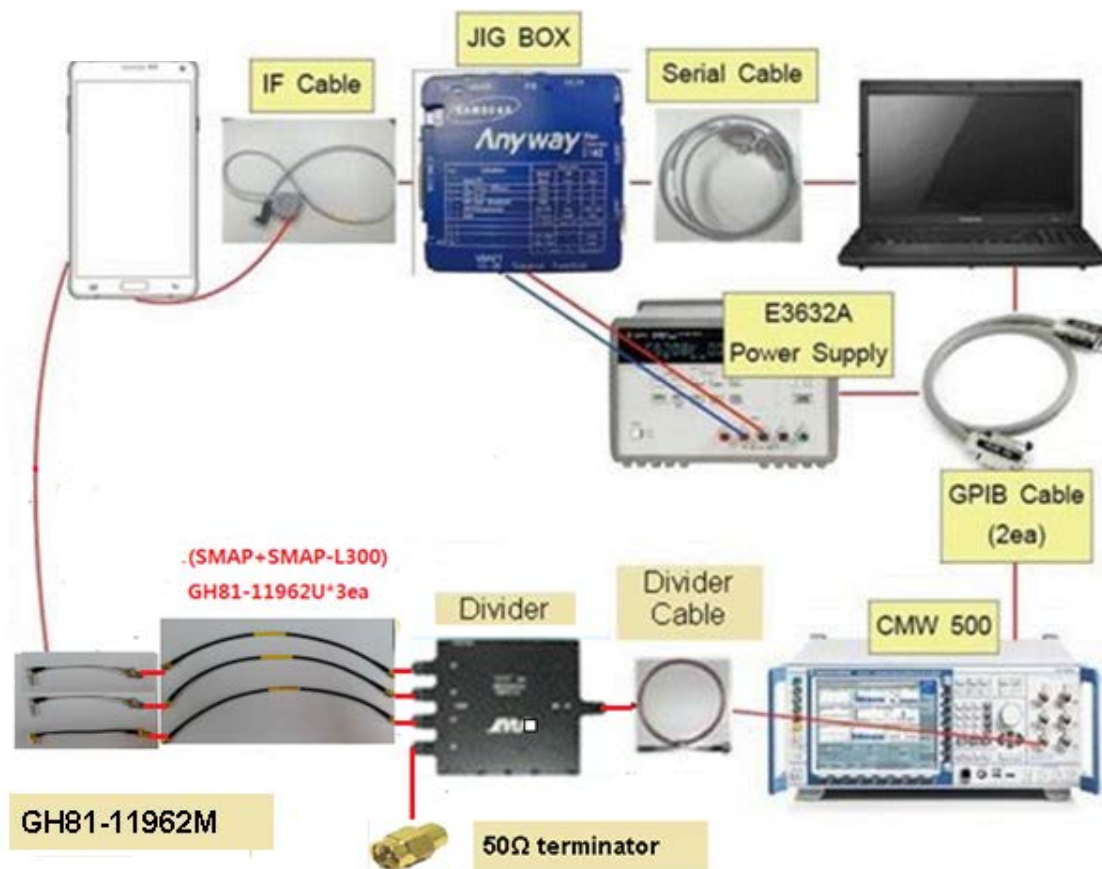
- Mobile Phone
- R&S CMW500
- E3632A Power Supply
- GPIB Cable(2ea)
- JIG BOX(Anyway S103)
- Type-7Pin IF Cable(GH81-10952A)
- Adapter (GH81-11888K)
- UART Serial Cable
- RF Cable(GH81-11962U) 3EA
- 1.2T RF Cable (GH81-11962M) 3EA
- 50Ω terminator(GH81-11962E 1EA)
- 4Port Divider(GH81-11962A 1EA)
- Divider Cable(GH81-11962B 1EA)

#### ❖ Table of test cables

|                      |                                                                                                           |                                                                                                              |                                                                                                                |
|----------------------|-----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| IF Cable             | GH81-10952A                                                                                               |                                                                                                              |                                                                                                                |
|                      | USB 2.0, Type-7Pin                                                                                        |                                                                                                              |                                                                                                                |
| RF Cable<br>(Manual) | GH81-11962M                                                                                               | GH81-11962U                                                                                                  |                                                                                                                |
|                      | 1.2T ,102mm<br>3EA<br> | SMAP, 300mm,<br>3EA<br>  |                                                                                                                |
| 4 Port Divider       | GH81-11962A                                                                                               | GH81-11962B                                                                                                  | GH81-11962E                                                                                                    |
|                      | Divider<br>1EA<br>     | Divider Cable<br>1EA<br> | 50Ω terminator<br>1EA<br> |

## 6. Level 1 Repair

### ❖ Setting



---

## 6. Level 1 Repair

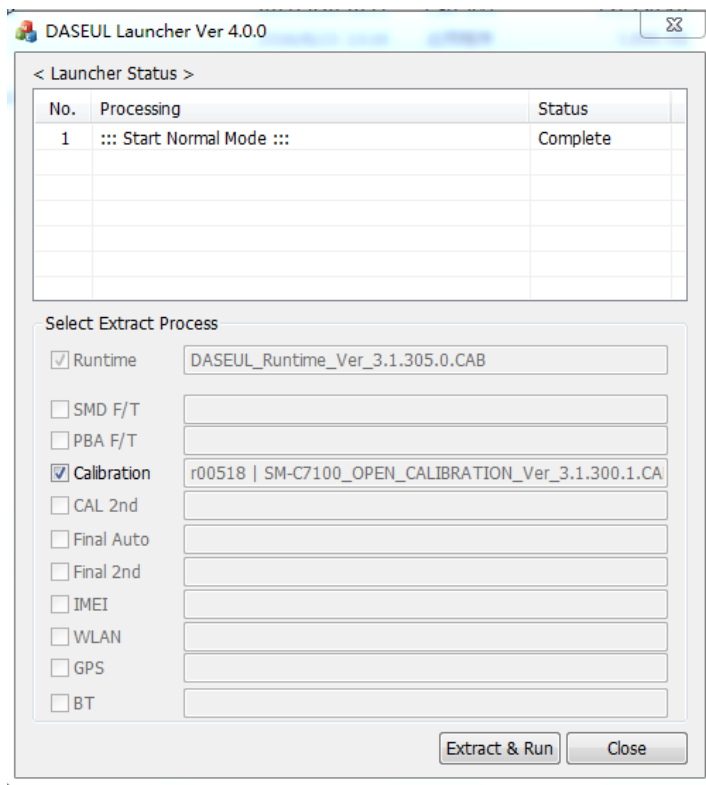
---

### 6-3-2. RF Calibration Program

1. Run the RF Calibration Program Launcher, '[DASEUL\\_Launcher\\_vx.x.xx.exe](#)'.

|                                                                                   |                                             |                 |        |            |
|-----------------------------------------------------------------------------------|---------------------------------------------|-----------------|--------|------------|
|  | DASEUL_CAL_ALL_Component_r00518.CAB         | 2017/5/24 16:44 | Cab 文件 | 95,726 KB  |
|  | DASEUL_CAL_ALL_Runtime_3.1.305.0_r00518.CAB | 2017/5/26 16:11 | Cab 文件 | 131,230 KB |
|  | DASEUL_Launcher_v4.0.0.exe                  | 2016/8/23 13:16 | 应用程序   | 5,896 KB   |
|  | DASEUL_Runtime_Ver_3.1.305.0.CAB            | 2017/5/23 10:23 | Cab 文件 | 37,514 KB  |
|  | SM-C7100_OPEN_CALIBRATION_Ver_3.1.300.1.CAB | 2017/6/11 8:39  | Cab 文件 | 264 KB     |

2. Check the '[Calibration](#)' menu, and select '[Extract & Run](#)'.





## 6. Level 1 Repair

3. Check the 'CAL' and open the [model file](#), then select 'Start' button.

**Select Sequence Files & Login**  
Select sequence files & the resolution. Change the permission, Join, etc

Select The Sequence File  
Deploy Path : C:\DIST\DAEUL

|                                     |              |                                             |
|-------------------------------------|--------------|---------------------------------------------|
| <input type="checkbox"/>            | SMD F/T      |                                             |
| <input type="checkbox"/>            | PBA F/T      |                                             |
| <input checked="" type="checkbox"/> | CAL          | SM-C7100_OPEN_CALIBRATION_Ver_3.1.300.1.seq |
| <input type="checkbox"/>            | CAL2nd       |                                             |
| <input type="checkbox"/>            | FINAL        |                                             |
| <input type="checkbox"/>            | FINAL2nd     |                                             |
| <input type="checkbox"/>            | FINAL MANUAL |                                             |
| <input type="checkbox"/>            | IMEI         |                                             |
| <input type="checkbox"/>            | WLAN         |                                             |
| <input type="checkbox"/>            | GPS          |                                             |
| <input type="checkbox"/>            | BT           |                                             |

☐ Skip Configuration Dialog      Permission : Operator

Resolution : 1024 x 768      [Change Permission](#)      [Exit](#)      [Start](#)

4. Change the Line Type to 'Block Cell' and disable 'Smart Cloud Cell'.

**Set System Configuration**  
Set System Configuration Dialog...

Test Process  
[Process] [Master] [Slave]  
SMD F/T ☐ ☐  
PBA F/T ☐ ☐  
Calibration ☒ ☐  
Calibration 2ND ☐ ☐  
Final Auto ☐ ☐  
Final Auto 2ND ☐ ☐  
Final Manual ☐ ☐  
IMEI Write ☐ ☐  
IMEI Check ☐ ☐  
MDL +2nd Check ☐ ☐  
MDL Rework ☐ ☐  
IMEI Read ☐ ☐  
STA Write ☐ ☐  
STA Check ☐ ☐  
STA Reset ☐ ☐  
WLAN ☐ ☐  
GPS ☐ ☐  
BT ☐ ☐  
WLAN ☐ ☐  
Power Off-On before WLAN ☐ ☐  
Bluetooth ☐ ☐  
LCIA ☐ ☐  
Merge 2G3G Block Rad. ☐ ☐  
[Process Order](#)

Test Condition  
Calibration  
Real CAL Cycle: on every 20 default CALs  
Calibration Mode : FDT  
CAL2nd Mode : FDT  
Final  
Supply RF Signal by Conduction  
[Reset Loss Correction Count](#)  
Test Mode : Signaling  
WLAN  
Test Mode : WLAN  
IMEI  
Use RFMSM ☐  
Use Second PC ☐  
Save ODS ☐  
Merge Felica Cal ☐  
OQC Reset ☐  
IBI Reset ☐  
OQC SKD USER D/L ☐

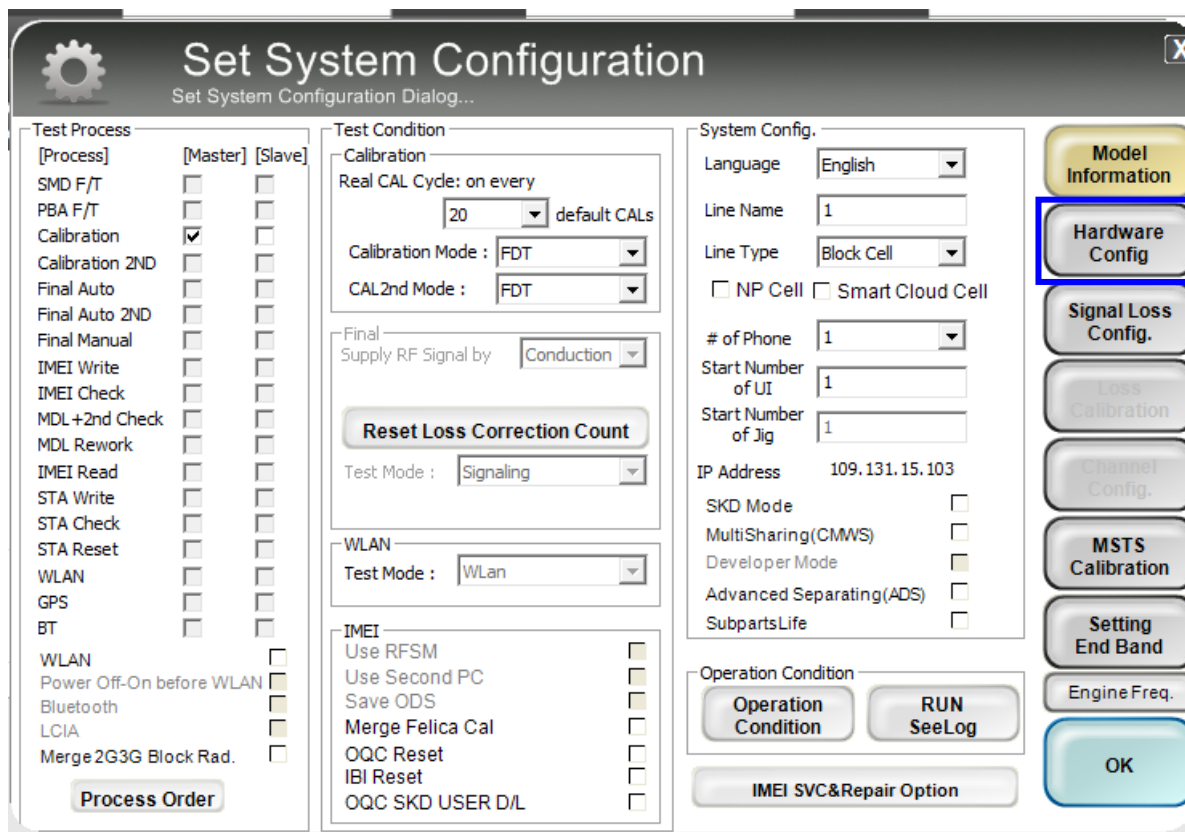
System Config.  
Language English  
Line Name 1  
Line Type Block Cell  
☐ NP Cell ☐ Smart Cloud Cell  
# of Phone 1  
Start Number of UI 1  
Start Number of Jig 1  
IP Address 109.131.15.103  
SKD Mode ☐  
MultiSharing(CMWS) ☐  
Developer Mode ☐  
Advanced Separating(ADS) ☐  
SubpartsLife ☐  
Operation Condition  
[Operation Condition](#) [RUN SeeLog](#)  
[IMEI SVC&Repair Option](#)

[Model Information](#)  
[Hardware Config](#)  
[Signal Loss Config.](#)  
[Loss Calibration](#)  
[Channel Config.](#)  
[MSTS Calibration](#)  
[Setting End Band](#)  
[Engine Freq.](#)  
[OK](#)



## 6. Level 1 Repair

5. Set the GPIB address of MSTs(CMW500) and Power Supply(E3632A) to enter



The 'Set System Configuration' dialog box is shown. The 'Test Process' tab is active, showing a list of test processes. The 'Test Condition' tab is also visible, showing calibration settings. The 'System Config.' tab is active, showing various system configuration options. The 'Hardware Config' button is highlighted with a blue box.

**Test Process**

| [Process]                | [Master]                            | [Slave]                  |
|--------------------------|-------------------------------------|--------------------------|
| SMD F/T                  | <input type="checkbox"/>            | <input type="checkbox"/> |
| PBA F/T                  | <input type="checkbox"/>            | <input type="checkbox"/> |
| Calibration              | <input checked="" type="checkbox"/> | <input type="checkbox"/> |
| Calibration 2ND          | <input type="checkbox"/>            | <input type="checkbox"/> |
| Final Auto               | <input type="checkbox"/>            | <input type="checkbox"/> |
| Final Auto 2ND           | <input type="checkbox"/>            | <input type="checkbox"/> |
| Final Manual             | <input type="checkbox"/>            | <input type="checkbox"/> |
| IMEI Write               | <input type="checkbox"/>            | <input type="checkbox"/> |
| IMEI Check               | <input type="checkbox"/>            | <input type="checkbox"/> |
| MDL +2nd Check           | <input type="checkbox"/>            | <input type="checkbox"/> |
| MDL Rework               | <input type="checkbox"/>            | <input type="checkbox"/> |
| IMEI Read                | <input type="checkbox"/>            | <input type="checkbox"/> |
| STA Write                | <input type="checkbox"/>            | <input type="checkbox"/> |
| STA Check                | <input type="checkbox"/>            | <input type="checkbox"/> |
| STA Reset                | <input type="checkbox"/>            | <input type="checkbox"/> |
| WLAN                     | <input type="checkbox"/>            | <input type="checkbox"/> |
| GPS                      | <input type="checkbox"/>            | <input type="checkbox"/> |
| BT                       | <input type="checkbox"/>            | <input type="checkbox"/> |
| WLAN                     | <input type="checkbox"/>            | <input type="checkbox"/> |
| Power Off-On before WLAN | <input type="checkbox"/>            | <input type="checkbox"/> |
| Bluetooth                | <input type="checkbox"/>            | <input type="checkbox"/> |
| LCIA                     | <input type="checkbox"/>            | <input type="checkbox"/> |
| Merge 2G3G Block Rad.    | <input type="checkbox"/>            | <input type="checkbox"/> |

**Test Condition**

Calibration  
Real CAL Cycle: on every 20 default CALs  
Calibration Mode : FDT  
CAL2nd Mode : FDT  
Final Supply RF Signal by : Conduction  
Reset Loss Correction Count  
Test Mode : Signaling  
WLAN Test Mode : WLAN  
IMEI Use RFMS Use Second PC Save ODS Merge Felica Cal OQC Reset IBI Reset OQC SKD USER D/L

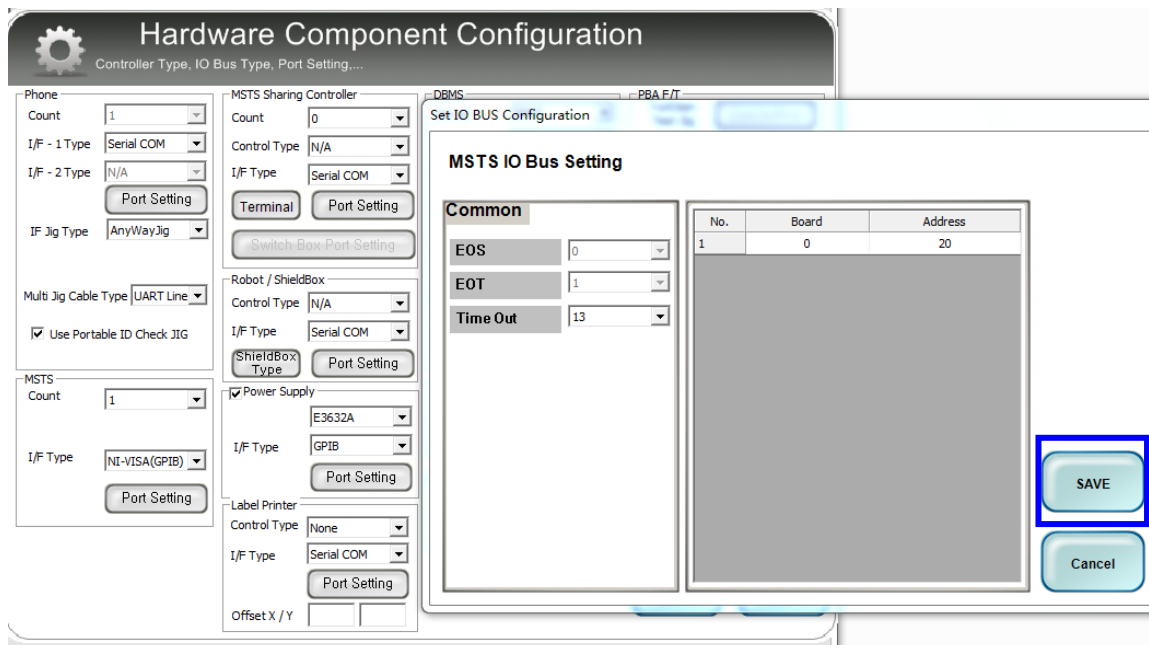
**System Config.**

Language : English  
Line Name : 1  
Line Type : Block Cell  
☐ NP Cell ☐ Smart Cloud Cell  
# of Phone : 1  
Start Number of UI : 1  
Start Number of Jig : 1  
IP Address : 109.131.15.103  
SKD Mode ☐  
MultiSharing(CMWS) ☐  
Developer Mode ☐  
Advanced Separating(ADS) ☐  
SubpartsLife ☐  
Operation Condition : Operation Condition RUN SeeLog  
IMEI SVC&Repair Option

**Hardware Config**

Model Information  
Signal Loss Config.  
Loss Calibration  
Channel Config.  
MSTS Calibration  
Setting End Band  
Engine Freq.  
OK

'Hardware Config' and 'Save'. (Check the GPIB address of equipments in advance)



The 'Hardware Component Configuration' dialog box is shown. The 'MSTS IO Bus Setting' tab is active, showing settings for the MSTs IO Bus. The 'SAVE' button is highlighted with a blue box.

**Hardware Component Configuration**

Controller Type, IO Bus Type, Port Setting...

Phone Count : 1  
I/F - 1 Type : Serial COM  
I/F - 2 Type : N/A  
Port Setting  
IF Jig Type : AnyWayJig  
Multi Jig Cable Type : UART Line  
Use Portable ID Check JIG ☒  
MSTS Count : 1  
I/F Type : NI-VISA(GPIB)  
Port Setting

**MSTS Sharing Controller**

Count : 0  
Control Type : N/A  
I/F Type : Serial COM  
Terminal Port Setting  
Switch Box Port Setting  
Robot / ShieldBox Control Type : N/A  
I/F Type : Serial COM  
ShieldBox Type Port Setting  
Power Supply E3632A  
I/F Type : GPIB  
Port Setting  
Label Printer Control Type : None  
I/F Type : Serial COM  
Port Setting  
Offset X / Y

**MSTS IO Bus Setting**

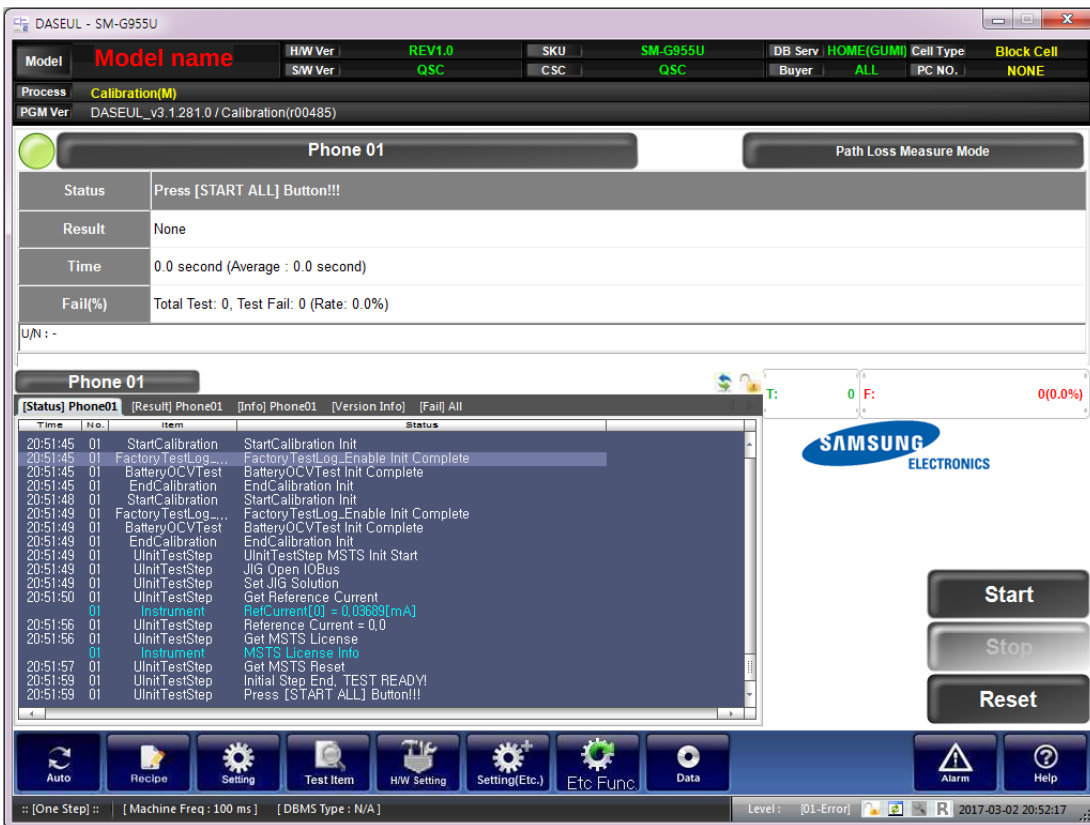
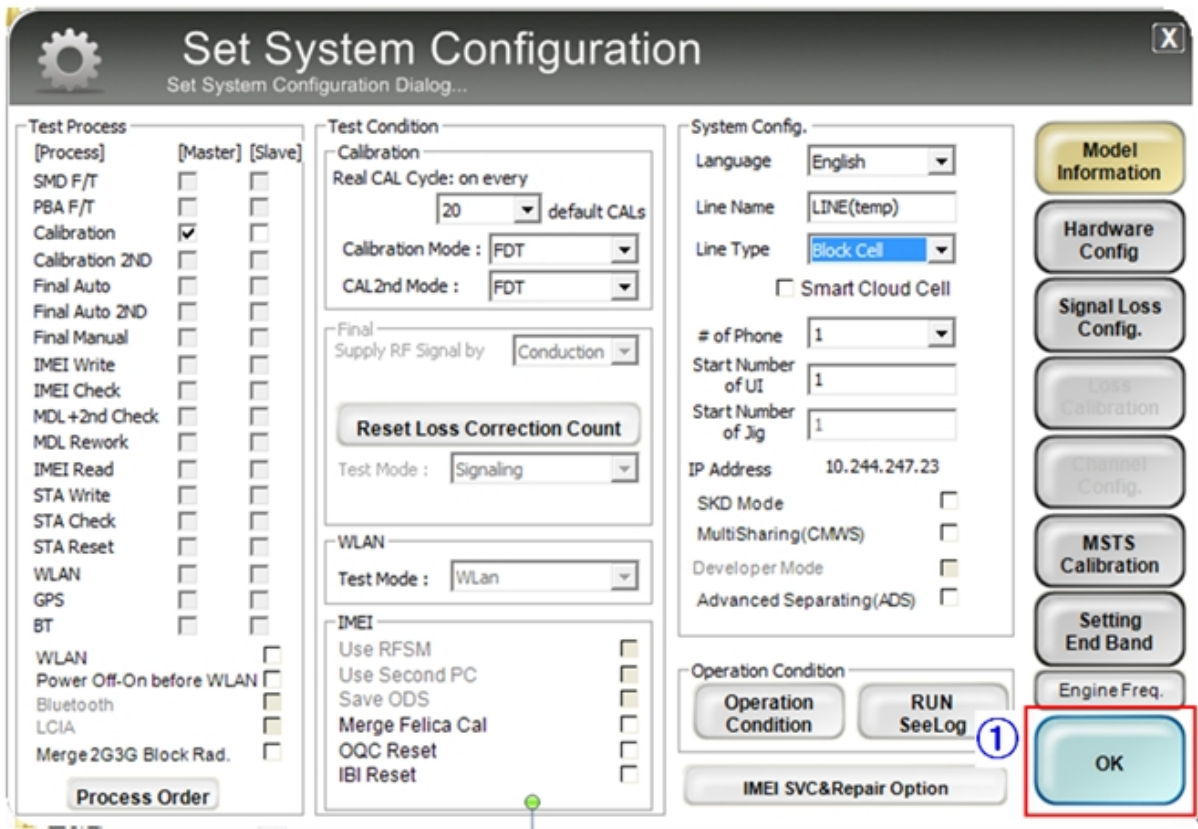
Common  
EOS : 0  
EOT : 1  
Time Out : 13

| No. | Board | Address |
|-----|-------|---------|
| 1   | 0     | 20      |

SAVE  
Cancel

## 6. Level 1 Repair

6. Press 'OK' to start RF Calibration after completing all settings.

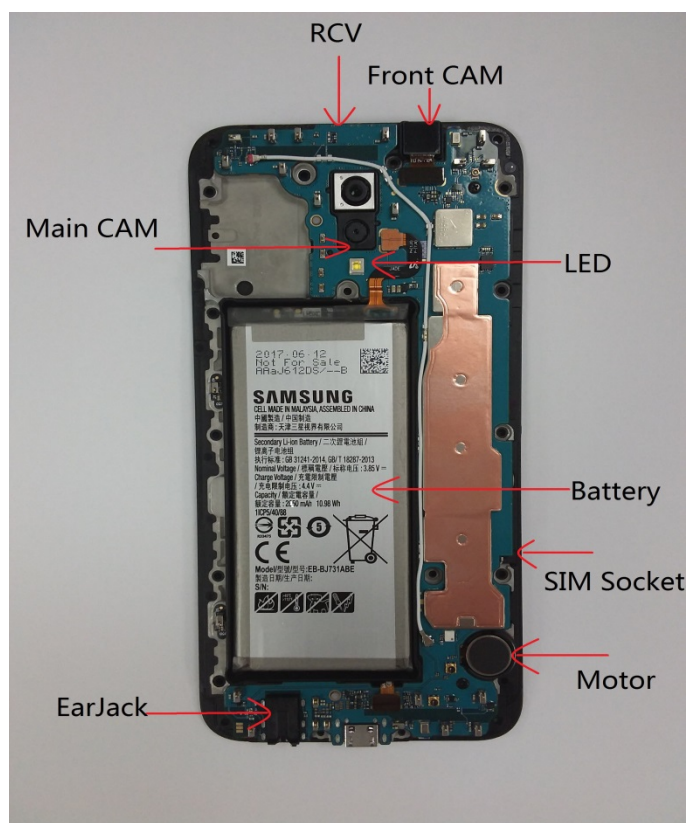
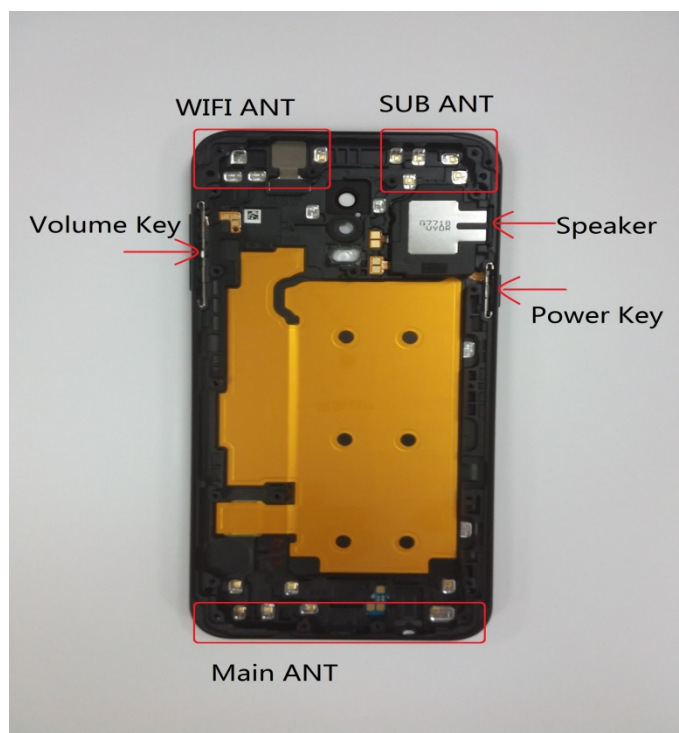


---

## 7. Level 2 Repair

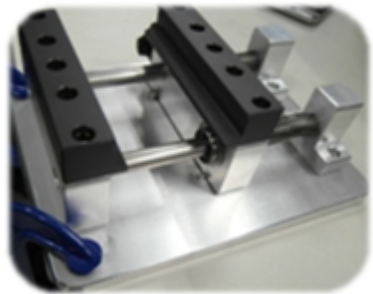
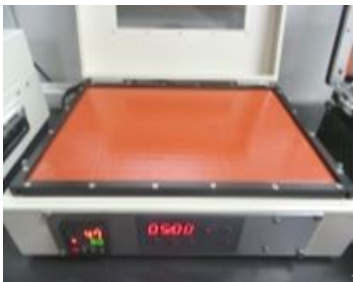
---

### 7-1. Components on the Rear Case



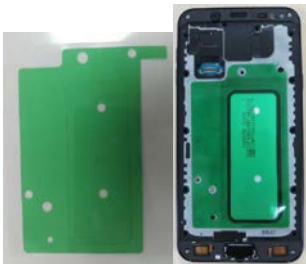
## 7. Level 2 Repair

### 7-2. Pre-requisite

|                                                                                     |                                                                                      |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|    |   |
| <b>Tweezers / Disass'y Stick / Screw Driver</b>                                     | <b>Anti-static Gloves</b>                                                            |
|    |    |
| <b>OCTA Disassembly Holder</b>                                                      | <b>Anti-static Mat</b>                                                               |
|  |  |
| <b>Hot Plate</b>                                                                    | <b>OCTA Disassembly Upper</b>                                                        |
|  |  |
| <b>Ethyl Alcohol</b>                                                                | <b>Cotton Swab</b>                                                                   |

## 7. Level 2 Repair

### 7-3. Parts which must be changed after repair

| BOM Description                                                                                                                                            | Picture                                                                                                                                                                                                            | Remarks                           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| A/S-TAPE_<br>MAIN_WINDOW_SVC<br>[GH02-14924A]<br>[GH02-14923A]<br><br>Touch Key Top Tape<br>GH81-14737A<br><br>A/S-TAPE HOTMELT-HR<br>SVC<br>[GH02-15149A] | <br>[TOP_ GH02-14924A]<br><br>[BOT_ GH02-14923A] | Replace when<br>repair Front part |
|                                                                                                                                                            |                                                                                                                                  |                                   |
|                                                                                                                                                            |                                                                                                                                  |                                   |

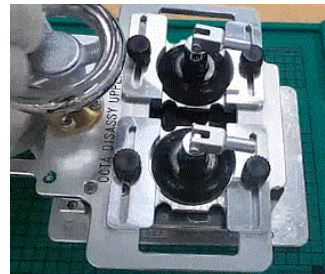


## 7. Level 2 Repair

### 7-4. Disassembly

※ Be careful to put on antistatic wrist strap!

1. Put the device in the chamber or hot plate as following below heating condition
2. Temp :  $70^{\circ}\pm 5$  , Time : 15 Minutes  
※ Please confirm the heating condition released lastly, and follow it.



Take out SET, when OCTA and Front ASS'Y Detach

[1] After heating for 15~30 minutes at  $70\sim 80^{\circ}\text{C}$ , take out the device from Mobile dryer.

[2] Detach the Back Cover from SET using Back Cover Disassembly Holder.



[3] Loosen 20 screws.

※ Caution

Take care of scratch and damage.

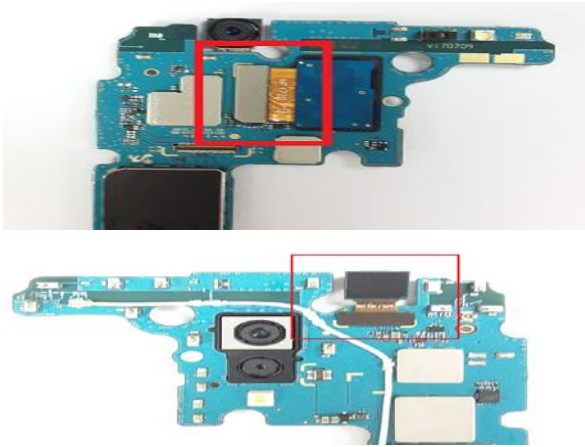


[4] Disassemble rear from bracket assy.(from top to bottom)

※ Caution

Be careful not to Damage

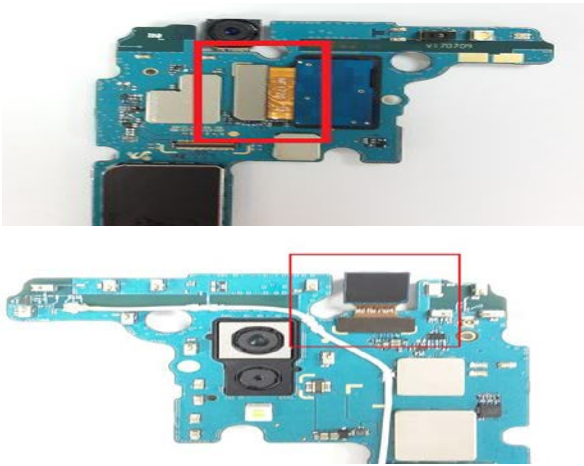
7. Level 2 Repair

|                                                                                                                                                                                                                                         |                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
|                                                                                                                                                        |                      |
| <p>[5]</p> <p>1) Disassemble Home Key Connector/Battery Connector</p> <p>2) Separate PBA from Brocket. Detach 2 screws point</p> <p>※ Caution</p> <p>1) Be careful not to damage.</p> <p>2) Separate PBA after detach the PBA Conn.</p> | <p>[6] Disassemble Mega Cam and Front Cam on PBA.</p> <p>※ Caution</p> <p>Be careful not to Damage</p> |
|                                                                                                                                                      |                    |
| <p>[7]Disassemble RCV, Motor</p>                                                                                                                                                                                                        | <p>[8]Disassemble SPK from the REAR</p>                                                                |

## 7. Level 2 Repair

### 7-5. Assembly

※ Be careful to put on antistatic wrist strap!

|                                                                                                   |                                                                                      |
|---------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|                  |    |
| <p>[1] Assemble the Main CAM and VT CAM to the PBA</p>                                            | <p>[2] Assemble Motor, RCV</p>                                                       |
|                |  |
| <p>[3] Assemble PBA to Bracket with 2 point screws<br/>※ Caution<br/>Be careful not to Damage</p> | <p>[4] Assemble Homekey connector to the PBA</p>                                     |



## 7. Level 2 Repair



**[5] Assemble Rear to Bracket assy ( from bottom to Top)**

※ **Caution**

Be careful not to Damage

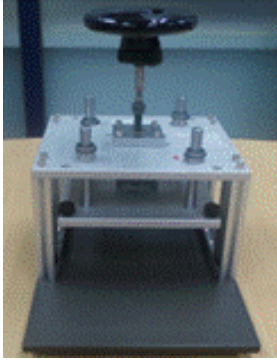
**[6] Assemble Screw 20 points**



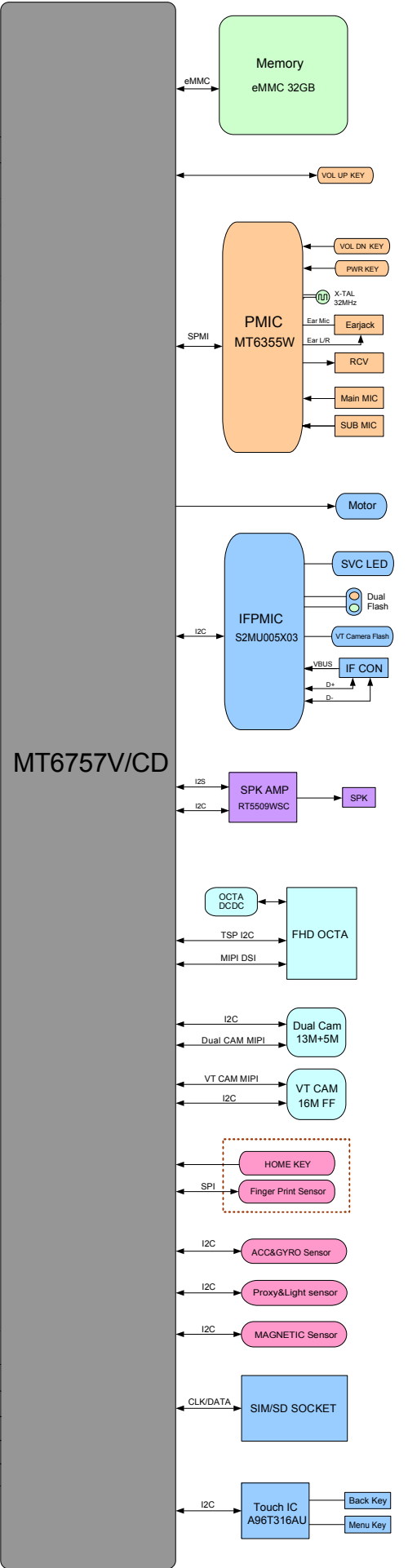
**[7] Assemble HR tape Touch key TOP tape to the Bracket and window tape to OCTA**

**[8] Assemble OCTA connector & OCTA Conn CAP**

## 7. Level 2 Repair

|                                                                                                                |                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | <div data-bbox="863 237 1422 461"><p>1. Press the device using pressing jig</p><ul style="list-style-type: none"><li>- Force : 1N</li><li>- Time : 1minute</li></ul><p>※ Please confirm the pressing condition released lastly, and follow it.</p></div>  |
| <p><b>[9]Assemble OCTA to Bracket</b></p> <p>※ <b>Caution</b></p> <p>Be careful not to Damage the home key</p> | <p><b>[10]Press SET using OCTA PRESS Jig (1N for 1 minute)</b></p>                                                                                                                                                                                                                                                                          |

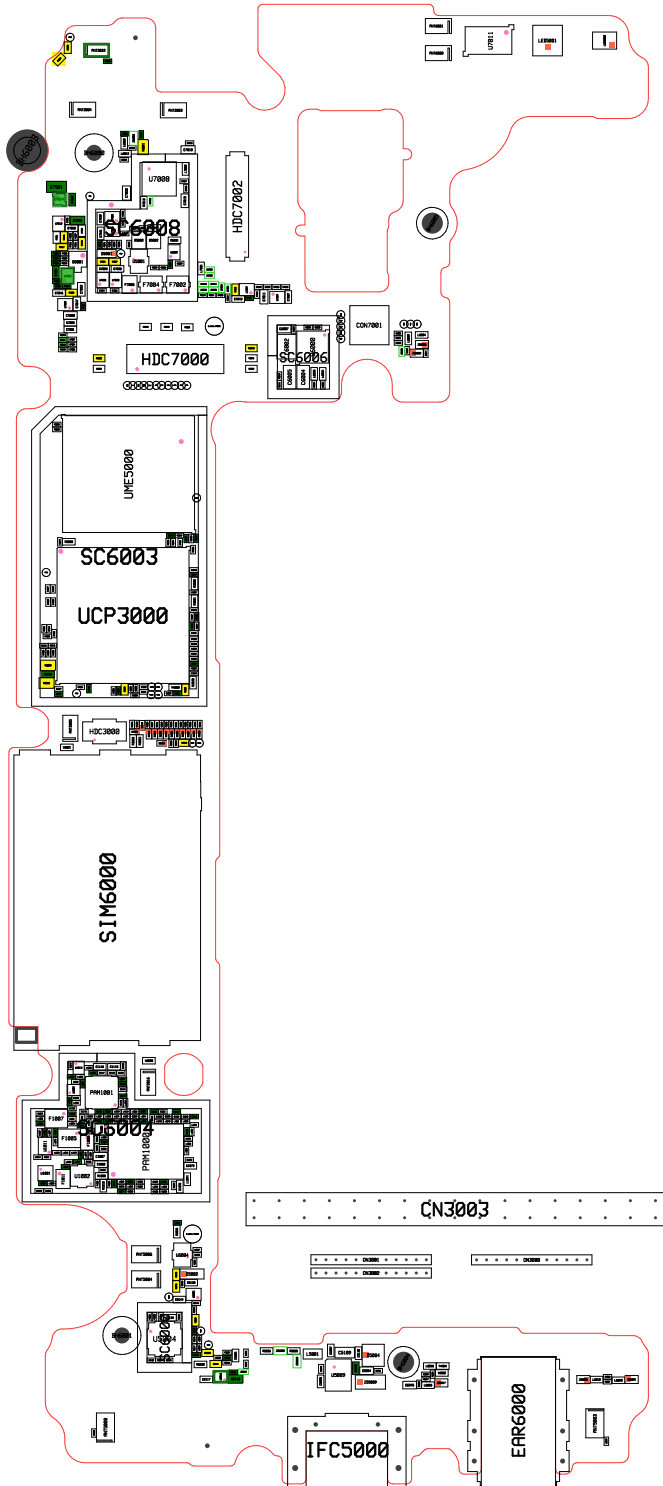
## SM-C7108 SVC Manual Block Diagram

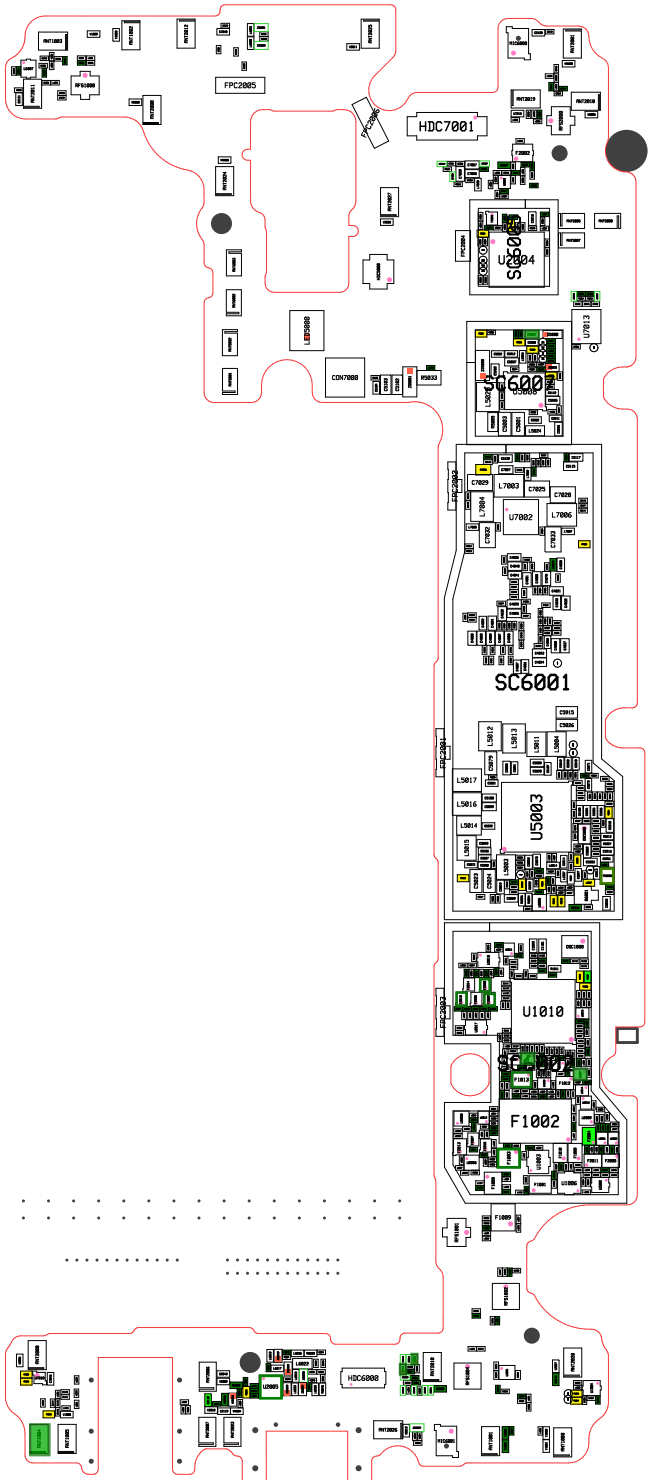


## Information of RF Bands supported on SM-C7108 Model

GSM850 : Rx : 869.2 ~ 893.8 MHz / Tx : 824.2 ~ 848.8 MHz  
GSM1900 : Rx : 1930.2 ~ 1989.8 MHz / Tx : 1850 .2 ~ 1909 .8 MHz  
GSM900 : Rx : 925.2 ~ 959.8 MHz / Tx : 880.2 ~ 914.8 MHz  
GSM1800 : Rx : 1805.2 ~ 1879.8 MHz / Tx : 1710 .2 ~ 1784 .8 MHz  
WCDMA Band 1 : Rx : 2112.4 ~ 2167 .6 MHz / Tx : 1922.4 ~ 1977.6 MHz  
WCDMA Band 2 : Rx : 1930 ~ 1990 MHz / Tx : 1850 ~ 1910 MHz  
WCDMA Band 5 : Rx : 871.4 ~ 891.6 MHz / Tx : 826.4 ~ 846.6 MHz  
WCDMA Band 8 : Rx : 927.4 ~ 957.6 MHz / Tx : 882.4 ~ 912.6 MHz  
CDMA BC0: RX: 869~894MHz / TX: 824~894MHz  
LTE Band 1 : Rx : 2110 ~ 2170 MHz / Tx : 1920 ~ 1980 MHz  
LTE Band 3 : Rx : 1805 ~ 1880 MHz / Tx : 1710 ~ 1785 MHz  
LTE Band 5 : Rx : 869 ~ 894 MHz / Tx : 824 ~ 849 MHz  
LTE Band 7 : Rx : 2620 ~ 2690 MHz / Tx : 2500 ~ 2570 MHz  
LTE Band 8 : Rx : 925 ~ 960 MHz / Tx : 880 ~ 915 MHz  
LTE Band 38: 2570~2620 MHz  
LTE Band 39: 1880~1920 MHz  
LTE Band 40: 2300~2400 MHz  
LTE Band 41 : 2555MHz~2655MHz  
LTE Band 34 : 2010MHz~2025MHz  
TDSCDMA B34: 2010 ~ 2025 MHz  
TDSCDMA B39: 1880 ~ 1920 MHz  
BT/BLE/ANT+ : 2402 ~ 2480 MHz  
WLAN :  
2402 ~ 2480 MHz  
5180 ~ 5240 MHz / 5260 ~ 5320 MHz / 5500 ~ 5700 MHz  
5745 ~ 5825 MHz  
DTS : 2412 ~ 2472 MHz  
UNII : Ch144,142,138 not support

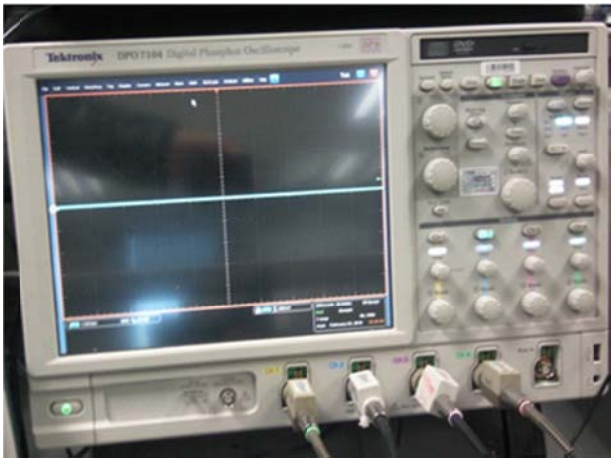
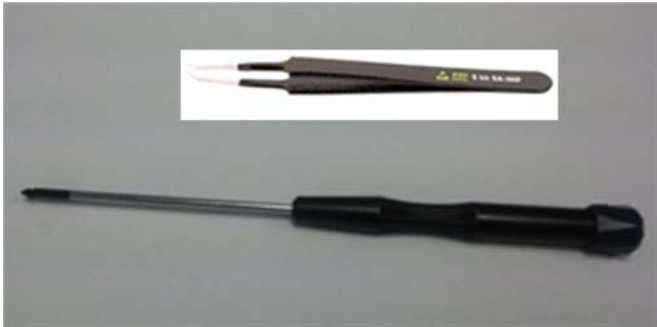
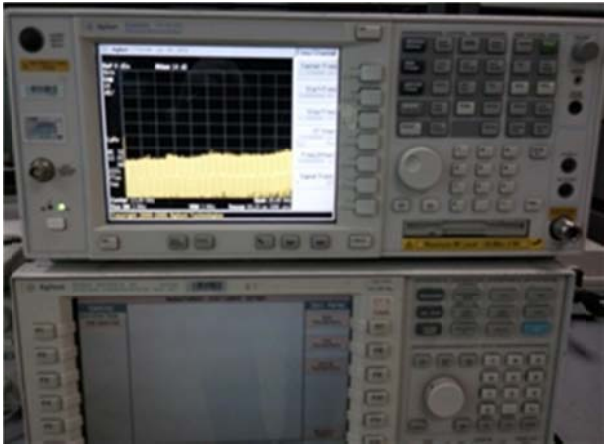
GSM850 : Rx : 869.2 ~ 893.8 MHz / Tx : 824.2 ~ 848.8 MHz  
GSM1900 : Rx : 1930.2 ~ 1989.8 MHz / Tx : 1850.2 ~ 1909.8 MHz  
GSM900 : Rx : 925.2 ~ 959.8 MHz / Tx : 880.2 ~ 914.8 MHz  
GSM1800 : Rx : 1805.2 ~ 1879.8 MHz / Tx : 1710.2 ~ 1784.8 MHz  
WCDMA Band 1 : Rx : 2112.4 ~ 2167.6 MHz / Tx : 1922.4 ~ 1977.6 MHz  
WCDMA Band 2 : Rx : 1930 ~ 1990 MHz / Tx : 1850 ~ 1910 MHz  
WCDMA Band 5 : Rx : 871.4 ~ 891.6 MHz / Tx : 826.4 ~ 846.6 MHz  
WCDMA Band 8 : Rx : 927.4 ~ 957.6 MHz / Tx : 882.4 ~ 912.6 MHz  
CDMA BC0: RX: 869~894MHz / TX: 824~894MHz  
LTE Band 1 : Rx : 2110 ~ 2170 MHz / Tx : 1920 ~ 1980 MHz  
LTE Band 3 : Rx : 1805 ~ 1880 MHz / Tx : 1710 ~ 1785 MHz  
LTE Band 5 : Rx : 869 ~ 894 MHz / Tx : 824 ~ 849 MHz  
LTE Band 7 : Rx : 2620 ~ 2690 MHz / Tx : 2500 ~ 2570 MHz  
LTE Band 8 : Rx : 925 ~ 960 MHz / Tx : 880 ~ 915 MHz  
LTE Band 38: 2570~2620 MHz  
LTE Band 39: 1880~1920 MHz  
LTE Band 40: 2300~2400 MHz  
LTE Band 41 : 2555MHz~2655MHz  
LTE Band 34 : 2010MHz~2025MHz  
TDSCDMA B34: 2010 ~ 2025 MHz  
TDSCDMA B39: 1880 ~ 1920 MHz  
BT/BLE/ANT+ : 2402 ~ 2480 MHz  
WLAN :  
2402 ~ 2480 MHz  
5180 ~ 5240 MHz / 5260 ~ 5320 MHz / 5500 ~ 5700 MHz  
5745 ~ 5825 MHz  
DTS : 2412 ~ 2472 MHz  
UNII : Ch144,142,138 not support





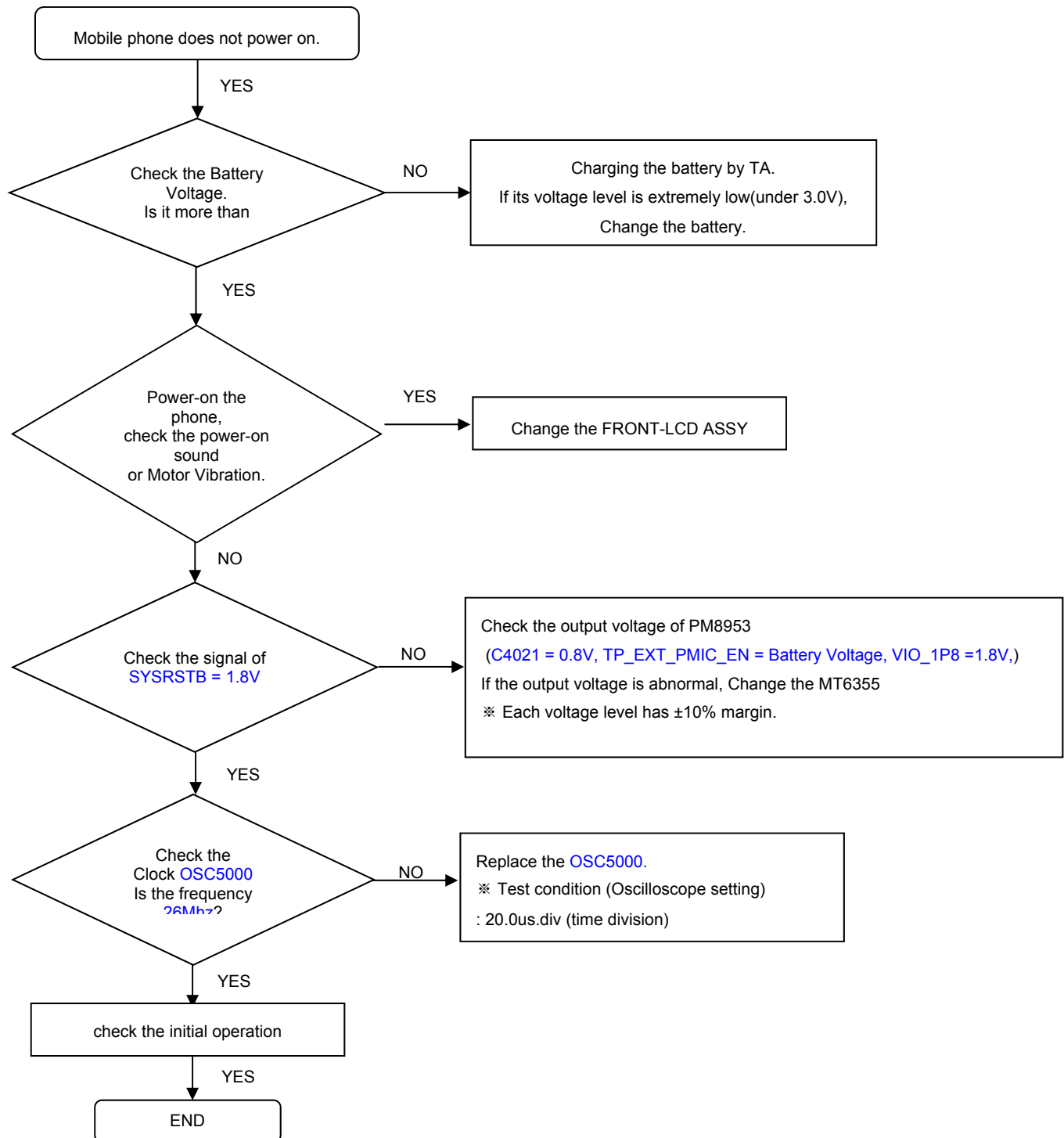
## 8. Level 3 Repair

### 8-3. Flow Chart of Troubleshooting

|                                                                                     |                                                                                      |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|    |    |
| <b>Oscilloscope</b>                                                                 | <b>Digital Multimeter</b>                                                            |
|   |   |
| <b>Power Supply</b>                                                                 | <b>+ driver, ESD Safe Tweezer</b>                                                    |
|  |  |
| <b>8960 &amp; Spectrum Analyzer</b>                                                 | <b>Soldering iron</b>                                                                |

## 8. Level 3 Repair

### 8-4-1. Power On

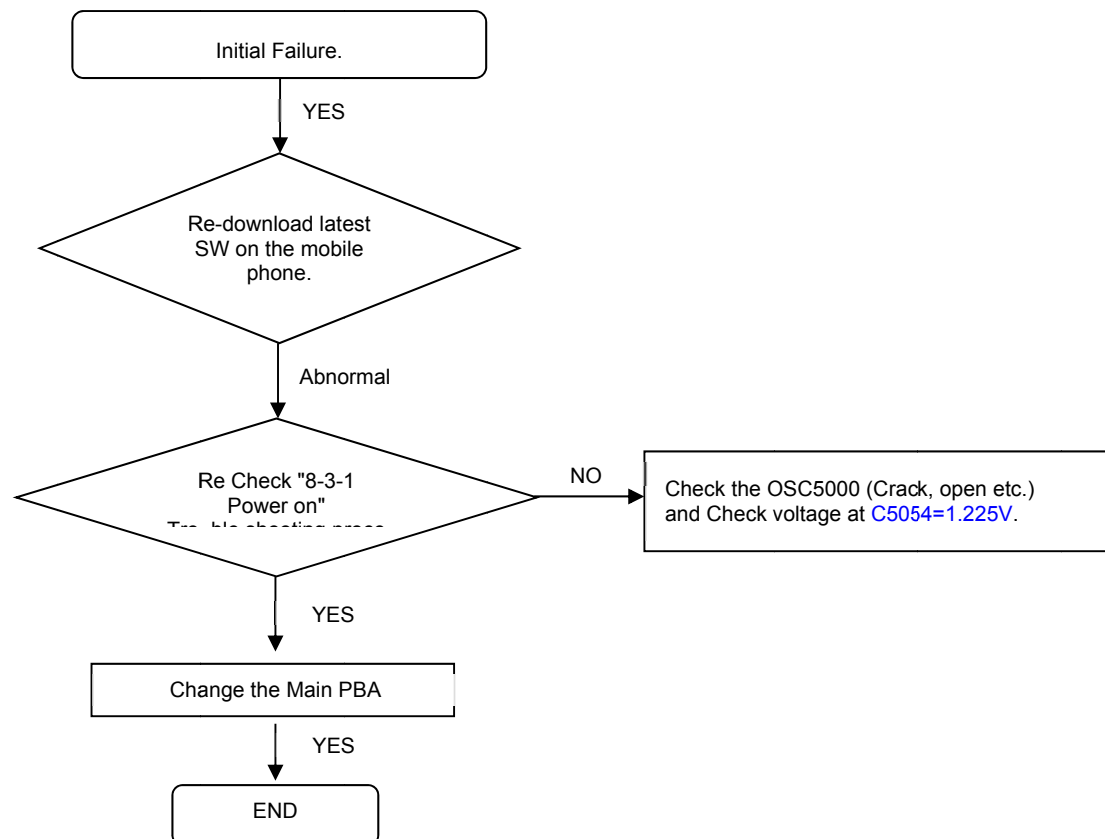


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## 8. Level 3 Repair

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### 8-4-2. Initial



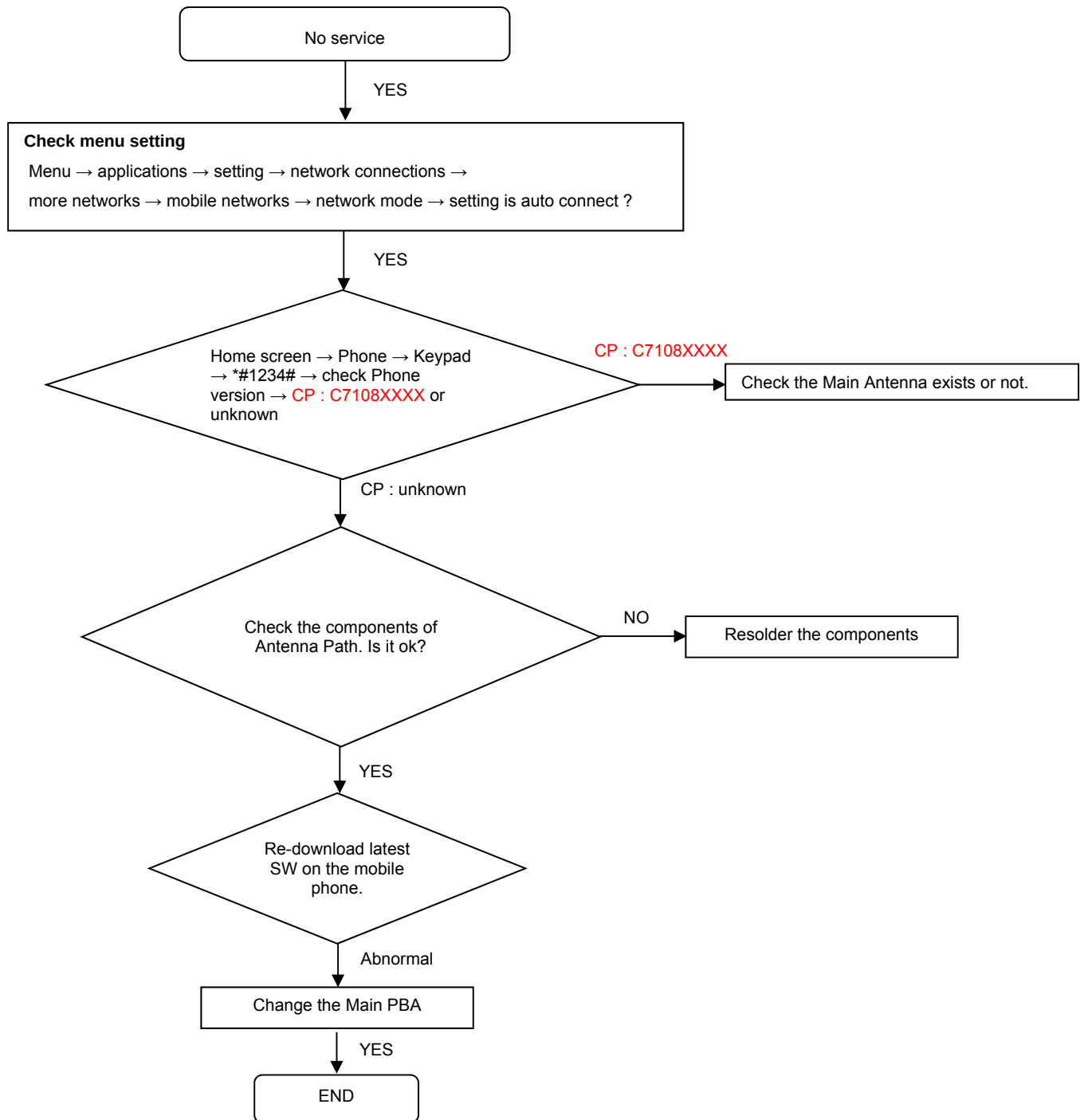


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## 8. Level 3 Repair

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### 8-4-3. No Service

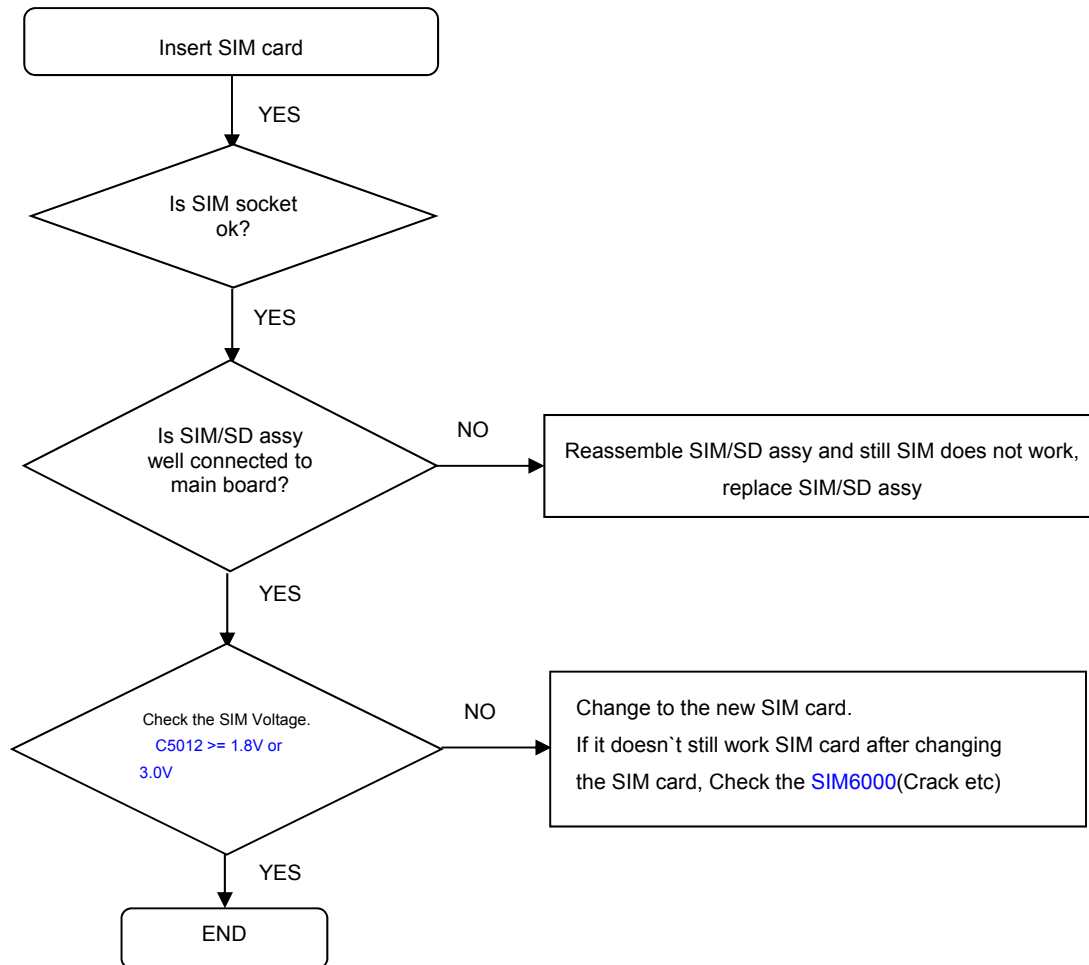


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## 8. Level 3 Repair

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### 8-4-4. Sim Part



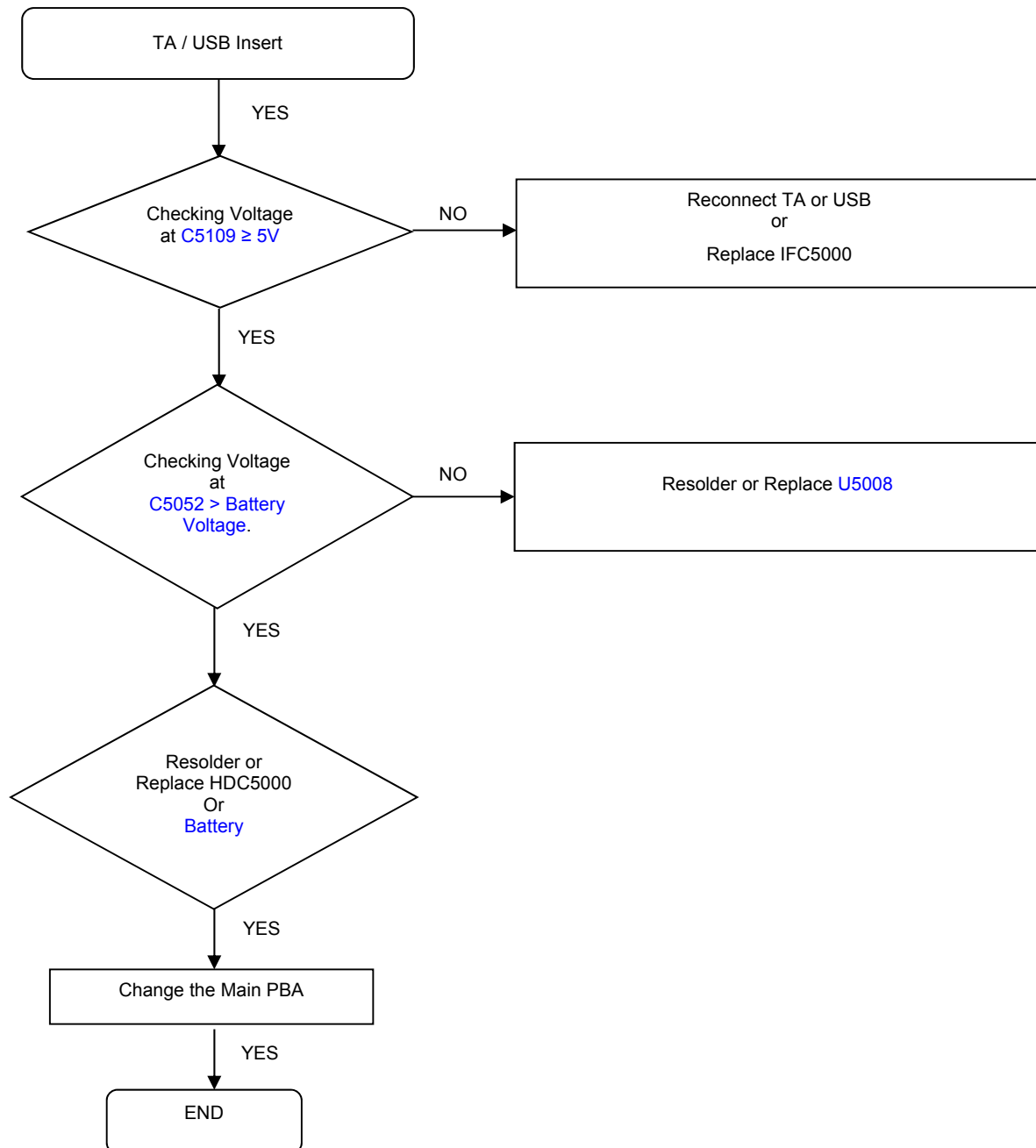
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## 8. Level 3 Repair

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### 8-4-5. Charging Part

- First of all, Please check the battery status.

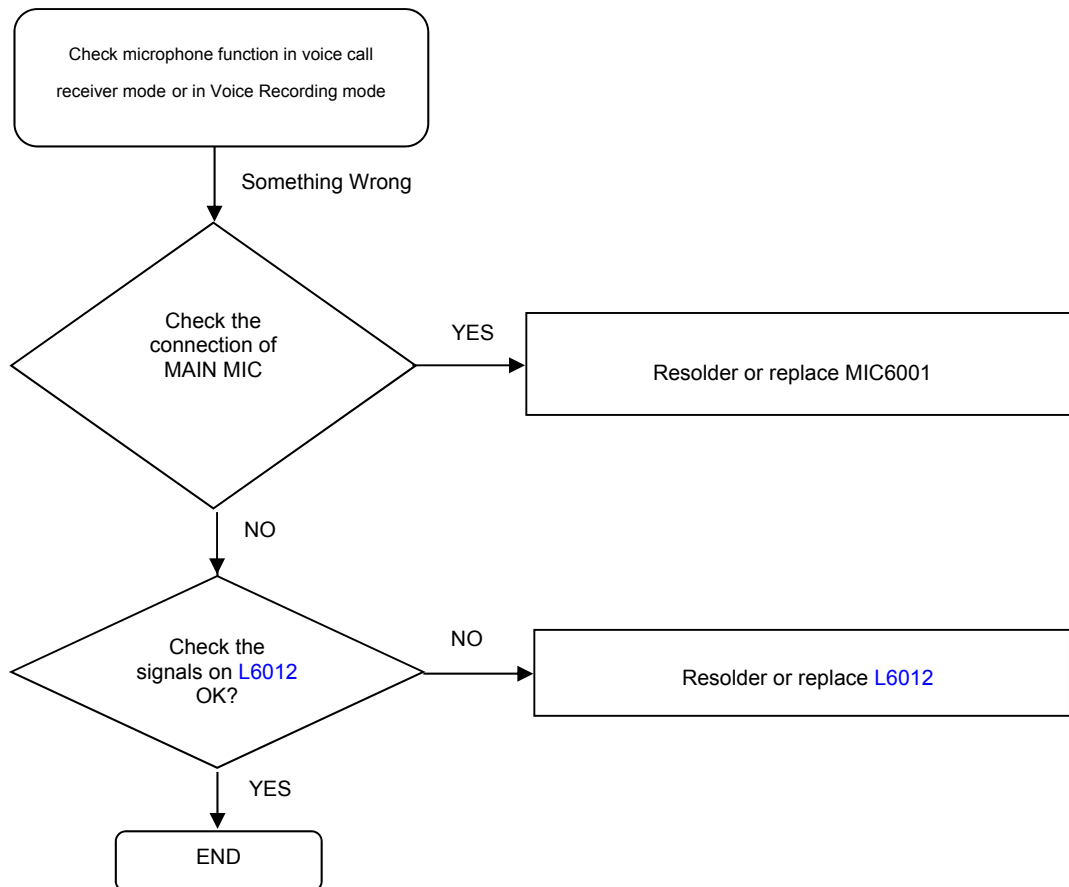


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## 8. Level 3 Repair

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### 8-4-6. Microphone Part

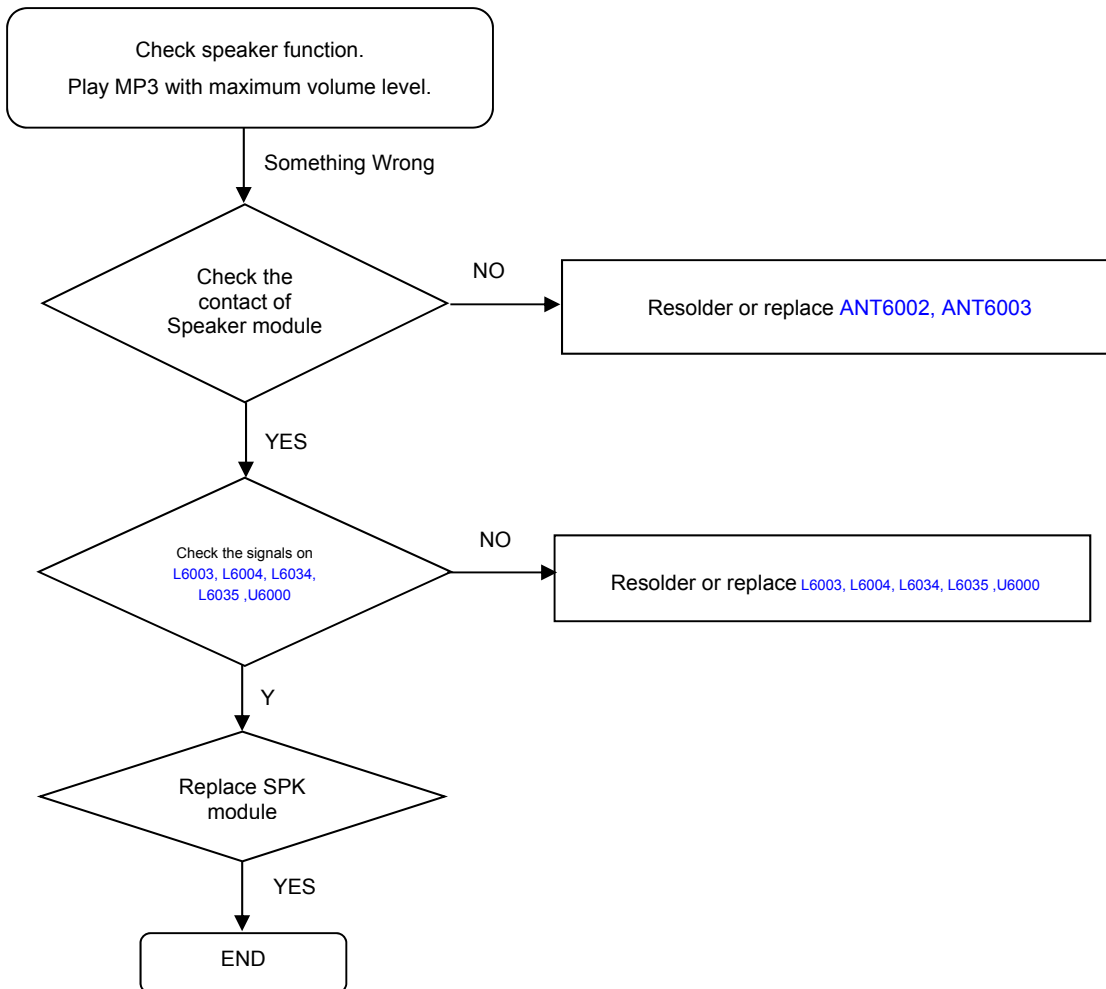


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## 8. Level 3 Repair

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### 8-4-7. Speaker Part

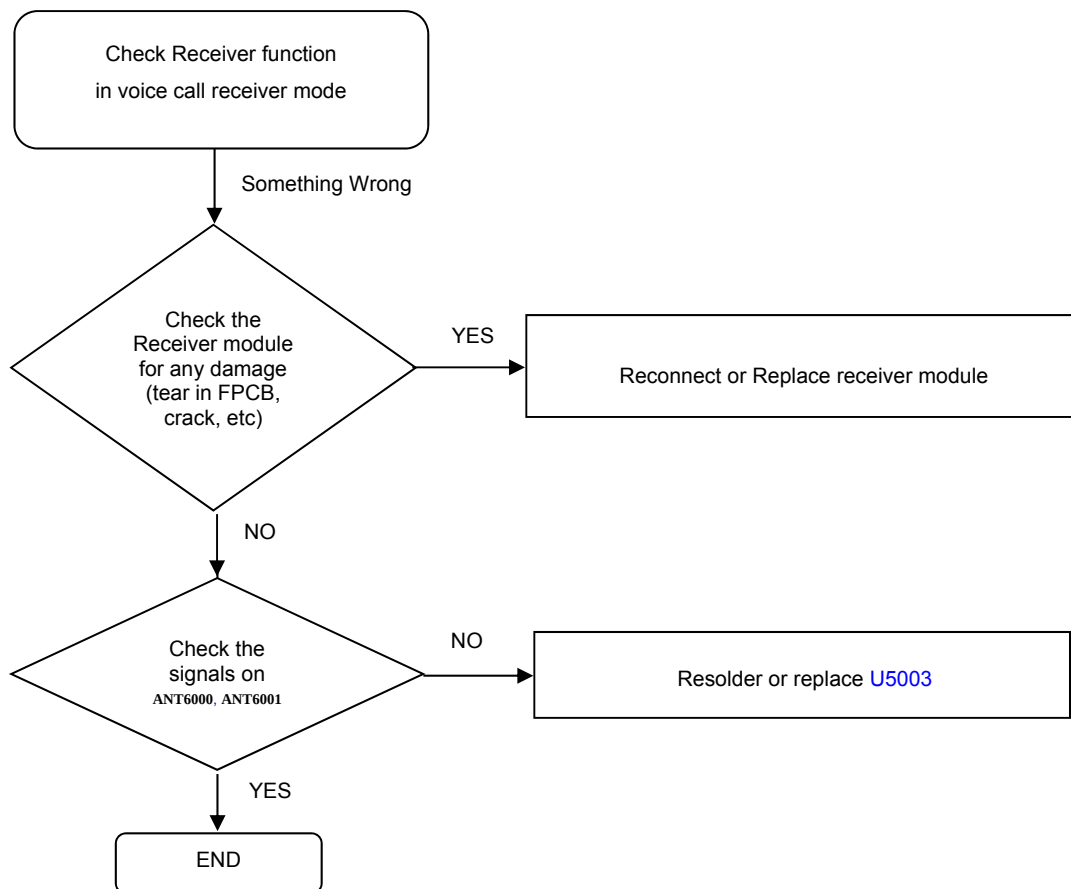


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## 8. Level 3 Repair

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### 8-4-8. Receiver Part

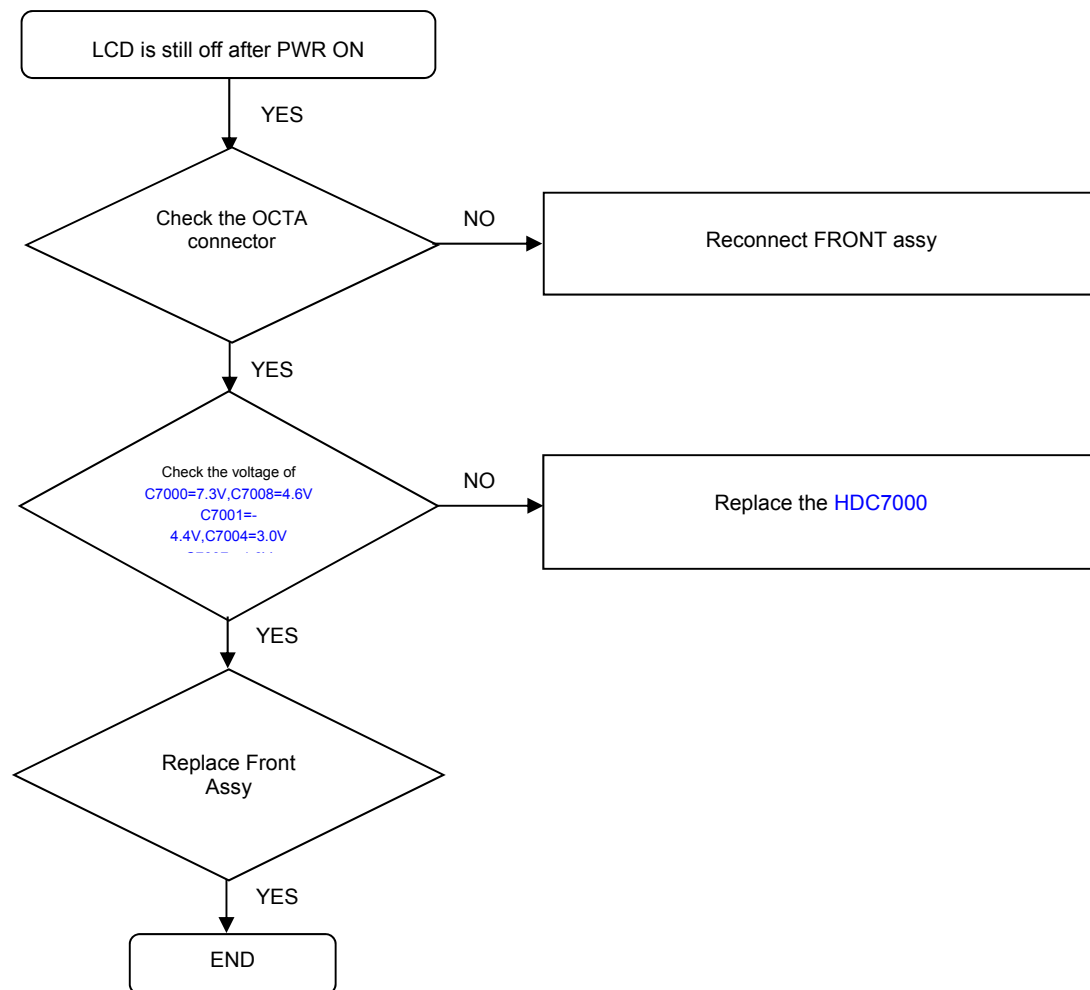


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## 8. Level 3 Repair

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### 8-4-9. OCTA

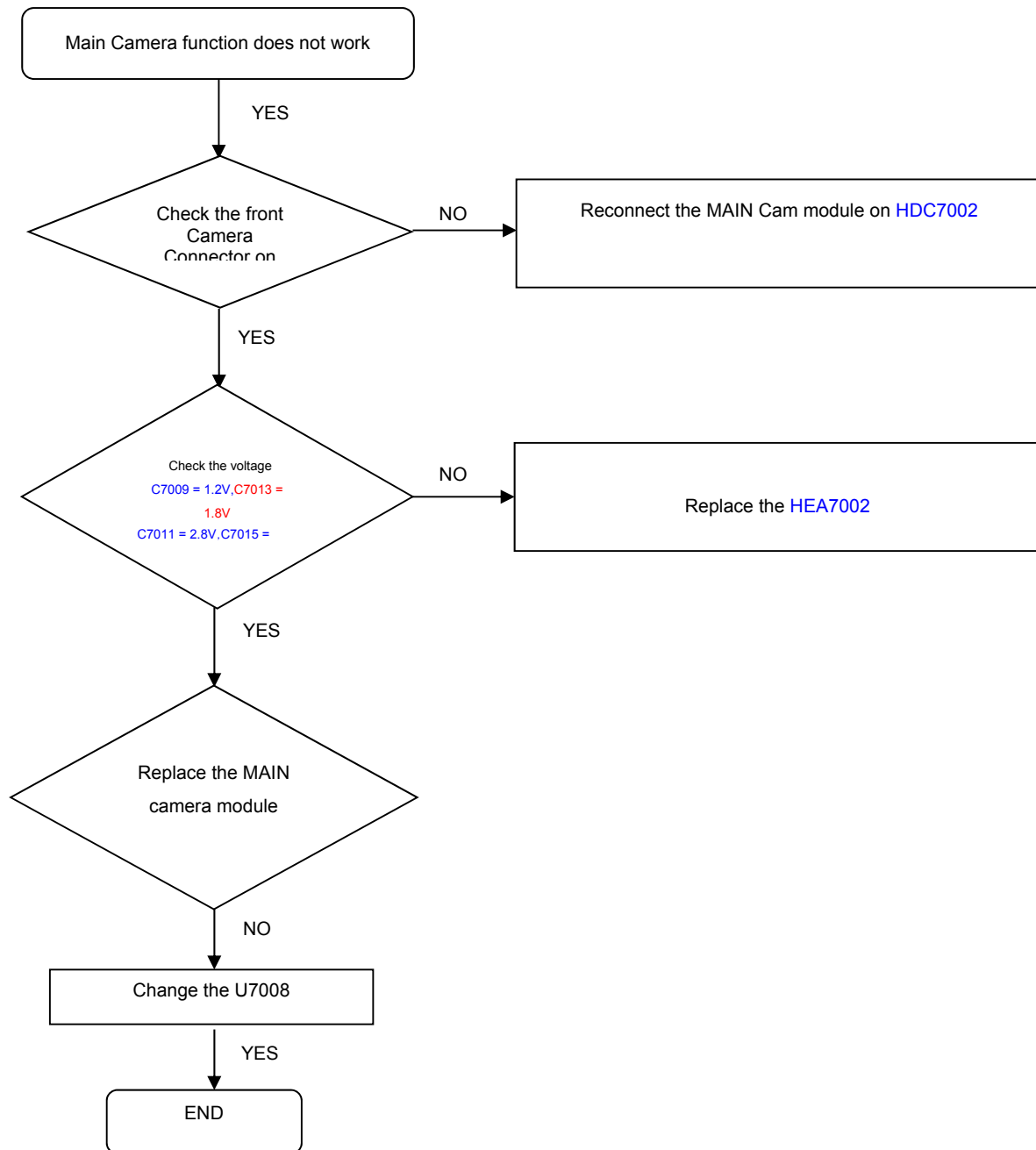


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## 8. Level 3 Repair

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### 8-4-10. MAIN CAMERA



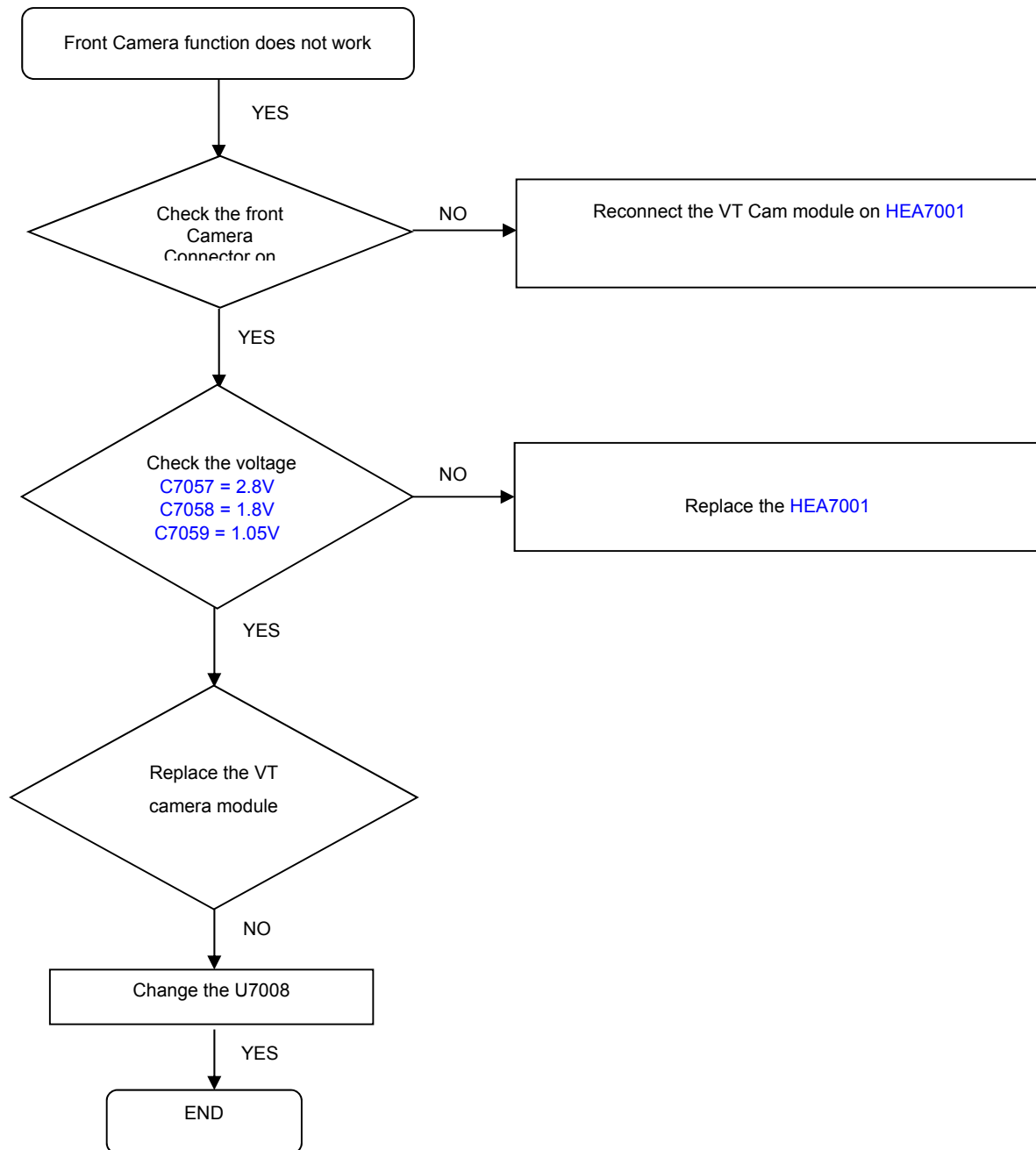


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## 8. Level 3 Repair

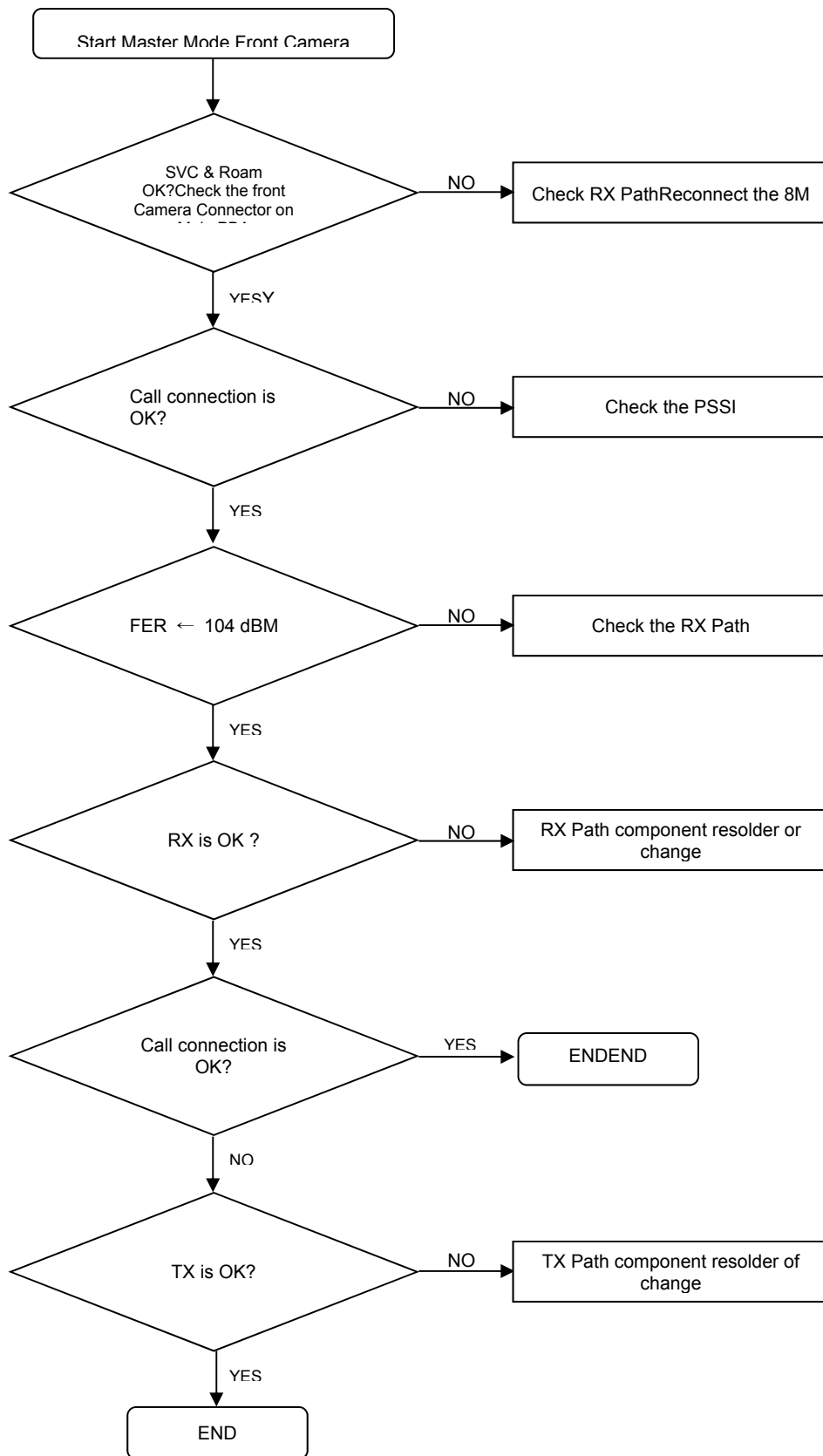
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### 8-4-11. VT CAMERA



## 8. Level 3 Repair

### 8-4-12. RF Section(LTE,WCDMA,TDSCDMA,CDMA,GSM)



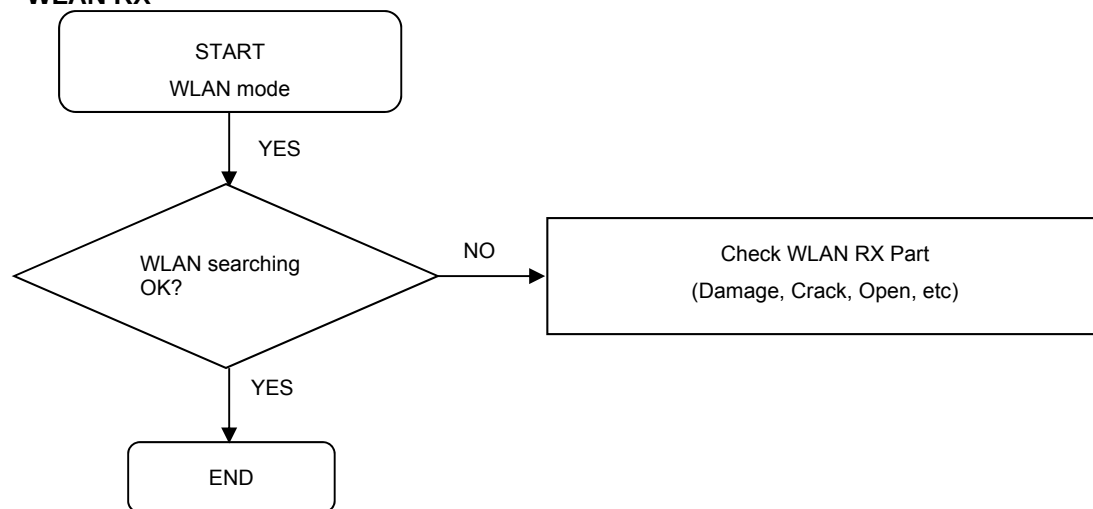
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## 8. Level 3 Repair

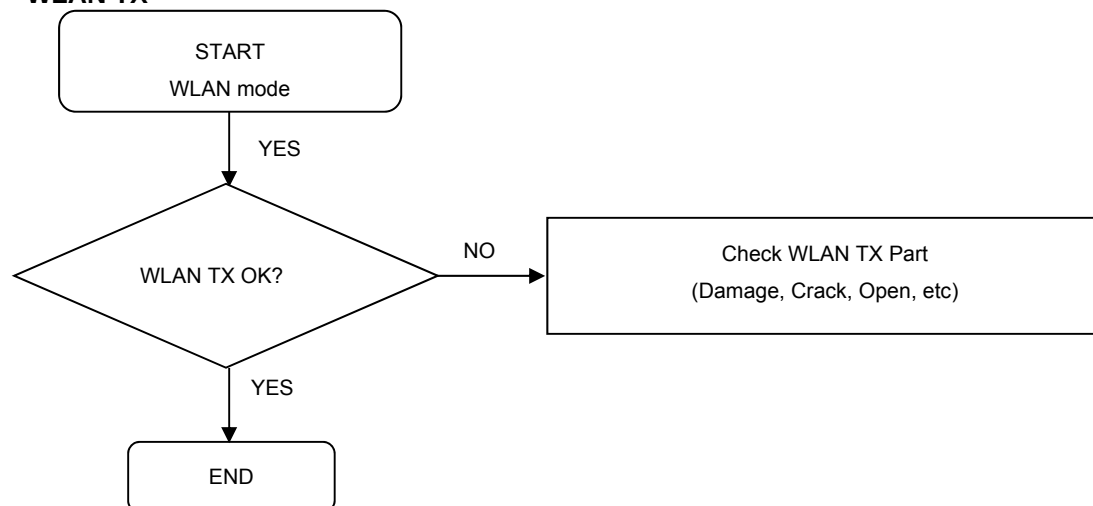
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### 8-4-13. RF Section(WLAN / BT)

#### - WLAN RX



#### - WLAN TX

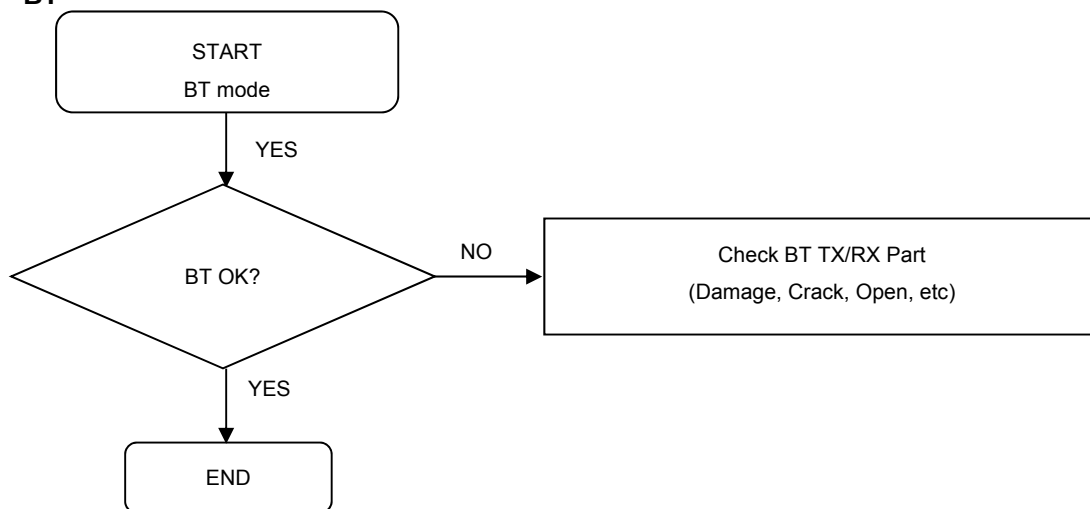


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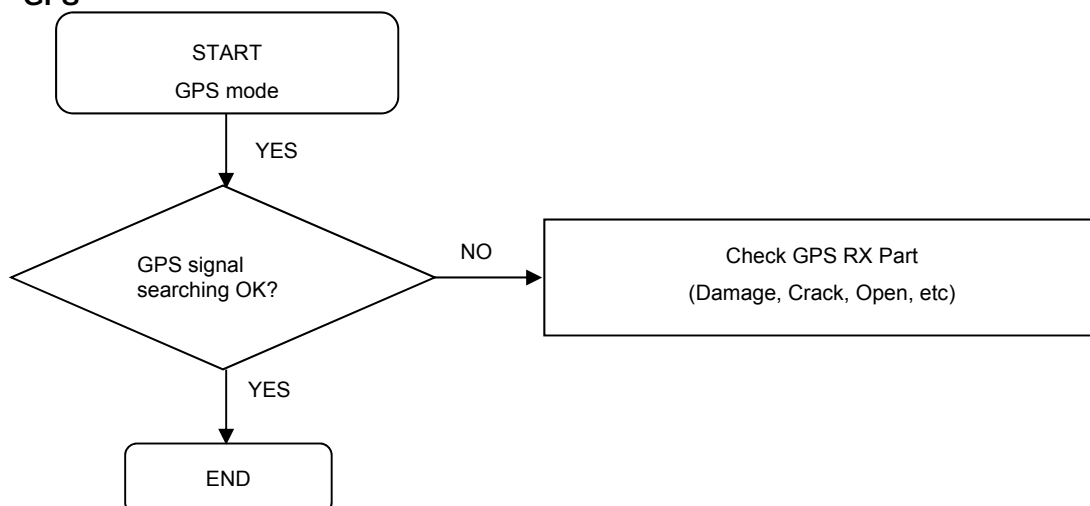
## 8. Level 3 Repair

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### - BT



### - GPS

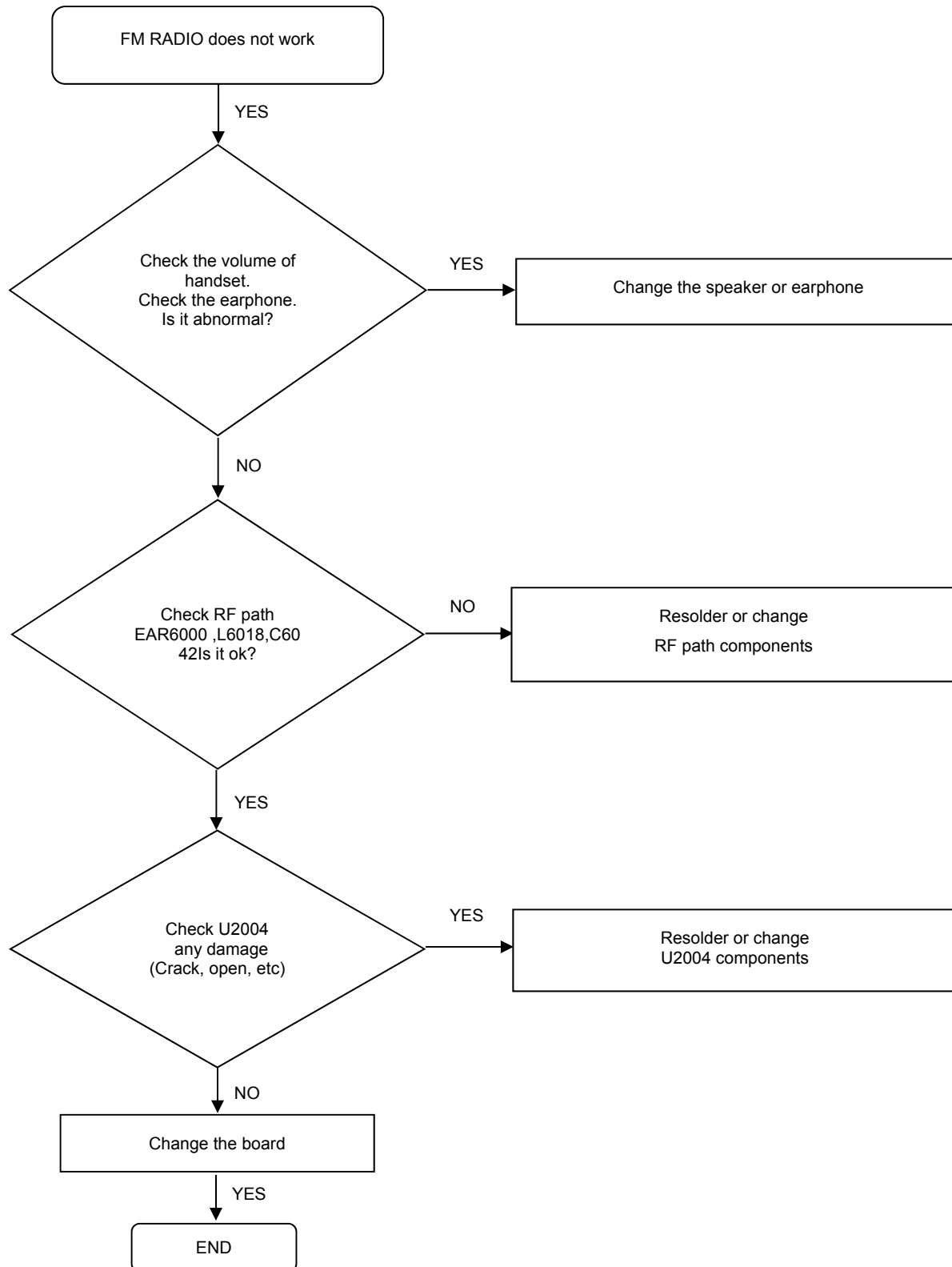


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## 8. Level 3 Repair

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### 8-4-14. FM RADIO



## 8. Level 3 Repair

### 8-4. Service Schematics

#### - NC Point(Top View)

#### UCP3000(AP)

|    | 781 | 1 | 2              | 3              | 4              | 5    | 6   | 7    | 8    | 9    | 10             | 11   | 12   | 13   | 14   | 15   | 16   | 17             | 18   | 19 | 20   | 21   | 22   | 23   | 24 | 25 | 26 | 27 | 28 | 29             | 30             | 31             | 32 |    |
|----|-----|---|----------------|----------------|----------------|------|-----|------|------|------|----------------|------|------|------|------|------|------|----------------|------|----|------|------|------|------|----|----|----|----|----|----------------|----------------|----------------|----|----|
| A  |     |   | AVD08<br>R_X01 |                | CAB            | CAB  |     |      |      |      | DQ18           | DQ21 |      | DQ17 | DQ22 |      | DQ1  | DQ6            |      |    |      | DQ25 |      | DQ19 |    |    |    |    |    | AVD03<br>R_M01 | DV55           |                | A  |    |
| B  |     |   | WR_C7<br>R_W01 | WR_C7<br>R_W01 | CA6            | CA2  |     | CA3  |      |      | DV55           | DQ26 | DQ29 | DQ28 | DQ2  | DV55 | DQ3  |                |      |    |      | DQ27 | DQ38 | DQ31 |    |    |    |    |    | AVD04<br>R_X02 | AVD05<br>R_X03 |                | B  |    |
| C  |     |   | WR_C7<br>R_W01 | WR_C7<br>R_W01 | DV55           |      |     | DV55 | CAS  | DV55 | CAB            | CAB  | DV55 | DQ23 | DV55 | DQ24 | DV55 | TM_M01         | DQ21 |    | DV55 | DQ29 | DV55 |      |    |    |    |    |    |                | AVD04<br>R_X02 | AVD05<br>R_X03 |    | C  |
| D  |     |   | WR_C7<br>R_W01 | WR_C7<br>R_W01 | WR_C7<br>R_W01 | DV55 | CAB | CA7  | DV55 |      | DQ1            | DQ2  | DQ2  | DQ2  | DQ2  | DQ2  | DQ2  | WR_C7<br>R_W01 | DQ21 |    | DV55 | DQ29 | DV55 |      |    |    |    |    |    |                | AVD04<br>R_X02 | AVD05<br>R_X03 |    | D  |
| E  |     |   | WR_C7<br>R_W01 | WR_C7<br>R_W01 | WR_C7<br>R_W01 | DV55 |     |      | DV55 |      | DV55           | DV55 | DV55 | DQ22 | DV55 | DV55 | DV55 | TM_M01         |      |    |      |      |      |      |    |    |    |    |    |                | AVD04<br>R_X02 | AVD05<br>R_X03 |    | E  |
| F  |     |   | WR_C7<br>R_W01 | WR_C7<br>R_W01 | WR_C7<br>R_W01 | DV55 |     |      | DV55 |      | AVD01<br>R_X04 |      | DQ2  | DQ2  | DQ2  | DQ2  | DQ2  | WR_C7<br>R_W01 | DQ21 |    | DV55 | DQ29 | DV55 |      |    |    |    |    |    |                | AVD04<br>R_X02 | AVD05<br>R_X03 |    | F  |
| G  |     |   | WR_C7<br>R_W01 | WR_C7<br>R_W01 | WR_C7<br>R_W01 | DV55 |     |      | DV55 |      | DV55           | DQ22 |      | DQ2  | DQ2  | DQ2  | DQ2  | WR_C7<br>R_W01 | DQ21 |    | DV55 | DQ29 | DV55 |      |    |    |    |    |    |                | AVD04<br>R_X02 | AVD05<br>R_X03 |    | G  |
| H  |     |   | WR_C7<br>R_W01 | WR_C7<br>R_W01 | WR_C7<br>R_W01 | DV55 |     |      | DV55 |      | DV55           | DQ22 |      | DQ2  | DQ2  | DQ2  | DQ2  | WR_C7<br>R_W01 | DQ21 |    | DV55 | DQ29 | DV55 |      |    |    |    |    |    |                | AVD04<br>R_X02 | AVD05<br>R_X03 |    | H  |
| J  |     |   | WR_C7<br>R_W01 | WR_C7<br>R_W01 | WR_C7<br>R_W01 | DV55 |     |      | DV55 |      | DV55           | DQ22 |      | DQ2  | DQ2  | DQ2  | DQ2  | WR_C7<br>R_W01 | DQ21 |    | DV55 | DQ29 | DV55 |      |    |    |    |    |    |                | AVD04<br>R_X02 | AVD05<br>R_X03 |    | J  |
| K  |     |   | WR_C7<br>R_W01 | WR_C7<br>R_W01 | WR_C7<br>R_W01 | DV55 |     |      | DV55 |      | DV55           | DQ22 |      | DQ2  | DQ2  | DQ2  | DQ2  | WR_C7<br>R_W01 | DQ21 |    | DV55 | DQ29 | DV55 |      |    |    |    |    |    |                | AVD04<br>R_X02 | AVD05<br>R_X03 |    | K  |
| L  |     |   | WR_C7<br>R_W01 | WR_C7<br>R_W01 | WR_C7<br>R_W01 | DV55 |     |      | DV55 |      | DV55           | DQ22 |      | DQ2  | DQ2  | DQ2  | DQ2  | WR_C7<br>R_W01 | DQ21 |    | DV55 | DQ29 | DV55 |      |    |    |    |    |    |                | AVD04<br>R_X02 | AVD05<br>R_X03 |    | L  |
| M  |     |   | WR_C7<br>R_W01 | WR_C7<br>R_W01 | WR_C7<br>R_W01 | DV55 |     |      | DV55 |      | DV55           | DQ22 |      | DQ2  | DQ2  | DQ2  | DQ2  | WR_C7<br>R_W01 | DQ21 |    | DV55 | DQ29 | DV55 |      |    |    |    |    |    |                | AVD04<br>R_X02 | AVD05<br>R_X03 |    | M  |
| N  |     |   | WR_C7<br>R_W01 | WR_C7<br>R_W01 | WR_C7<br>R_W01 | DV55 |     |      | DV55 |      | DV55           | DQ22 |      | DQ2  | DQ2  | DQ2  | DQ2  | WR_C7<br>R_W01 | DQ21 |    | DV55 | DQ29 | DV55 |      |    |    |    |    |    |                | AVD04<br>R_X02 | AVD05<br>R_X03 |    | N  |
| P  |     |   | WR_C7<br>R_W01 | WR_C7<br>R_W01 | WR_C7<br>R_W01 | DV55 |     |      | DV55 |      | DV55           | DQ22 |      | DQ2  | DQ2  | DQ2  | DQ2  | WR_C7<br>R_W01 | DQ21 |    | DV55 | DQ29 | DV55 |      |    |    |    |    |    |                | AVD04<br>R_X02 | AVD05<br>R_X03 |    | P  |
| R  |     |   | WR_C7<br>R_W01 | WR_C7<br>R_W01 | WR_C7<br>R_W01 | DV55 |     |      | DV55 |      | DV55           | DQ22 |      | DQ2  | DQ2  | DQ2  | DQ2  | WR_C7<br>R_W01 | DQ21 |    | DV55 | DQ29 | DV55 |      |    |    |    |    |    |                | AVD04<br>R_X02 | AVD05<br>R_X03 |    | R  |
| T  |     |   | WR_C7<br>R_W01 | WR_C7<br>R_W01 | WR_C7<br>R_W01 | DV55 |     |      | DV55 |      | DV55           | DQ22 |      | DQ2  | DQ2  | DQ2  | DQ2  | WR_C7<br>R_W01 | DQ21 |    | DV55 | DQ29 | DV55 |      |    |    |    |    |    |                | AVD04<br>R_X02 | AVD05<br>R_X03 |    | T  |
| U  |     |   | WR_C7<br>R_W01 | WR_C7<br>R_W01 | WR_C7<br>R_W01 | DV55 |     |      | DV55 |      | DV55           | DQ22 |      | DQ2  | DQ2  | DQ2  | DQ2  | WR_C7<br>R_W01 | DQ21 |    | DV55 | DQ29 | DV55 |      |    |    |    |    |    |                | AVD04<br>R_X02 | AVD05<br>R_X03 |    | U  |
| V  |     |   | WR_C7<br>R_W01 | WR_C7<br>R_W01 | WR_C7<br>R_W01 | DV55 |     |      | DV55 |      | DV55           | DQ22 |      | DQ2  | DQ2  | DQ2  | DQ2  | WR_C7<br>R_W01 | DQ21 |    | DV55 | DQ29 | DV55 |      |    |    |    |    |    |                | AVD04<br>R_X02 | AVD05<br>R_X03 |    | V  |
| W  |     |   | WR_C7<br>R_W01 | WR_C7<br>R_W01 | WR_C7<br>R_W01 | DV55 |     |      | DV55 |      | DV55           | DQ22 |      | DQ2  | DQ2  | DQ2  | DQ2  | WR_C7<br>R_W01 | DQ21 |    | DV55 | DQ29 | DV55 |      |    |    |    |    |    |                | AVD04<br>R_X02 | AVD05<br>R_X03 |    | W  |
| Y  |     |   | WR_C7<br>R_W01 | WR_C7<br>R_W01 | WR_C7<br>R_W01 | DV55 |     |      | DV55 |      | DV55           | DQ22 |      | DQ2  | DQ2  | DQ2  | DQ2  | WR_C7<br>R_W01 | DQ21 |    | DV55 | DQ29 | DV55 |      |    |    |    |    |    |                | AVD04<br>R_X02 | AVD05<br>R_X03 |    | Y  |
| AA |     |   | WR_C7<br>R_W01 | WR_C7<br>R_W01 | WR_C7<br>R_W01 | DV55 |     |      | DV55 |      | DV55           | DQ22 |      | DQ2  | DQ2  | DQ2  | DQ2  | WR_C7<br>R_W01 | DQ21 |    | DV55 | DQ29 | DV55 |      |    |    |    |    |    |                | AVD04<br>R_X02 | AVD05<br>R_X03 |    | AA |
| AB |     |   | WR_C7<br>R_W01 | WR_C7<br>R_W01 | WR_C7<br>R_W01 | DV55 |     |      | DV55 |      | DV55           | DQ22 |      | DQ2  | DQ2  | DQ2  | DQ2  | WR_C7<br>R_W01 | DQ21 |    | DV55 | DQ29 | DV55 |      |    |    |    |    |    |                | AVD04<br>R_X02 | AVD05<br>R_X03 |    | AB |
| AC |     |   | WR_C7<br>R_W01 | WR_C7<br>R_W01 | WR_C7<br>R_W01 | DV55 |     |      | DV55 |      | DV55           | DQ22 |      | DQ2  | DQ2  | DQ2  | DQ2  | WR_C7<br>R_W01 | DQ21 |    | DV55 | DQ29 | DV55 |      |    |    |    |    |    |                | AVD04<br>R_X02 | AVD05<br>R_X03 |    | AC |
| AD |     |   | WR_C7<br>R_W01 | WR_C7<br>R_W01 | WR_C7<br>R_W01 | DV55 |     |      | DV55 |      | DV55           | DQ22 |      | DQ2  | DQ2  | DQ2  | DQ2  | WR_C7<br>R_W01 | DQ21 |    | DV55 | DQ29 | DV55 |      |    |    |    |    |    |                | AVD04<br>R_X02 | AVD05<br>R_X03 |    | AD |
| AE |     |   | WR_C7<br>R_W01 | WR_C7<br>R_W01 | WR_C7<br>R_W01 | DV55 |     |      | DV55 |      | DV55           | DQ22 |      | DQ2  | DQ2  | DQ2  | DQ2  | WR_C7<br>R_W01 | DQ21 |    | DV55 | DQ29 | DV55 |      |    |    |    |    |    |                | AVD04<br>R_X02 | AVD05<br>R_X03 |    | AE |
| AF |     |   | WR_C7<br>R_W01 | WR_C7<br>R_W01 | WR_C7<br>R_W01 | DV55 |     |      | DV55 |      | DV55           | DQ22 |      | DQ2  | DQ2  | DQ2  | DQ2  | WR_C7<br>R_W01 | DQ21 |    | DV55 | DQ29 | DV55 |      |    |    |    |    |    |                | AVD04<br>R_X02 | AVD05<br>R_X03 |    | AF |
| AG |     |   | WR_C7<br>R_W01 | WR_C7<br>R_W01 | WR_C7<br>R_W01 | DV55 |     |      | DV55 |      | DV55           | DQ22 |      | DQ2  | DQ2  | DQ2  | DQ2  | WR_C7<br>R_W01 | DQ21 |    | DV55 | DQ29 | DV55 |      |    |    |    |    |    |                | AVD04<br>R_X02 | AVD05<br>R_X03 |    | AG |
| AH |     |   | WR_C7<br>R_W01 | WR_C7<br>R_W01 | WR_C7<br>R_W01 | DV55 |     |      | DV55 |      | DV55           | DQ22 |      | DQ2  | DQ2  | DQ2  | DQ2  | WR_C7<br>R_W01 | DQ21 |    | DV55 | DQ29 | DV55 |      |    |    |    |    |    |                | AVD04<br>R_X02 | AVD05<br>R_X03 |    | AH |
| AJ |     |   | WR_C7<br>R_W01 | WR_C7<br>R_W01 | WR_C7<br>R_W01 | DV55 |     |      | DV55 |      | DV55           | DQ22 |      | DQ2  | DQ2  | DQ2  | DQ2  | WR_C7<br>R_W01 | DQ21 |    | DV55 | DQ29 | DV55 |      |    |    |    |    |    |                | AVD04<br>R_X02 | AVD05<br>R_X03 |    | AJ |
| AK |     |   | WR_C7<br>R_W01 | WR_C7<br>R_W01 | WR_C7<br>R_W01 | DV55 |     |      | DV55 |      | DV55           | DQ22 |      | DQ2  | DQ2  | DQ2  | DQ2  | WR_C7<br>R_W01 | DQ21 |    | DV55 | DQ29 | DV55 |      |    |    |    |    |    |                | AVD04<br>R_X02 | AVD05<br>R_X03 |    | AK |
| AL |     |   | WR_C7<br>R_W01 | WR_C7<br>R_W01 | WR_C7<br>R_W01 | DV55 |     |      | DV55 |      | DV55           | DQ22 |      | DQ2  | DQ2  | DQ2  | DQ2  | WR_C7<br>R_W01 | DQ21 |    | DV55 | DQ29 | DV55 |      |    |    |    |    |    |                | AVD04<br>R_X02 | AVD05<br>R_X03 |    | AL |
| AM |     |   | WR_C7<br>R_W01 | WR_C7<br>R_W01 | WR_C7<br>R_W01 | DV55 |     |      | DV55 |      | DV55           | DQ22 |      | DQ2  | DQ2  | DQ2  | DQ2  | WR_C7<br>R_W01 | DQ21 |    | DV55 | DQ29 | DV55 |      |    |    |    |    |    |                | AVD04<br>R_X02 | AVD05<br>R_X03 |    | AM |
| AN |     |   | WR_C7<br>R_W01 | WR_C7<br>R_W01 | WR_C7<br>R_W01 | DV55 |     |      | DV55 |      | DV55           | DQ22 |      | DQ2  | DQ2  | DQ2  | DQ2  | WR_C7<br>R_W01 | DQ21 |    | DV55 | DQ29 | DV55 |      |    |    |    |    |    |                | AVD04<br>R_X02 | AVD05<br>R_X03 |    | AN |

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## 9. Reference Abbreviate

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### Reference Abbreviate

- **AAC**: Advanced Audio Coding.
- **AVC** : Advanced Video Coding.
- **BER** : Bit Error Rate
- **BPSK**: Binary Phase Shift Keying
- **CA** : Conditional Access
- **CDM** : Code Division Multiplexing
- **C/I** : Carrier to Interference
- **DMB** : Digital Multimedia Broadcasting
- **EN** : European Standard
- **ES** : Elementary Stream
- **ETSI**: European Telecommunications Standards Institute
- **MPEG**: Moving Picture Experts Group
- **PN** : Pseudo-random Noise
- **PS** : Pilot Symbol
- **QPSK**: Quadrature Phase Shift Keying
- **RS** : Reed-Solomon
- **SI** : Service Information
- **TDM** : Time Division Multiplexing
- **TS** : Transport Stream

# SAMSUNG